Deforestation without Development?
Trade, Land Use, and Technology Investments*

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September 6, 2026

Most Recent Draft Here

Abstract

Does higher foreign demand for agricultural exports drive economic development through technology adoption, or simply fuel deforestation through land expansion? Adopting modern inputs requires costly investment in capital and technology that primarily benefits crop cultivation. However farmers may raise output instead by clearing forests to expand pasture. We study this choice during the 2000s commodity boom in Brazil, combining novel data on the near-universe of agricultural input purchases from cross-bank transactions with satellite data on farm-level land use. Using a shift-share shock, we find that foreign demand for agricultural exports causes farmers in the Amazon to reallocate land from capital-intensive soybean toward land-intensive pasture, reducing their purchases of machinery and technology as pasture replaces forest. We show that this response emerges as a privately optimal decision due to the asymmetric use of land and technology across agricultural activities, since abundant land can cause farmers to specialize in pasture, reducing the area under crop cultivation needed to justify technology investments. We then embed this mechanism into a general equilibrium model and provide a welfare rationale for forest protection beyond environmental externalities. Instead, we assume that credit frictions distort adoption below its social optimum and show that a perfectly enforced no-deforestation tax of around US$700 per hectare can raise the welfare gains from trade by more than 75 percent.

Keywords: Economic Development, International Trade, Agricultural Technology Adoption, Land Use, Deforestation, Credit Frictions, Environmental Policy.
JEL Codes: O11, O13, F14, F18, Q15, Q17, Q23, Q56.

*We thank Tiago Cavalcanti, Rafael Araújo, Christopher Rauh, Pierluca Pannella, Michael McMahon, Hanbaek Lee, Paul Bouscasse, Lidia Smitkova, Charles Parry, Anton Heil, Alba Patozi, Chuan-Han Cheng, Fergus McCormack, Razan Amine and seminar participants at the German Development Economics Conference (GDEC) in Kiel, the Asian Meeting of the Econometric Society (AMES) in Hong Kong, the Annual Congress of the European Economic Association (EEA) in Dublin, the EAYE Annual Meeting in Bilbao, the RCEA Conference in Madrid, the CERF Cavalcade at the Cambridge Judge Business School, and the Cambridge PhD Macro workshop for valuable comments and discussions. We acknowledge generous funding from the Cambridge Endowment for Research in Finance and the Keynes Fund PhD Awards Scheme. The views expressed in the paper are those of the authors and do not necessarily reflect those of the Banco Central do Brasil.

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1 Introduction

Trade is well established as a driver of technology adoption among manufacturing firms, in which access to foreign markets raises revenues and stimulates investment in capital and modern inputs (Lileeva & Trefler 2010, Aw et al. 2011, Bustos 2011). Yet in many low- and middle-income countries, agriculture, rather than manufacturing, accounts for a large share of output and employment. Whether trade promotes modernization in the same way in these economies is less clear. Like manufacturing firms, agricultural producers can expand output by investing to raise productivity. But farms may also expand by bringing uncultivated land into production, particularly where land supply is highly elastic (Boserup 1965). Agricultural expansion accounts for nearly 90 percent of global deforestation (FAO 2022). Trade may therefore encourage agricultural modernization, but it may instead (or simultaneously) induce expansion onto new land with potentially adverse consequences for productivity growth and the environment.

Motivated by these issues, we first ask if higher foreign demand for agricultural exports stimulates technology adoption, or simply fuels deforestation through land expansion. Second, given the relationship between modernization and land use, we ask if there is a role for environmental protection in shaping the welfare gains from trade beyond environmental externalities.

We use Brazil's experience during the agricultural commodity boom as a laboratory to address these question. Brazil offers a unique setting in which the elasticity of land supply varies dramatically within a single country. Inside the Amazon region (i.e., the Legal Amazon), forests are relatively abundant and property rights are weakly enforced, while for the rest of Brazil agricultural land is largely consolidated and the scope for land expansion is small. To analyze this setting, we construct a novel panel dataset by combining confidential records on the near universe of cross-bank electronic transfers with satellite imagery to obtain a rare window into both land use and input purchases at the farm level. Using a shift-share shock based on each farmer's exposure to non-Brazilian import demand, we find no empirical evidence that rising foreign demand stimulates agricultural technology adoption. Instead, within the Amazon, farmers reduce their expenditures on machinery and modern inputs as they reallocate land from capital-intensive soybean cultivation toward land-intensive pasture. Notably, our estimated effects persist even after controlling for each farmer's exposure to differential growth in foreign demand for beef and soy, ruling out that our results are driven by relatively higher demand for beef.

We show that this pattern emerges naturally from the asymmetric use of technological inputs and land across livestock production and crop cultivation. Returns to modernization accrue primarily to crop farming, whereas cattle ranching scales almost exclusively by converting forest into pasture. Moreover, freshly deforested land is largely unsuitable for immediate crop cultivation. To guide intuition, we start with a parsimonious model that takes these differences in technologies into account. We show that when farmers can clear forest at low cost, a decline in modernization can arise as profit-maximizing response to a proportional increase in foreign demand for livestock and crops. Farmers may specialize in pasture, reducing the area under crop cultivation needed to justify technology investments –a result that contrasts with standard models of trade and technology adoption.

To understand the role of deforestation for the welfare gains from trade, we then embed this mechanism into a rich general equilibrium model with heterogeneous farmers and endogenous technology adoption which is able to rationalize our empirical findings. Notably, the direct effect of deforestation on welfare through environmental externalities like carbon emissions, biodiversity losses, or reduced rainfall might be large. However, we deliberately shut down these channels, such that forests enter neither household utility nor firm productivity, and deforestation carries no social cost beyond the private cost of clearing. Instead, we show that environmental policy can increase the welfare gains from trade in the presence of a financial friction that distorts technology adoption below its social optimum —a well documented feature of Brazilian agriculture (e.g., Assunção et al. 2020). Specifically, an environmental policy draws high-productivity pasture farms into crop cultivation and raises technology adoption toward its efficient level, so that less deforestation increases the welfare gains from trade.

Our contribution is twofold. First, we provide novel farm-level evidence that rising export demand causes farmers to forgo technology adoption in favor of land expansion when uncultivated land is readily available. Specifically, we construct a novel dataset that combines confidential transaction records from the Brazilian central bank with geo-referenced farm boundaries and satellite imagery, allowing us to trace beef and soy exports back to individual farms and observe both land use changes and input purchases over time. Because farmers may raise yields through different types of investments, we split farm expenditures into machinery (e.g., tractors and harvesters), durable capital and irrigation (e.g., silos and irrigation systems), and technology inputs (e.g., agronomic services and precision agriculture).

We then estimate how land use and investment decisions respond to a shift-share shock that interacts each farmer's pre-determined destination exposure with non-Brazilian import demand, isolating foreign demand from domestic supply conditions. Specifically, agricultural trade in Brazil is almost entirely intermediated by designated export firms such that our share component chains exposure shares of farms to different exporters with the exposure shares of exporters to different export-destination countries. In response to our shift-share foreign demand shock, our regression results then suggest that exporters increase their supply disproportionately more from farms located within the legal Amazon where farmers respond by shifting land from capital-intensive soybean toward land-intensive pasture, while spending on machinery and technology falls. We also find that forest cover within the Amazon declines as farmers clear uncultivated land for pasture. In a complementary difference-in-differences design, we exploit the timing of farmers' use of rural credit and show that receiving rural credit raises spending on modern inputs and shifts land from pasture toward soybean, encouraging modernization. Our empirical results present strong evidence that agricultural modernization follows changes in farmers' land use mix. In contrast to pasture, cultivating soybean requires machinery and agronomic inputs, such that the demand for technology inputs falls in tandem with the share of land under crop cultivation.

In our second contribution, we show that a sufficiently strong deforestation tax can increase the welfare gains from agricultural trade without resorting to assumptions about carbon externalities, biodiversity losses, or climatic changes that affect firm productivity or household utility. Here, we build an open economy general equilibrium model to understand how an agricultural export boom drives deforestation and affects technology adoption. We assume the economy is populated by a non-agricultural sector with heterogeneous firms supplying consumption, intermediate, and technology goods alongside an agricultural sector where farmers produce both crops and livestock. Within agriculture, we introduce two sources of heterogeneity: farmers draw an idiosyncratic productivity shock and a cost of clearing forest. We use the clearing cost draw to capture variation in land scarcity and property rights, which allows us to capture geographic variation of land characteristics without explicitly modeling spatial units. Upon observing the respective draws, farmers allocate land on their owned property between crop cultivation and pasture. Drawing on agronomic evidence, we assume that crops and pasture differ in their underlying use of technology and land such that the returns to adopting modern technology accrue solely to crop cultivation, while only pasture can replace forest. Furthermore, we take into account that inherent land heterogeneity limits the substitutability between pasture and crops as well as between the established owned plot and newly deforested pasture. Farmers thus decide how to allocate the owned plot between crops and pasture, how many additional hectares to deforest, and whether to adopt modern crop technology.

A key feature of the equilibrium is that even high-productivity farms may find it optimal not to adopt the modern technology if their clearing cost is sufficiently low—a feature we observe directly in the data. We then show that under a commodity boom where prices for livestock and crops rise proportionally, farms drawing a low clearing cost (e.g., farms with access to abundant land or weakly enforced property rights) find it optimal to specialize further into pasture and abandon the modern technology, effectively de-modernizing their production.

We calibrate the model to the beginning of the commodity boom period in 2003 for Brazil using granular micro-data, aggregate statistics, and our own empirical estimates to discipline the parameter space in our model. We show that our framework is able to match a range of key untargeted moments that capture differences in land use, farm scale, and modernization between the Legal Amazon and the rest of Brazil. Furthermore, the implied share of modern firms in each sector matches proxies that we construct from Brazilian firm surveys and the agricultural census. Equipped with the calibrated model that can rationalize our empirical findings on the effect of foreign demand on land allocation and technology spending, we then ask how an environmental policy shapes the welfare gains from trade. Therefore we introduce a perfectly enforced per-hectare environmental tax on deforested pasture that is rebated lump-sum to the household. In our model, we assume that farms face financial restrictions, a well established fact for Brazilian agriculture, such that farmers must take out an intra-period loan to finance the fixed adoption cost. The interest is a transfer to households, so the private cost of adopting exceeds its social cost. Hence, in equilibrium, the agricultural sector adopts modern technology below the social optimum. An environmental tax then forces farmers to scale up crop cultivation, directly raising the return on technology investments. The policy consequently distorts the deforestation margin, but also draws highly productive pasture farms into modern agriculture and lifts technology adoption toward the efficient level.

In a counterfactual exercise we show that a perfectly enforced no-deforestation tax of around US$700 per hectare increases the welfare gains from trade by more than 75 percent. Importantly, welfare gains from trade only materialize after a reduction in aggregate deforestation of more than 90% (at a tax above US$170 per hectare). This is because a moderate tax distorts land allocation without raising crop returns enough to cover the fixed modernization cost, while a sufficiently strong policy forces farmers to reallocate enough land into crop cultivation to fully justify the investment.

Related Literature. Our paper connects to several strands of the literature. First, a growing body of work studies how international trade affects agricultural production and deforestation. Several papers document that commodity booms increase tropical forest loss (Pendrill et al. 2019, Abman & Lundberg 2020, Da Mata & Dotta 2021, Harding et al. 2021), while quantitative models of agricultural trade emphasize that trade promotes modernization and can be land-saving. Farrokhi & Pellegrina (2023) show that trade lowers the cost of intermediate inputs, promoting the adoption of modern technology and reducing the total amount of land devoted to agriculture. Pellegrina (2022) studies how soy adaptation reshaped the spatial organization of Brazilian agriculture, and Farrokhi et al. (2025) build a dynamic trade model in which trade reduces deforestation globally by shifting production toward more productive regions. Carreira et al. (2024) disentangle trade from technology at the municipality level and find that soy seed adoption rather than Chinese export demand drove deforestation. We depart from the literature by using novel farm level data, as opposed to regional or municipality level data, that allows us to trace input purchases and land use changes to individual farms, directly observing the farmer's choice between adopting modern technology and expanding onto new land. We show that the reallocation of land between crops and livestock within a farm can dictate the farmer's modernization decisions, due to their asymmetric use of technology and land, which the literature has not taken into account.

Second, our paper contributes to a broad literature that views international trade as a driver of economic growth and modernization. A large body of evidence finds that openness raises income and productivity across countries (Frankel & Romer 1999, Alcalá & Ciccone 2004), that wider exposure to foreign markets leads firms to adopt modern technology, invest in R&D, and upgrade product quality (Bustos 2011, Aw et al. 2011, Lileeva & Trefler 2010, Verhoogen 2008, Bloom et al. 2016), and that trade raises aggregate productivity by reallocating activity toward the most productive firms (Melitz 2003, Bernard et al. 2007). A parallel body of growth theory formalizes how trade can accelerate innovation and the diffusion of technology (Grossman & Helpman 1991, Rivera-Batiz & Romer 1991, Sampson 2016, Perla et al. 2021), and quantitative models trace sizable welfare gains to sectoral linkages and trade in intermediate goods (Caliendo & Parro 2015). Taken together, these studies find that larger markets raise the return to modern technology, so that trade and modernization advance together. Almost all of the firm level evidence comes from manufacturing. However for the agricultural sector we show that taking into account cheap land to deforest and extensify production can lead to an optimal reallocation away from adopting modern technology. Our idea that agriculture responds to trade differently from manufacturing is also linked to earlier work on structural transformation by e.g, Matsuyama (1992) who shows that in an open economy, a comparative advantage in agriculture can slow growth when manufacturing has learning by doing externalities.

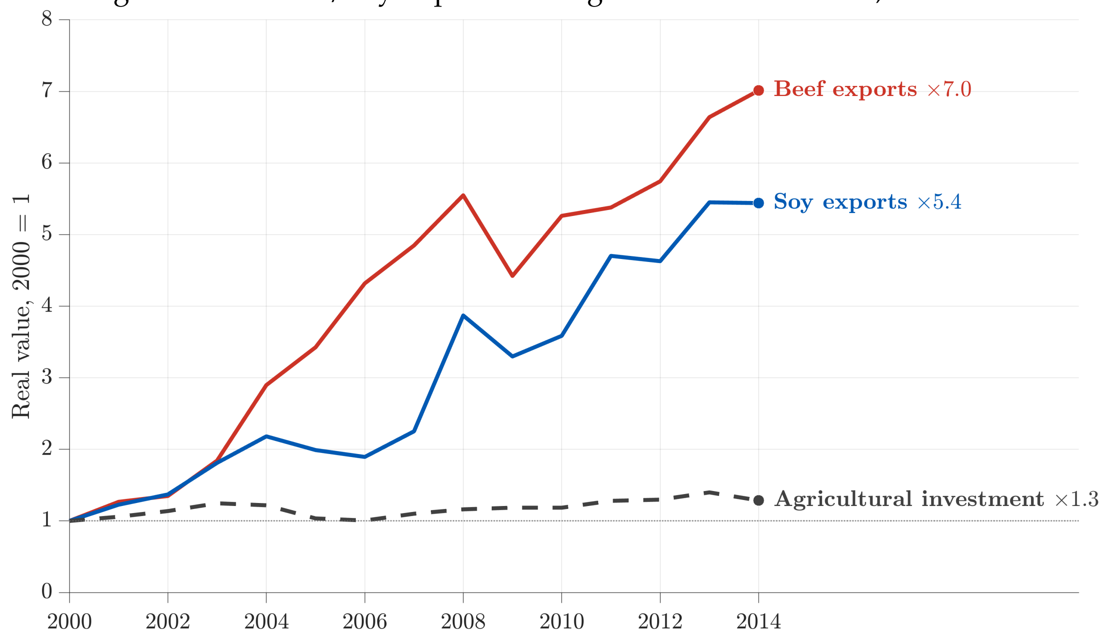
Third, we relate to a large literature on environmental policies and deforestation, which finds that policy instruments such as satellite monitoring, higher enforcement, and restrictions on rural credit can reduce deforestation (Assunção et al. 2015, Assunção, McMillan, Murphy & Souza-Rodrigues 2023, Assunção et al. 2020, Assunção, Gandour & Rocha 2023). Souza-Rodrigues (2019) provides a structural framework for estimating the demand for deforestation in the Amazon and finds that taxes on agricultural land can be highly cost-effective in preserving the rainforest. Dominguez-Iino (2026) finds a downstream tax on agribusiness a poorly targeted substitute for a tax on farmers, because agribusiness buyers with monopsony power pass little of the tax through to the upstream frontier where emissions are most intense. Araujo (2024) studies how deforestation feeds back onto agricultural productivity through altered rainfall, and Araujo et al. (2026) estimate the efficient level of forest conservation accounting for the carbon externality. We depart from this literature by finding a welfare rationale for an environmental tax without relying on an externality such as carbon or rainfall, but rather due to incentivizing farmers to adopt modern technology.

Fourth, we relate to the broad literature on the drivers and consequences of tropical deforestation (see Balboni et al. 2023, for a recent survey). Early work established that clearing follows the economic returns to land, with roads, transport costs, and output prices shaping environmental outcomes (Chomitz & Gray 1996, Barbier & Burgess 1997, Pfaff 1999), while later work emphasizes the political incentives of local governments (Burgess et al. 2012). Closest to our work is Szerman et al. (2022) who show that a productivity gain from electrification favors crops and leads credit-constrained farmers to reallocate land away from land-intensive grazing and into capital-intensive crop cultivation, lowering total land demand, with conservation benefits comparable to the most prominent conservation packages ever implemented in Brazil. In our paper, we use a similar mechanism arguing that modernization gains predominantly occur to crop cultivation, not livestock. We also take into account that crops are largely unsuitable for cleared land. By combining the asymmetric use of technological inputs and land, we show that a foreign export boom can optimally drive farmers to extensify rather than modernize.

2 Background

Over the past two decades, Brazil became the world's largest exporter of both beef and soybeans. During the commodity boom between 2000 and 2014, real agricultural exports for beef and soy grew more than sevenfold and fivefold respectively, driven largely by rising demand from China, where rapid income growth translated into higher consumption of livestock products requiring soy feed (Nepstad et al. 2006, Costa et al. 2016) (see Figure 1). Yet export growth was not accompanied by a corresponding increase in agricultural demand for investment goods. Instead, agricultural spending on capital goods remained relatively flat, a fact difficult to reconcile with standard models of export driven growth, where expanded access to foreign markets raises firm revenues and stimulates technology adoption (e.g., Lileeva & Trefler 2010, Bustos 2011, Aw et al. 2011). We would therefore expect the agricultural sector to increase purchases of capital goods alongside rising exports.

Figure 1: Real Beef/Soy Exports and Agricultural Investment, 2000–2014



Notes: All series are normalized to one in 2000 and labels report the 2014 value relative to 2000. Soy exports are the sum of FAOSTAT export values for soya beans, soya bean cake, and soya bean oil. Beef exports are the sum of FAOSTAT export values for boneless cattle meat, bone-in cattle meat, beef and veal preparations, edible cattle offal, live cattle, and salted, dried, or smoked bovine meat. Export values are deflated using the US consumer price index and expressed in constant 2003 US dollars. Agricultural investment is gross fixed capital formation in agriculture, forestry, and fishing in Brazil, from the FAOSTAT Capital Stock domain, expressed in constant 2015 reais.

The export boom also coincided with one of the largest expansions of agricultural land in modern history. Between 2000 and 2020, crop and pasture area expanded substantially in the Amazon and Cerrado biomes (Zalles et al. 2019, Souza Jr et al. 2020). In the Amazon alone, 18.9 million hectares of forest were cleared between 2000 and 2006, driven predominantly by the expansion of cattle pasture (Barona et al. 2010). However, crops and pasture differ fundamentally in their land suitability and use of technological inputs. We document these differences in the following two observations, which are well established in natural sciences but largely overlooked in the economics literature.

Observation 1: *Modernization gains predominantly accrue to crop production instead of livestock.*

It is well understood that crop yields respond strongly to purchased inputs such as machinery, chemical fertilizers, and improved seed varieties (e.g., Rada 2013, Goldsmith 2008). Just improved seed varieties alone can raise grain yields by roughly two percent a year (Umburanas et al. 2022). Mechanization can then further amplify productivity by facilitating practices like double-cropping, allowing a single plot to be harvested multiple times within a year (e.g., Goldsmith & Montesdeoca 2018). However, beef production does not offer comparable returns to capital due to inherent physiological limits on cattle growth and feed conversion. Roughly seventy percent of a cow's energy is allocated to basic biological maintenance rather than to gain (e.g., Ferrell & Jenkins 1985) which means that livestock eventually reaches a physiological limit on how much excess meat can be generated per animal, regardless of the investment into feed or management (e.g., Owens et al. 1995, Webb & Casey 2010). Consequently, the returns to modernization predominantly concentrate within crop farming.

Observation 2: *Pasture is the main driver of deforestation, while crops are largely unsuitable for freshly cleared land.*

The literature also documents that freshly cleared forests are often poorly suited for crop cultivation due to acidic soils that are short of essential nutrients (e.g., Sanchez et al. 1982, Lopes 1992). Converting tropical forests into cropland therefore takes years of costly soil correction before a first harvest, whereas cattle can graze deforested land as soon as it is cleared. Pasture is thus often the first use of newly cleared forest, while crops remain on consolidated land (e.g., Macedo et al. 2012). Reflecting this asymmetry, cattle pasture occupies more than 60 percent of deforested land in the Brazilian Amazon, compared to under five percent for crop cultivation (Almeida et al. 2016).

Both observations are also visible in aggregate statistics on agricultural investment and land use. Between 2000 and 2012, Brazilian crop farming purchased on average roughly ten times as much machinery and equipment compared to livestock. On the other hand, out of roughly 40 million hectares of forest converted for agricultural use, only 3 million hectares transitioned into cropland while 37.6 million hectares were converted into pasture (see Appendix Figure F.1).

3 A parsimonious framework

To guide intuition, we start by setting up a simple parsimonious framework where a representative farmer maximizes profit over crop and livestock production given technologies that reflect the asymmetric use in land and technology. Importantly, we also take into account that inherent land heterogeneity prevents land from being perfectly substitutable between established farmland (i.e., the owned plot) and newly deforested pasture. We then use this framework to demonstrate how cheap land for pasture expansion competes with crop production and can depress capital spending during an export boom.

Specifically, we assume that crops y_c and beef y_p are supplied by a representative farmer who takes the respective prices p_c p_p as given, hires capital at an exogenous rental rate r, hires labor at a wage that we normalize to one, and clears forest at a constant cost c per area unit. The farmer is endowed with a plot of owned land of unit size that can be split between crop use l_c and pasture use l_p,

$$l_c + l_p \;\leq\; 1. \tag{1}$$

In line with *Observation 1* and *Observation 2*, we further assume that crops can only be produced on owned land l_c, capital k, and labor n_c, whereas cattle is produced with labor n_p on a land composite $\bar{l}_p$ that bundles owned pasture l_p and freshly cleared forest d,

$$y_c \;=\; l_c^{\gamma_c}\, k^{\nu_c}\, n_c^{1-\gamma_c-\nu_c}, \qquad y_p \;=\; \bar{l}_p^{\gamma_p}\, n_p^{1-\gamma_p} \;\text{ where }\; \bar{l}_p \;=\; \left(l_p^{\frac{\varepsilon_p-1}{\varepsilon_p}} + d^{\frac{\varepsilon_p-1}{\varepsilon_p}}\right)^{\frac{\varepsilon_p}{\varepsilon_p-1}}, \tag{2}$$

Here, we assume that only crops use capital, reflecting *Observation 1*, and freshly cleared land only enters pasture, reflecting *Observation 2*. Key to our mechanism is to allow owned pasture l_p and freshly cleared forest d to be imperfect substitutes in cattle production with elasticity $\varepsilon_p > 1$, which captures well-known differences in land characteristics such as ranching infrastructure and market access (see, e.g., Margulis 2004, Chomitz & Gray 1996). The representative farmer then solves

$$\pi \;=\; \max_{l_c,\, l_p,\, k,\, d,\, n_c,\, n_p} \; p_c\, y_c + p_p\, y_p \;-\; r\, k \;-\; (n_c + n_p) \;-\; c\, d \qquad \text{s.t.} \qquad l_c + l_p \leq 1. \tag{3}$$

Furthermore, we assume that the farmer faces isoelastic foreign demand for crops and livestock with a common elasticity $\eta > 1$ such that in equilibrium

$$p_c \;=\; \rho\, p^w\, y_c^{-1/\eta} \quad \text{and} \quad p_p \;=\; (1-\rho)\, p^w\, y_p^{-1/\eta} \tag{4}$$

where p^w captures the common component in exogenous agricultural world prices and $\rho \in (0, 1)$ governs the relative demand for crops versus livestock.

We apply this setup to an increase in the world price for agricultural goods p^w. Note that a rise in the price of beef alone would shift the farmer toward cattle through a change in the relative price, which we deliberately rule out so that any reallocation comes from the differential use in land and technology between crops and pasture.

We are interested in whether the export boom, captured by an increase in p^w, leads the farmer to intensify production, by reallocating land into capital-intensive crop cultivation, or to extensify production, by raising cattle on deforested land d. Here we deliberately assume that freshly cleared forest substitutes imperfectly for owned pasture (e.g., due to the lack of fencing, water, and road access) with an elasticity $\varepsilon_p < \infty$. The farmer therefore raises cattle by partly combining cleared forest d with owned pasture l_p, which in turn competes with land dedicated to crops l_c. This means a farmer whose clearing cost c is sufficiently low to induce extensification consequently also reallocates land away from crops, which can lower crop revenues and capital spending altogether. We formalize this mechanism in Proposition 1.

Proposition 1. *Let $s_c \equiv l_c/(l_c + l_p + d)$ be the crop share of farmed land and rk^* the farmer's spending on capital goods. Then there exist clearing costs $0 < \bar{c} < c^\star$ such that a boom $dp^w > 0$*

(i) (Extensification.) *if* $\gamma_c < \gamma_p$ *and* $1 < \varepsilon_p < 1 + \gamma_c(\eta - 1)$*, shifts land use toward pasture whenever clearing is cheap,* $c < c^{\star}$*,*

$$\frac{\partial l_c^*}{\partial p^w} < 0, \qquad \frac{\partial d^*}{\partial p^w} > 0, \qquad \frac{\partial s_c^*}{\partial p^w} < 0;$$

(ii) (De-modernization.) *if* $\gamma_c < \gamma_p$ *and* $1 < \varepsilon_p < \gamma_c(\eta - 1)$*, reduces capital spending where clearing is cheap and raises it elsewhere,*

$$\frac{\partial (rk^*)}{\partial p^w} < 0 \quad \text{if } c < \bar{c}, \qquad \frac{\partial (rk^*)}{\partial p^w} > 0 \quad \text{if } c > \bar{c}.$$

The proof is in Appendix J.1.

To understand the mechanism of Proposition 1, note that the equilibrium change in crop revenue satisfies

$$d\ln(p_c\, y_c) \;=\; \underbrace{\frac{\eta}{1+\gamma_c(\eta-1)}}_{\text{direct price effect}}\; d\ln p^w \;+\; \underbrace{\frac{\gamma_c(\eta-1)}{1+\gamma_c(\eta-1)}}_{\text{land reallocation}}\; d\ln l_c^*. \tag{5}$$

Hence, crop revenue increases with both the world price and the crop area l_c. With an increase in p^w, crop revenue therefore only falls if the reallocation away from l_c is large enough to outweigh the increase in p_c. This is important because investment co-moves with crop revenue, by the first-order condition $rk = \nu_c\, p_c\, y_c$. A farmer reallocates land away from crops if and only if the boom raises the value of a hectare in pasture more than a hectare in crops, which ties the underlying clearing cost c directly to the level of crop modernization k.

Suppose that clearing forest is cheap (e.g., in a situation where property rights are weakly enforced and land is abundant and cheap to clear), so the farmer finds it optimal to extensify and deforests d. By assumption, the cleared forest d only partly substitutes for owned pasture, and thus also raises the value of the owned pasture l_p operated alongside the new land d. Formally, combining the first-order conditions for owned pasture l_p and cleared forest d, an additional hectare of owned pasture raises the farmer's profit from cattle by $c\,(d/l_p)^{1/\varepsilon_p}$, which increases in d. The reallocation therefore depends on the clearing cost only through the amount of cleared land the farmer operates alongside owned pasture. In the limit of free clearing, the reallocation takes the closed form

$$\lim_{c \to 0} \frac{d\ln l_c^*}{d\ln p^w} \;=\; -\,\frac{\eta\,\left[1+\gamma_c(\eta-1)-\varepsilon_p\right]}{\varepsilon_p}, \tag{6}$$

which is negative if and only if $\varepsilon_p < 1+\gamma_c(\eta-1)$ and thus recovers exactly the parametric condition of Proposition 1(i). Given a low enough clearing cost c, land therefore moves toward pasture whenever cleared forest substitutes sufficiently poorly for owned pasture relative to foreign demand feedback for crops, $d\ln p_c = -\frac{1}{\eta}\, d\ln y_c$ from (4), and to the land share of the crop activity γ_c. Note that the price feedback is weak when world demand is elastic (high η), because otherwise a decline in the crop supply would raise p_c such that the farmer may no longer find it optimal to specialize in cattle production.

Under this condition, we then show in Proposition 1 that below a threshold $c^{\star}$, the boom raises the value of a hectare by more in pasture than in crops such that the farmer reallocates land away from crops, $\partial l_c^* / \partial p^w < 0$. Below an even stricter threshold $\bar{c} < c^\star$, the loss from the smaller crop area exceeds the gain from the higher price such that crop revenue falls, and the farmer consequently spends less on capital goods, $\partial(rk^*)/\partial p^w < 0$. Importantly, we think the parametric restriction for Proposition 1 is a relatively mild assumption. Pellegrina (2022) estimates land shares of $\gamma_c = 0.50$ for soy, 0.49 for wheat, and 0.41 for corn, so the threshold is approximately $(1+\eta)/2$ and rises with the elasticity of world demand, which is plausibly high for homogeneous export commodities (e.g., Broda & Weinstein 2006).[1] Under an illustrative $\eta = 5$, part (i) requires $\varepsilon_p < 3$ and part (ii) $\varepsilon_p < 2$.

Also, note that here we completely abstract from any gains from specialization within the owned plot. The literature documents substantial gains from concentrating agricultural production in a single use, arising from indivisible equipment, use-specific knowledge, and homogeneous land suitability within plots (e.g., Holmes & Lee 2012). When a boom raises agricultural prices, farms with low clearing costs that already operate large pastures would reallocate owned land even more aggressively toward pasture under such a specialization force, strengthening the extensification channel and relaxing the parametric restriction on ε_p. We introduce this channel in our general equilibrium model in Section 9.3.

In Proposition 1, we establish that taking into account the differential use of land and technology between crop and livestock production, a proportional commodity boom can have very different effects on technology adoption within the agricultural sector depending on the cost of deforestation and the degree of substitutability between land already under agricultural production and land available for expansion. In particular, a decline in modernization may arise as a profit-maximizing decision when cheap uncultivated land is available, a result that contrasts with the standard literature on trade and technology adoption.

4 Data and Sample

We exploit confidential data on the near universe of cross-bank electronic transfers from the Central Bank of Brazil (BCB) between 2010 and 2020. Linking these transaction records to farm-level land use outcomes and export exposure, however, presents several challenges. First, agricultural establishments in Brazil are often not registered as formal firms with a corporate tax identifier (cnpj). Instead, land typically belongs to individuals identified only by a personal tax identifier (cpf), which means we must trace transactions of individuals rather than firms. Since those personal accounts may also be used for non-agricultural purchases, we exclude transactions directed to suppliers in sectors that are unlikely to represent farm inputs, such as gambling, tobacco, domestic services or entertainment.[2] Second, agricultural trade in Brazil is predominantly intermediated by large trading companies like Bunge, Cargill, and JBS, so measuring exports directly at the farm level would be futile. We instead link farms to downstream exporters through payment relationships to measure each farm's exposure

[1] Costinot et al. (2016) estimate η at 5.4 for ten major crops.

[2] Specifically, we exclude payments to suppliers classified under the following 2-digit cnae divisions: 12 (tobacco products), 58 (publishing), 59 (film, video, and sound recording), 79 (travel agencies and tour operators), 90 (arts, creative activities, and entertainment), 92 (gambling and betting), 93 (sports, recreation, and leisure), 96 (other personal services), 97 (domestic services), and 99 (international organizations).

to changes in foreign demand. Third, while Brazil's Environmental Rural Registry (CAR) provides geo-referenced boundaries for over 6 million rural properties, the registry available to the BCB does not identify the individuals or firms operating each property. We overcome this limitation by linking CAR properties to borrowers in Brazil's Rural Credit System, which requires farmers to register their property when obtaining credit. Fourth, the individual operating a farm is often not the landowner but rather a manager (or farmer) whose identifier would not appear in property registries in the first place. We address this by assuming that input purchases are made from the same account holder that receives rural credit, allowing us to link transaction flows to specific properties even when the farmer differs from the owner and is not registered as a formal firm. Combining these data sources, we construct a novel dataset that traces beef and soy exports back to individual farms and observes both their land use changes and their input purchases over time.

Brazilian Payment System. We use confidential information on the universe of cross-bank electronic transfers processed through the central clearing houses of the Central Bank of Brazil (BCB). We draw on data from the *Sistema de Transferência de Reservas* (STR) and the *Sistema de Transferência de Fundos* (SITRAF) between 2003 and 2020. STR and SITRAF process *Transferências Eletrônicas Disponíveis* (TEDs), which are real-time, gross settlement wire transfers used for the majority of payments in Brazil during our sample period. Following Kamkar et al. (2026), we exclude transactions involving financial institutions, government transfers, and tax payments. Specifically, we remove interbank transfers facilitating liquidity management, transfers related to social security contributions, judicial deposits, and the internal movement of funds within financial institutions. We further restrict the sample to non-financial firms that transact a minimum aggregate value of 10 BRL annually to eliminate technicalities such as penny checks. In 2019, the cleaned sample covers approximately 9.8 million unique firms and 155.7 million unique transaction pairs. The data captures the universe of cross-bank TEDs which make up more than 90% of all cross-bank transactions.[3] Transactions where the sender and recipient hold accounts at the same bank remain unobserved. Kamkar et al. (2026) show that the share of unobserved within-bank transactions corresponds to the Herfindahl-Hirschman Index of bank concentration. They estimate that unobserved flows account for less than 15 percent of non-local transactions and roughly 25 percent of local transactions. Hence, we argue that the observed cross-bank transactions offer a meaningful window into trade relationships in Brazil.

Land Use Outcomes. We exploit the Environmental Rural Registry (CAR) which contains geo-referenced boundaries for over 6 million rural properties in Brazil. Furthermore, we match the property boundaries with annual satellite imagery from MapBiomas, which provides land cover classification at a 30$\times$30 meter resolution for the entire Brazilian territory. By overlaying the georeferenced CAR shapes onto the raster data, we calculate the annual area allocated to specific uses within each property, including forest formation, pasture and soybean.

Rural Credit System. We identify agricultural producers and their associated CAR property using

[3]Our data does not contain paper-based cheques (processed through *Centralizadora da Compensação de Cheques e Outros Papéis*, COMPE), *Documentos de Ordem de Crédito* (DOCs), which are low-value next-day transfers, *Transferências Especiais de Crédito* (TECs), a system used for batch payments, or payments from the instant payment system Pix, which was launched at the end of the observed period in November 2020. The share of unobserved payments in the total transaction volume is small, around 8 percent on average between 2006 and 2020 (Kamkar et al. 2026).

Brazil's Rural Credit System *Sistema de Operações do Crédito Rural e do PROAGRO* (SICOR) from 2013 to 2020. More specifically, we exploit the requirement that borrowers must register their property in the Environmental Rural Registry (CAR) to link borrower tax identifiers (cnpj or cpf) with CAR properties.[4] Importantly, we assume that a farmer observed on a specific CAR property in SICOR also operated on the same property anytime between 2010 and 2020 as long as the property had more than 5% of its property area covered by farming activities. We also use data from *Registro Comum de Operações Rurais* (RECOR) between 2000 and 2012 to identify all cpfs/cnpjs that are responsible for farming. Even though RECOR contains no links to CAR properties, we still want to capture transactions between informal agricultural producers (those who are not registered with a cnpj) since, for example, soy is a major input into the production of beef. This gives us a broad sample of Brazil's agricultural sector that captures all producers that have a record in the rural credit system since 2000 with over 8 million unique tax identifiers (borrowers) and around 2.3 million unique CAR properties.

Export Data. We exploit granular customs data from the TRASE initiative to measure agricultural exports for the beef and soy supply chains. TRASE combines per-shipment trade records with sanitary inspection logs to construct a firm-by-year-by-destination panel, covering 2004 to 2020 for soy and 2010 to 2020 for beef (excluding 2018). Since TRASE identifies exporters by name, we manually match the entities to their corporate tax identifiers (cnpj) using public records. Most beef and soy exports are intermediated by trading companies, so the set of exporters is largely different from the set of agricultural producers, and we recover the link between the two through the payment data described above.[5] Figure F.2 (Appendix) plots the cumulative distribution of average yearly export values for beef and soy exporters. The figure highlights the concentration in agricultural exports, where a few dominant intermediaries, such as Bunge, Cargill, and ADM for soy, and JBS, Minerva, and Marfrig for beef, control the majority of trade volume.

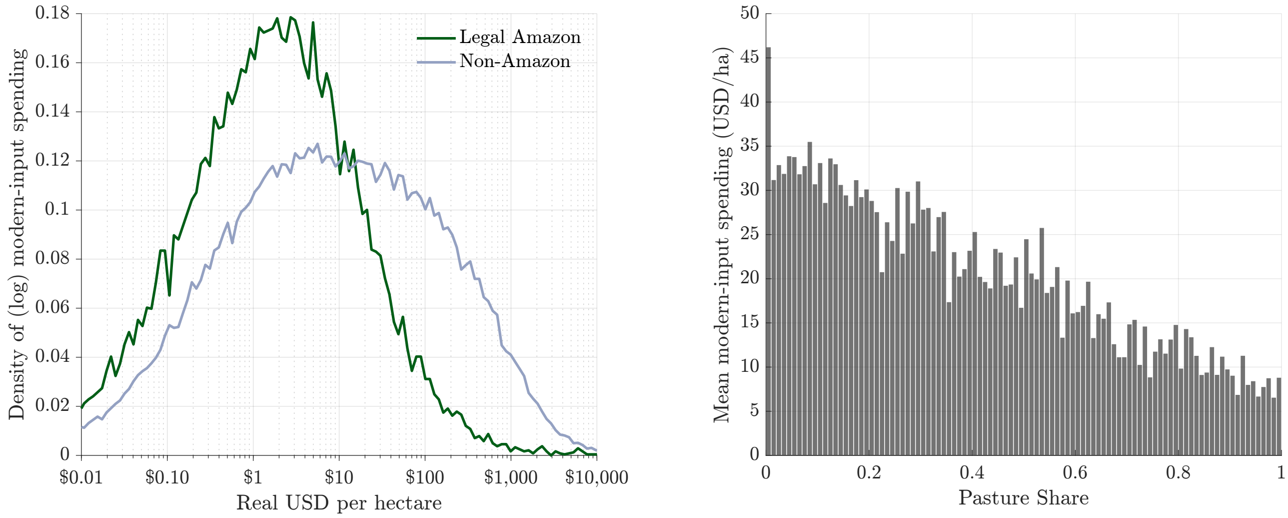
First, we consider all farmers in our panel regardless of any exporter link and aggregate all payments directed to their suppliers classified as producing goods in agricultural machinery, equipment, and technology services. Then we compute each farmers mean annual spending per hectare of property area in real USD.

We then document in Figure 2 that on average, the level of spending into modern inputs like capital, machinery and technology is an order of magnitude lower in the Amazon compared to the rest of Brazil. Additionally, reflecting *Observation 1*, we also show that the average spending on modern inputs falls from roughly US$46 to US$9 per hectare (a decline of over 80 percent) as the pasture share rises from zero to one. In the left panel of Figure 2 we plot the distribution of per-hectare spending among farmers, separately for the Legal Amazon and the rest of Brazil. In the Legal Amazon, per-hectare spending concentrates between one and three dollars, while outside the Legal Amazon the distribution shifts rightward by roughly an order of magnitude, with substantial mass above one hundred dollars per hectare. In the right panel of Figure 2, we plot average per-hectare spending

[4]The borrower represents the active agent managing the farm's operations and purchasing inputs, whereas the land owner may be an absentee landlord detached from daily production decisions or be the borrower herself.

[5]Of the 1,523 exporters in TRASE, 89 are classified as agricultural producers themselves and we drop them, accounting for 1.3 percent of total export value. We then match 92 percent of the remaining TRASE exporters to at least one upstream farmer through the payment data, covering 97 percent of total beef and soy export value.

Figure 2: Modern input spending per hectare, by region and by pasture share



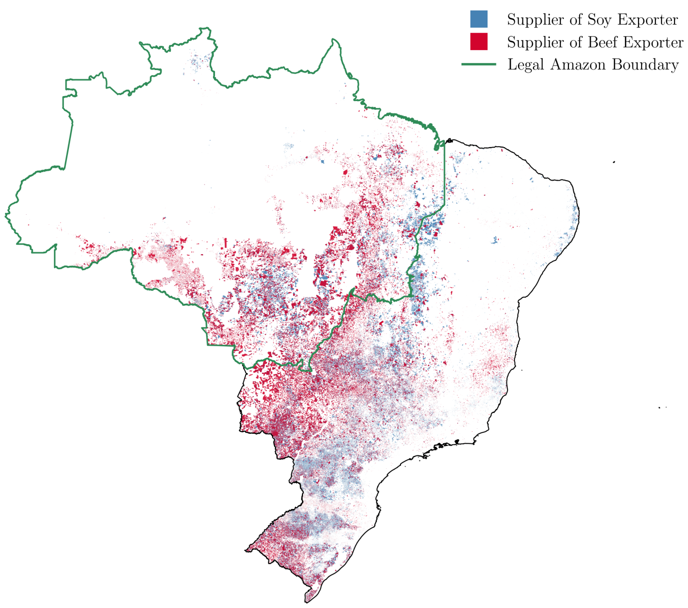
Notes. The sample includes all farmers with at least 5 percent of property area under crops or pasture who are observed making payments between 2010 and 2020. We classify modern inputs as payments to suppliers of agricultural machinery, capital goods and irrigation equipment, and technology goods and services (Table E.1), deflated to constant USD, and compute spending per hectare of property area. *Left panel:* density of log spending per hectare among farmers with positive spending; 20,017 farmers (40,082 unique CAR properties) in the Legal Amazon and 113,552 farmers (260,339 unique properties) in the rest of Brazil. *Right panel:* mean spending per hectare by pasture share of farmed area (100 bins), including farmers with no observed modern-input spending; 640,321 farmers (949,589 unique CAR properties), of whom 20.7 percent record positive spending in modern inputs. Data is Winsorized at the 99th percentile ($2,576/ha).

binned by pasture share.

Next, we investigate farmer-exporter linkages. Figure 3 maps all CAR properties in our sample that are linked to beef or soy exporters through the payment linkages described above, covering over 460,000 individual properties. Since each farmer may operate multiple properties, the map displays the underlying property-level geography rather than the farmer-level panel used in estimation. The final regression sample contains 128,272 unique farmers and 352,040 farmer-year observations between 2010 and 2020, where each farmer's land-use outcomes aggregate across all CAR properties associated with that farmer's tax identifier. We classify a farmer as being in the Legal Amazon if more than 50 percent of the farmer's CAR properties fall within the Legal Amazon boundary. The Amazon subsample contains 30,295 farmers (99,499 observations) and the non-Amazon subsample 98,009 farmers (252,541 observations).

Table F.1 in Appendix F reports summary statistics at the farmer-year level, separately for the Amazon and non-Amazon subsamples. The data reveal substantial heterogeneity across the two regions. Farmers in the Amazon operate on larger properties, roughly 3,000 hectares on average compared to under 1,800 outside, and allocate a larger share of their land to pasture (about 46% vs. 21%) and forest (about 19% vs. 11%). Amazon farmers are also more heavily exposed to beef exporters, with nearly 60% of their export exposure flowing through beef on average, compared to about 30% outside the Amazon. Spending on machinery, capital goods, and technology services per hectare is an order of

Figure 3: Spatial Distribution of Agricultural Producers Linked to Exporters



Notes: The map plots the geo-referenced locations of all agricultural properties in our sample between 2010 and 2020 that are linked to beef (red) or soy (blue) exporters via transaction records. The green line denotes the Legal Amazon boundary. The sample includes 202,674 properties linked to beef exporters, 347,487 linked to soy exporters, and 88,394 linked to both.

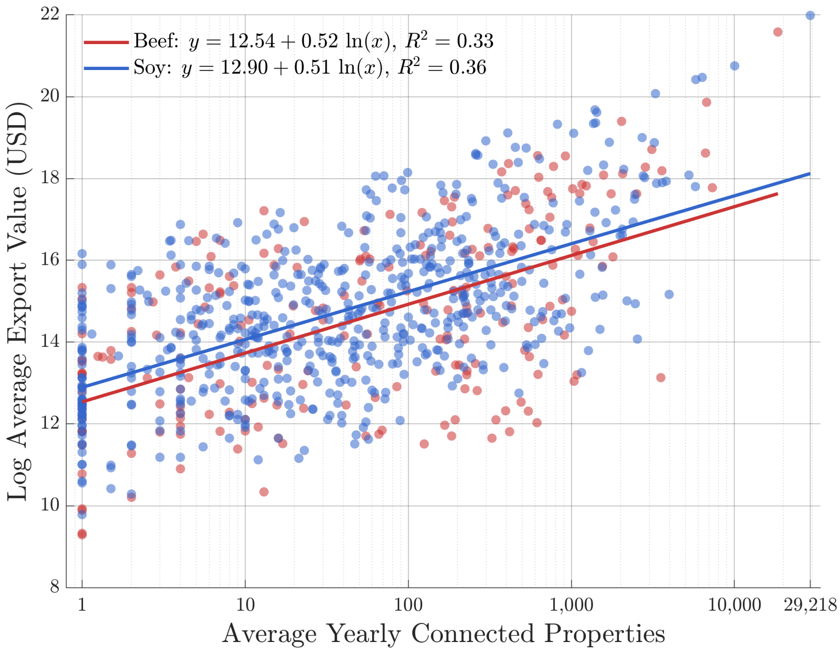
magnitude lower in the Amazon than outside, consistent with the low capital intensity of pasture-based production described in Section 2. Outside the Amazon, farmers allocate a larger share of their land to soybeans (about 34% vs. 20%) and spend substantially more per hectare on equipment and services. Table F.2 in Appendix F reports summary statistics split by beef and soy linkages.

Figure 4 plots each exporter's average yearly export value against the average number of upstream properties it sources from. The relationship is strikingly log-linear, with export volumes rising less than proportionally in the number of connected properties.[6] At the top end of the distribution, the largest exporters source from over 10,000 properties, while at the bottom, smaller exporters maintain only a handful of supplier relationships. One implication is that larger exporters source proportionally less export value per supplier relationship, with an exporter maintaining ten times as many supplier relationships generating only about three times the export value. We show in Appendix Figures F.3–F.3c that there is no systematic correlation between exporter size and the average land composition (forest, pasture, or soybean share) of connected farmers, which is relevant for identification since it suggests that exporter-level demand variation is not confounded with farmer-level land characteristics.

Beyond the exporter relationships, it is useful to understand how farmers allocate their spending across different types of suppliers. Appendix Figure F.4 reports the share of total farm-level spending

[6]The estimated elasticity of export value with respect to the number of connected properties is approximately 0.5 for both beef and soy. An elasticity below one implies that larger exporters source less per property than smaller ones, consistent with decreasing returns to scale in sourcing, though the R^2 of 0.33–0.36 suggests substantial heterogeneity around this relationship.

Figure 4: Exporter Size and Upstream Sourcing



Notes: The figure plots each exporter's log average yearly export value (USD) against the average number of upstream properties it sources from (log scale) between 2010 and 2020. Red dots represent beef exporters and blue dots represent soy exporters. The lines show log-linear fits with reported R^2.

directed to different supplier industries, separately for farmers linked to beef and soy exporters. For both groups, the largest spending category is agriculture and livestock, which captures inter-farm transactions such as purchases of feed, seed, and live animals and accounts for roughly half of all spending. The next-largest categories are transportation and logistics, wholesale trade, and chemicals and fertilizers. The three spending categories that we study in our empirical analysis, machines, capital goods and irrigation, and technology goods and services, each account for roughly 1 to 2 percent of total annual farm spending. The cnae codes assigned to each of these categories are reported in Table E.1 in Appendix C. Although small in share, these categories represent the margins most directly relevant to technology adoption and capital deepening, and they correspond to precisely the types of expenditure that should rise with export demand under standard models of export-led growth.

5 Empirical Specification

Brazilian agricultural trade runs almost entirely through intermediaries, with a handful of trading firms such as ADM, Bunge, and Cargill (often referred to as the *ABC* firms) handling most of the export value alongside a long tail of much smaller intermediaries. With few exceptions, these firms do not produce agricultural goods themselves but instead specialize in logistics and processing, purchasing beef and soy from potentially thousands of upstream farmers and selling them on international markets. Agricultural export shocks therefore propagate upstream from these intermediaries to farmers through payment relationships.

We observe transaction flows between farmers and export intermediaries in the payment data. Goods flow upstream from farmer to exporter, while payments flow in the opposite direction, so incoming payments measure each farmer's sales exposure to individual intermediaries. For each farmer-exporter pair, we measure the share of exporter x in farmer i's total payments received from exporters in the first year farmer i appears in the data,

$$w_{ix} = \frac{\text{Payments}_{x \to i,\, t_0}}{\sum_{x'} \text{Payments}_{x' \to i,\, t_0}} \tag{7}$$

where $\text{Payments}_{x \to i,\, t_0}$ denotes the total transaction value from exporter x to farmer i, and t_0 is the first year farmer i appears in the data. We then chain these weights w_{ix} with each exporter's destination composition, measured in the first year t_0' that the exporter appears in our sample, to obtain farmer-level destination shares

$$s_{ic}^k = \sum_x w_{ix} \cdot \underbrace{\frac{\text{Exports}^k_{x \to c,\, t_0'}}{\sum_{c'} \text{Exports}^k_{x \to c',\, t_0'}}}_{s_{xc}^k}, \qquad k \in \{\text{beef, soy}\} \tag{8}$$

where s_{xc}^k is the share of exporter x's commodity-k exports destined for country c, fixed at the first year t_0' the exporter appears in the data. The farmer-level share s_{ic}^k thus measures farmer i's pre-determined exposure to destination country c for commodity k, traced through the farmer's initial exporter linkages. We report the distribution of initial years at which the exporter destination shares s_{xc}^k are fixed in Appendix Table A.2.

Using these pre-determined shares, we construct a farmer-specific demand shock as the share-weighted growth rate of non-Brazilian import demand

$$\Delta z_{it} = \sum_k \sum_c s_{ic}^k \cdot \Delta \ln g_{ct}^k \tag{9}$$

where g_{ct}^k denotes total non-Brazilian imports of commodity k by destination country c in year t, explicitly excluding Brazilian exports to avoid contamination by domestic supply-side factors. The growth rate $\Delta \ln g_{ct}^k$ captures year-on-year fluctuations in foreign demand at each destination, and the pre-determined shares s_{ic}^k translate these into farmer-specific shocks. The instrument pools beef and soy exposure into a single demand shock, so that farmers connected to both beef and soy exporters receive a shock that reflects their total agricultural export exposure.[7]

We study land-use and expenditure outcomes at the farmer-year level. For land use, Y_{it} is the share of a farmer's total registered property area allocated to forest, pasture, or soybean, aggregated across all CAR properties the farmer operates in a given year. For expenditure, Y_{it} measures the farmer's total outgoing payments to upstream input suppliers, decomposed into three categories based on the 7-digit cnae sector classification of the recipient (see Appendix Table E.1 for the complete mapping).

1. *Machines.* Spending on the core machinery fleet used in agricultural production, including agricultural tractors, harvesters, planters, earthmoving equipment, vehicles and trucks, and the maintenance and repair of agricultural equipment.

[7] We decompose the instrument into separate product-specific shocks $\Delta z_{it}^{\text{beef}}$ and $\Delta z_{it}^{\text{soy}}$ in Appendix D and show that both commodities generate qualitatively similar patterns.

2. *Capital Goods & Irrigation.* Physical infrastructure and equipment that support farm operations beyond the machinery fleet, including irrigation systems, water management infrastructure, electrical equipment such as generators and transformers, and specialized transport equipment.

3. *Technology Goods & Services.* The knowledge-intensive margin of farm investment, including computing and electronic equipment, IT and software services, engineering and agronomic consulting, and scientific research and development.

Farmer-level spending in these categories is lumpy and often zero, so instead of using standard logs we use $\log(1 + Y_{it})$ as our baseline transformation for spending outcomes, which preserves zeros while approximating log-linearity for positive values, allowing us to estimate semi-elasticities that incorporate both the extensive margin (whether a farmer spends at all) and the intensive margin (how much). We verify robustness to the inverse hyperbolic sine transformation $\ln(Y_{it} + \sqrt{Y_{it}^2 + 1})$, which has similar properties to $\log(1+Y_{it})$ but is exactly invariant to the units of measurement (see Burbidge et al. 1988), to extensive margin dummies $\mathbb{1}[Y_{it} > 0]$ (Appendix B), and to broader and narrower cnae-based supplier classifications (Appendix E).

Following Aghion et al. (2024), our baseline specification regresses these outcomes on leads and lags of the demand shock

$$Y_{it} = \sum_{\tau=-5}^{8} \alpha_\tau \, \Delta z_{i,t-\tau} \; + \; \mu_i \; + \; \delta_t \; + \; \gamma_t' \mathbf{C}_i \; + \; X_{it}' \lambda \; + \; \varepsilon_{it} \tag{10}$$

where μ_i and δ_t are farmer and year fixed effects. The vector $\mathbf{C}_i$ contains three time-invariant farmer-level characteristics, each interacted with a full set of year fixed effects and absorbed in the estimation. These are (i) the sum of farmer i's destination shares $\bar{s}_i = \sum_{k,c} s_{ic}^k$, which controls for incomplete shares following Borusyak et al. (2025), (ii) the fraction of farmer i's export exposure flowing through beef exporters, which absorbs differential trends between farmers whose shocks are driven by different product compositions, and (iii) the shares of farmer i's CAR property area allocated to forest and to soybeans in the first year t_0 a farmer appears in the sample, each separately interacted with year fixed effects, which absorb differential trends between farmers with different initial land compositions.[8] The control vector X_{it} includes the number of CAR properties associated with farmer i, which accounts for variation in farm scale that may affect both land use and spending decisions, and two measures of local weather conditions from McNally et al. (2017), the annual minimum rainfall anomaly and the standard deviation of monthly rainfall anomalies, defined as deviations from the 1982–2016 monthly average. These climate controls absorb weather-driven fluctuations in agricultural productivity that could otherwise confound the effect of demand shocks on land use and investment. Standard errors are clustered at the farmer level. The coefficients $\{\alpha_\tau\}$ trace out the dynamic response of Y_{it} to a unit increase in the demand shock.

We estimate equation (10) splitting the sample into farmers located inside and outside the Legal Amazon. The Legal Amazon represents the active deforestation frontier, where uncultivated land is

[8] We verify robustness to alternative initial characteristics, controlling instead for initial soybean share only, initial pasture share only, initial forest share only, or no initial characteristic at all (Appendix E).

relatively abundant, property rights are often weakly enforced, and the cost of clearing vegetation is low relative to the cost of mechanization, irrigation, or yield-enhancing input adoption (Barreto 2021). Outside the Amazon, agricultural land is already largely allocated, property rights are better established, and the scope for extensive-margin adjustment is correspondingly smaller.

6 Identification

The main threat to identification is reverse causality arising from unobserved supply-side shocks that simultaneously determine both exports and farmer-level outcomes. A positive productivity shock, such as favorable weather or a disease-free season, would mechanically increase a farm's output, expanding the volume of exports flowing through its downstream intermediaries while also changing the farm's land allocation and investment decisions. In standard heterogeneous-producer models, more productive firms self-select into supply chains connected to larger exporters, generating a positive correlation between export exposure and outcomes that is driven by selection on productivity rather than by a causal effect of foreign demand (e.g., Melitz 2003). Ordinary least squares estimates of the relationship between exports and farmer-level outcomes would therefore conflate the causal impact of foreign demand with these unobserved supply-side factors.

Our identification strategy addresses this concern through the construction of the demand shock Δz_{it} in equation (9). The shift component, $\Delta \ln g_{ct}^{k}$, captures year-on-year growth in non-Brazilian imports of commodity k at each destination, explicitly excluding Brazilian exports to avoid contamination by domestic supply conditions. The share component, s_{ic}^{k}, is fixed at the first year each farmer-exporter pair appears in the data, so the identifying variation arises from the interaction of pre-determined farmer-level exposure with time-varying fluctuations in foreign demand at each destination. Identification requires that non-Brazilian import demand at each destination is orthogonal to the idiosyncratic supply conditions of individual Brazilian farmers, conditional on the fixed effects and controls in equation (10).

The reduced-form specification provides a transparent test of this assumption. Because all leads and lags of Δz enter a single regression, the coefficients at $\tau < 0$ measure whether *future* demand shocks predict *current* outcomes. Under the identification assumption, they should not. As we show below, the lead coefficients are insignificant across our main outcomes. The shift-share design can be justified either through the exogeneity of many independent destination-level shocks or through the exogeneity of a few dominant shocks that carry the identifying variation (Borusyak et al. 2025). Our instrument draws on 143 destination countries (Table A.1 in Appendix A reports full diagnostics and discusses the concentration of importance weights; Appendix E shows that dropping any top-5 destination leaves results unchanged). The effective number of independent demand shocks is approximately 9, which is relatively low and reflects the concentrated structure of global beef and soybean trade. The largest importance weight falls on China at 29%, reflecting its dominant role in global demand for beef and soybeans. Chinese import demand for these commodities is driven by structural income growth and the associated dietary transition toward higher protein consumption, rather than by conditions in

Brazilian agriculture, plausibly satisfying the exogeneity requirement for the dominant shifter.[9]

7 Results

We show a visual representation of our findings in Figures 5–7, which plots the estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10), separately for farmers within and outside the Legal Amazon. A one-unit increase in Δz_{it} corresponds roughly to a doubling of foreign demand. The land use coefficients give the resulting change in the area share in percentage points, while the spending and payment coefficients give the resulting percent change and can be read as elasticities with respect to foreign demand.[10]

We begin by investigating how the demand shock transmits from exporters to farmers through the payment system. Therefore we regress the log of total payments farmer i receives from all exporters he is exposed to in t_0. Specifically, we estimate equation (10) with $Y_{it} = \ln \sum_{x \in \mathcal{X}i^0}$ Payments$x \rightarrow i, , t$, where $\mathcal{X}_i^0$ is the set of exporters linked to farmer i in the first year farmer i appears in the data (t_0), capturing payments along precisely the channel through which the demand shock propagates to the farm. Figure 5 shows that, in the Amazon, a doubling of foreign demand raises payments from linked exporters by roughly 33 percent at $\tau = 1$–2. Outside the Amazon, the pass-through is significantly smaller at around 10 percent at impact. Our results therefore suggest that exporters facing higher foreign demand scale up their agricultural purchases disproportionately more from suppliers located within the Amazon.

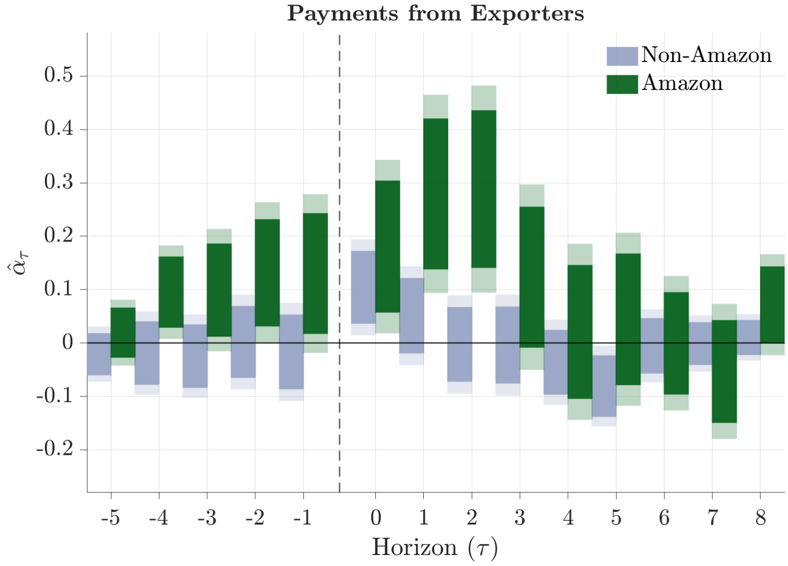
Next, we examine the effect of foreign demand on farmers' land use and modern input spending. The top two panels of Figure 6 show that farmers in the Amazon respond to higher export demand by switching away from capital-intensive soybean cultivation and toward land-intensive pasture. More specifically, the share of land dedicated to soybeans contracts by roughly 2 percentage points within three years of the shock, while the pasture share expands by a comparable magnitude over the same horizon, and both effects persist for several years.[11] Our results suggest a shift away from the technology frontier, since soybean production requires mechanized planting, chemical inputs, and precision agriculture, whereas pasture management uses little beyond fencing and basic veterinary care. When we decompose the pooled instrument into separate beef and soy shocks (Appendix D, Figure D.2), we find that even a pure soy demand shock reduces the soybean share in the Amazon, suggesting that the reallocation toward pasture is not driven by the composition of the shock but by the relative returns to land expansion versus modern production. Unsurprisingly, given the limited

[9] In Appendix E we probe the sensitivity of our results to this concentration by excluding dominant destinations one at a time, dropping the three largest exporters, excluding farmers connected to both beef and soy exporters, fixing the baseline year at 2010 for all farmers, and using a shorter event window of $\tau = -3$ to $+6$. Results are stable across all exercises.

[10] Given that Brazilian agricultural exports grew sevenfold between 2000 and 2020, a doubling over a few years is well within the range of observed variation.

[11] On the average Amazon farm of about 3,000 hectares (Table F.1), a 2 percentage point reallocation corresponds to roughly 60 hectares per farm. This figure varies considerably across the size distribution, with the largest farms in the sample exceeding 300,000 hectares, where a 2 percentage point reallocation translates into roughly 6,000 hectares. For comparison, Scott et al. (2025) find that a permanent 17 percent increase in beef prices roughly doubles steady-state deforestation rates in the Amazon, while short-run responses are near zero.

Figure 5: Agricultural Export Demand and Payments from Exporters



Notes: The figure plots the estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$, separately for farmers in the Legal Amazon (green) and outside it (blue). The outcome is $\ln \sum_{x \in \mathcal{X}_i^0} \text{Payments}_{x \to i,\, t'}$, where $\mathcal{X}_i^0$ is the set of exporters linked to farmer i in the first year farmer i appears in the data (t_0). Solid bars denote 95% confidence intervals and shaded areas denote 99% confidence intervals, based on standard errors clustered at the farmer level.

pass-through to farmers outside the Amazon, we observe little reallocation between pasture and soybean.

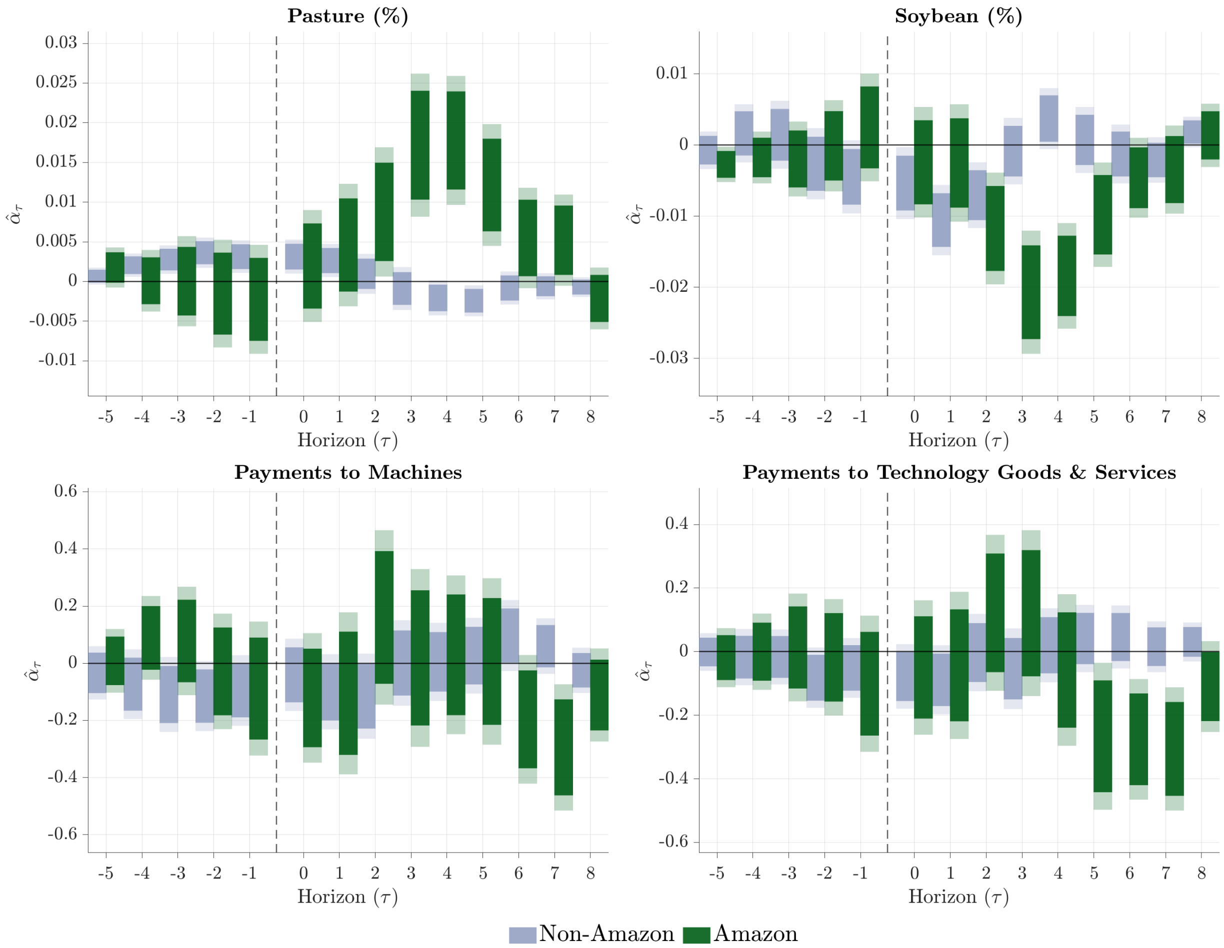
The bottom two panels of Figure 6 show that spending on modern inputs in the Amazon declines following the demand shock. Specifically, payments to *Technology Goods & Services*, which capture purchases from IT, engineering, agronomic consulting, and R&D, fall by roughly 25 percent at peak, and machinery spending declines by a similar magnitude.[12] These spending declines emerge only around $\tau = 5$ and persist through $\tau = 7$, well after land reallocation from soy to pasture begins at $\tau = 2$ and peaks at $\tau = 3$–4. The timing is consistent with spending adjusting to the earlier land-use change, since pasture management requires fewer tractors, harvesters, irrigation systems, and agronomic services than soybean cultivation. In the Legal Amazon, we also observe a decline in the number of suppliers from whom farmers purchase machines and technology (Appendix Figure B.3). Again consistent with the weaker pass-through outside of the Legal Amazon, spending on modern inputs does not respond to the demand shock.

Figure 7 shows the environmental externality. In the Amazon, the forest share declines by roughly 0.4 percentage points beginning around $\tau = 3$, tracking the expansion of pasture. Forest is cleared to accommodate the shift toward extensive production that replaces soybean. Outside the Amazon, the forest share is flat throughout, consistent with the absence of any land-use response in that subsample.

Taken together, our results suggest that when foreign demand for agricultural commodities rises, exporters increase their purchases disproportionately from Amazon farmers within the first two

[12] The peak decline in technology goods and services is approximately 0.30 in $\log(1 + Y)$ at $\tau = 5$–7, implying a reduction of about $1 - e^{-0.30} \approx 26$ percent. Machinery spending follows with a peak of 0.29 at $\tau = 7$. Results for *Capital Goods & Irrigation*, reported in Appendix Figure C.1, are qualitatively similar.

Figure 6: Agricultural Export Demand and Farm-Level Responses

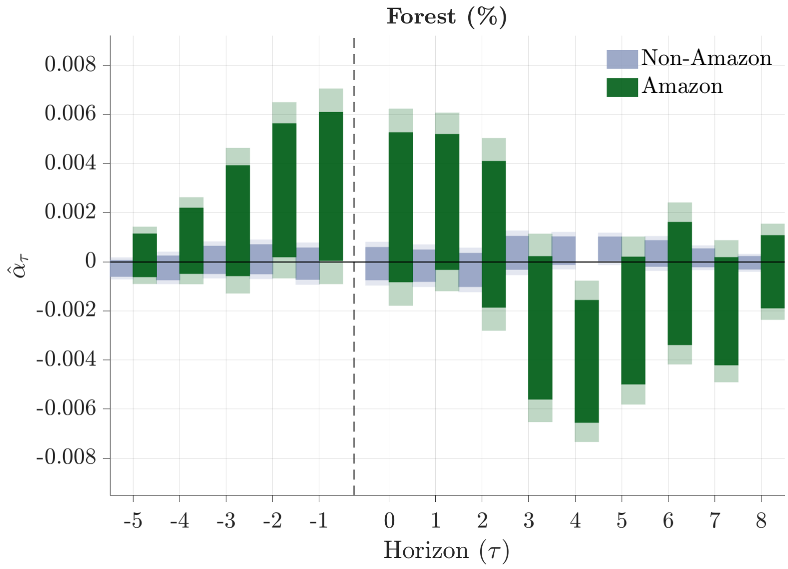


Notes: Each panel plots the estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$, separately for farmers in the Legal Amazon (green) and outside it (blue). Solid bars denote 95% confidence intervals and shaded areas denote 99% confidence intervals, based on standard errors clustered at the farmer level. Panels (a) and (b) report effects on pasture and soybean land-use shares, where each share is computed by aggregating satellite-measured land cover across all CAR properties a farmer operates and dividing by total registered property area. Panels (c) and (d) report effects on $\log(1 + Y_{it})$ where Y_{it} is total annual spending on suppliers classified under Machines and Technology Goods & Services (see Appendix Table E.1 for the full classification). Capital Goods & Irrigation results are reported separately in Appendix Figure C.1.

years. In response, farmers in the Amazon shift production from capital-intensive soybean cultivation toward land-intensive pasture over the following two to four years, specializing into pasture at the expense of forest and reducing the area under crop cultivation needed to justify technology investments. Specifically, farmers decrease their spending on technology goods persistently by around 25 percent, five to seven years after the shock. In Appendix E, we report a comprehensive set of robustness exercises, including alternative sample restrictions, specification variants, and instrument modifications.

Our payment data captures only cross-bank transactions, meaning that payments between a farmer and a supplier who hold accounts at the same bank are unobserved. Our classification of suppliers into three investment categories could also miss or misattribute relevant purchases. To assess whether these features of the data affect our ability to detect spending increases, we estimate a complementary

Figure 7: Agricultural Export Demand and Forest Cover



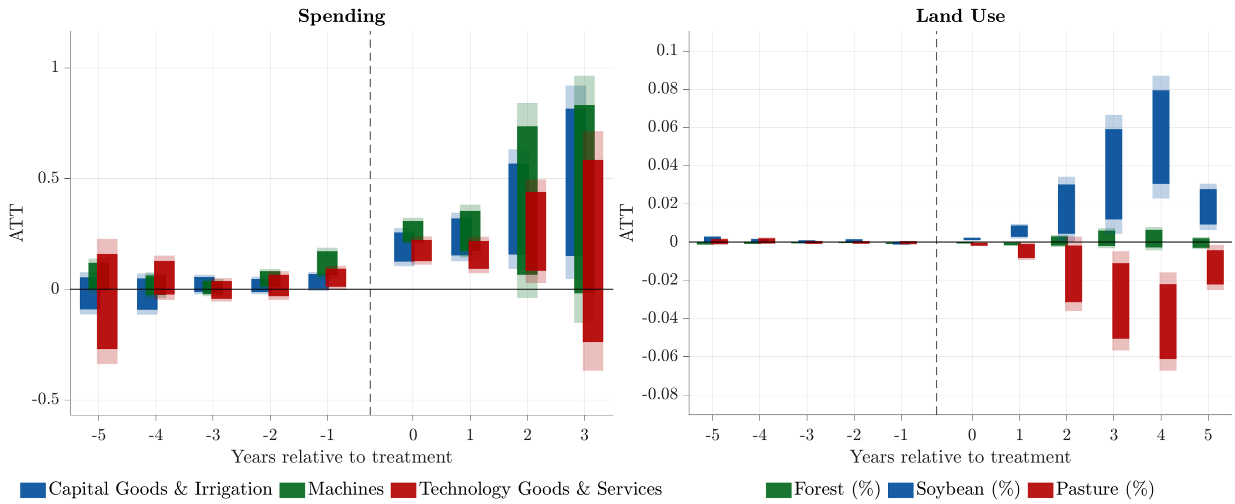
Notes: The figure plots the estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$, separately for farmers in the Legal Amazon (green) and outside it (blue). The outcome is the forest land-use share, computed by aggregating satellite-measured forest cover across all CAR properties a farmer operates and dividing by total registered property area. Solid bars denote 95% confidence intervals and shaded areas denote 99% confidence intervals, based on standard errors clustered at the farmer level.

specification where we exploit farmers' first use of rural credit. Specifically, Brazil's rural credit system provides subsidized loans dedicated to agricultural investment, which should mechanically increase purchases of machinery, irrigation, and capital goods. Moreover, in our parsimonious framework (Section 3), a reduction in the cost of capital r would trivially raise technology spending and shift land away from pasture, since cheaper capital increases the return to capital-intensive crop cultivation. Therefore, we estimate a Callaway & Sant'Anna (2021) difference-in-differences specification around the first time the a farmer receives subsidized rural credit, using not-yet-treated farmers as the control group.[13]

Figure 8 reports the results. The left panel shows that spending on machines, capital goods and irrigation, and technology all increase immediately when farmers first access rural credit, by roughly 20 percent at $\tau = 0$, and continue to grow over subsequent years. The right panel shows that access to rural credit also shifts how farmers use their land. After obtaining credit, farmers steadily reduce the share of land allocated to pasture and expand the share under soybean.

[13]The sample for this exercise consists of all farmers in the BCB payment data between 2010 and 2020 who appear in the rural credit system (SICOR from 2013 onward, RECOR from 2000 to 2012), merged with the same land-use outcomes from CAR and MapBiomas used in the main analysis. This is a broader sample than the export panel, as it does not condition on linkages to beef or soy exporters. We use doubly-robust inverse probability weighting with 1,000 bootstrap replications.

Figure 8: Callaway-Sant'Anna DiD: Spending and Land Use



Notes: Both panels plot event study estimates from a Callaway-Sant'Anna difference-in-differences design, where treatment is defined as the first year a farmer obtains rural credit. The left panel reports effects on $\log(1+Y_{it})$ where Y_{it} is total annual spending on suppliers classified under Capital Goods & Irrigation (navy), Machines (green), and Technology Goods & Services (red), with the event window spanning $\tau = -5$ to $+3$. The right panel reports effects on land-use shares for forest (green), soybean (blue), and pasture (red), where each share aggregates across all CAR properties a farmer operates, with the event window spanning $\tau = -5$ to $+5$. Solid bars denote 95% confidence intervals and shaded areas denote 99% confidence intervals. Estimation uses the doubly-robust inverse probability weighting method with not-yet-treated farmers as the control group, 1,000 bootstrap replications, and 99% confidence level, clustering at the farmer level.

8 Robustness: Relative Beef and Soy Demand

Next we investigate whether our results are driven by a *proportional* increase in foreign demand for beef and soy or by a *relative* shift for foreign demand toward beef. The main concern for the interpretation of our baseline results is that our estimates could simply be driven by systematically higher growth rates of non-Brazilian beef imports such that farmers specialize into pasture due to differences in relative demand rather than the availability of cheap land to convert to pasture. Recall that in our parsimonious framework in Section 3, we show that even under a proportional commodity boom, farmers may find it optimal to reallocate land away from soybeans and decrease their capital spending when cheap uncultivated land is available for deforestation. To address this concern, we construct a control variable that captures any growth differential between foreign demand in soy and beef farmer i is exposed to

$$\Delta q_{it} = \sum_c s_{ic}^{\text{beef}} \, \Delta \ln g_{ct}^{\text{beef}} \; - \; \sum_c s_{ic}^{\text{soy}} \, \Delta \ln g_{ct}^{\text{soy}}, \tag{11}$$

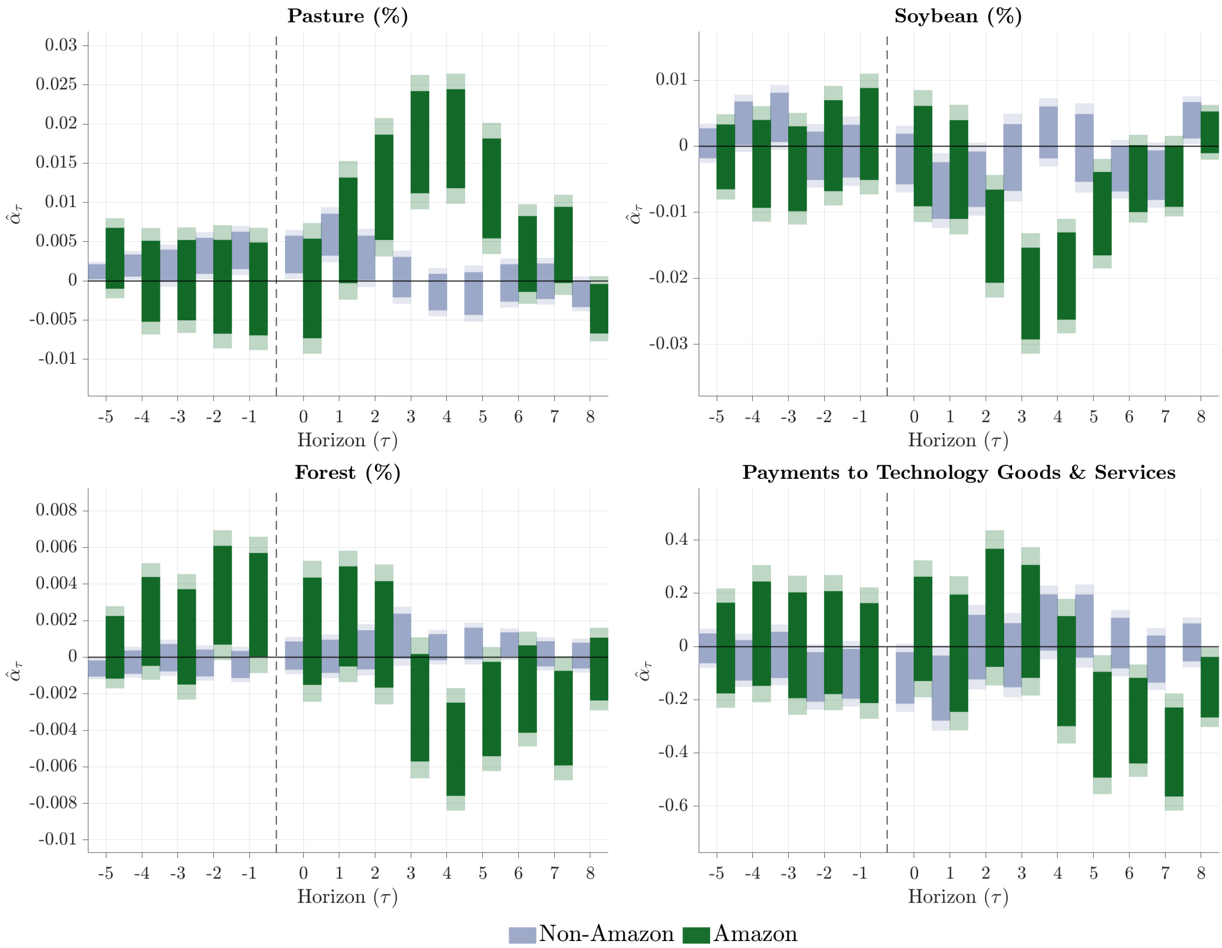
and then add the control to equation (10) at every τ such that

$$Y_{it} = \sum_{\tau=-5}^{8} \alpha_\tau \, \Delta z_{i,t-\tau} \; + \; \sum_{\tau=-5}^{8} \beta_\tau \, \Delta q_{i,t-\tau} \; + \; \mu_i \; + \; \delta_t \; + \; \gamma_t' \mathbf{C}_i \; + \; X_{it}' \lambda \; + \; \varepsilon_{it}. \tag{12}$$

where β_τ captures the dynamic effects of the relative demand shift.

Conditional on the control in (11), the identifying variation for our coefficients of interest α_τ is then identified from the differential exposure to common shocks, completely absorbing any relative

Figure 9: Farm-Level Responses to a Proportional Demand Shock



Notes: Each panel plots the estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (12) for $\tau = -5, \ldots, +8$, separately for farmers in the Legal Amazon (green) and outside it (blue). Conditional on the control in equation (11), the coefficients capture the response to demand growth common to beef and soy. Solid bars denote 95% confidence intervals and shaded areas denote 99% confidence intervals, based on standard errors clustered at the farmer level. Panels (a)–(c) report effects on pasture, soybean, and forest land-use shares, where each share is computed by aggregating satellite-measured land cover across all CAR properties a farmer operates and dividing by total registered property area. Panel (d) reports effects on $\log(1 + Y_{it})$ where Y_{it} is total annual spending on suppliers classified under Technology Goods & Services (see Appendix Table E.1 for the full classification).

demand effects at each horizon. We plot the estimated coefficients α_τ for the pasture, soybean, and forest shares and for technology spending in Figure 9, which documents that the decline in capital spending within the Amazon persists in both sign and magnitude. Our robustness exercise therefore suggests that our baseline results are indeed driven by the common component in beef and soy demand, rather than any relative shift toward a particular commodity.

9 Economic Environment

In line with our parsimonious framework from Section 3, our empirical analysis shows that an export boom leads farmers in the Amazon to reallocate land away from crop cultivation and reduce spending on modern technology. We now embed the mechanism from Section 3 into a general equilibrium model with heterogeneous farms that differ in their productivity and their cost of converting forest into pasture. We build a static, single-country open-economy model populated by a representative household and two sectors: agriculture and non-agriculture. Within both sectors, production is organized under monopolistic competition, and heterogeneous firms and farms make endogenous decisions over entry and technology adoption. To close the open economy, domestic sectoral output and foreign imports are combined via a constant elasticity of substitution aggregator into a sectoral composite good. This composite good is then used to satisfy all final consumption, intermediate inputs, and fixed investment costs. Domestic varieties not consumed at home are exported, facing constant-elasticity foreign demand, with a flexible exchange rate that clears balanced trade.

Within the agricultural sector, farms draw clearing cost to convert forest into pasture which captures variations in land scarcity and property rights without specifically modeling geographic regions. Upon the productivity and clearing cost draw, farmers choose whether to pay a fixed cost of adopting modern technology, how to allocate their land between crop cultivation and livestock, and how many additional hectares to deforest for pasture. A key feature of the model is that even high-productivity farms may find it optimal not to adopt the modern technology if their clearing cost is sufficiently low—a feature we observe directly in the data and our model matches as an untargeted moment. We calibrate the model to Brazilian microdata and show that the model can both rationalize our empirical findings and match a range of key untargeted moments capturing differences in land use and technology adoption between the Legal Amazon and the rest of Brazil.

In a counterfactual exercise, we then argue that an environmental policy can increase the welfare gains from trade without assuming externalities like carbon emissions, rainfall spillovers, or a loss of biodiversity that enters the utility of households. Instead, we assume that farmers face financial frictions (a well-established fact for Brazilian agriculture, e.g., in Assunção et al. 2020) and must finance the fixed cost of technology adoption through an intra-period loan. Following Bigio & La'o (2020), we model this friction as an interest rate spread on borrowed funds. All accrued interest payments are ultimately rebated lump-sum to the representative household, meaning the financing wedge acts as an internal transfer rather than a wasted resource. The social benefit of adoption therefore exceeds its private benefit, and a constrained social planner would prefer higher adoption rates such that the decentralized economy invests in modern technology below the social optimum. The deforestation tax then penalizes pasture expansion and induces producers to maintain larger crop operations. Since modernization gains apply asymmetrically to crops, this land reallocation directly raises the private return on technology investments.

Importantly, in our calibrated model, an environmental tax must be sufficiently strong to increase the welfare gains from trade. A moderate policy merely distorts the land allocation without raising crop returns enough to cover the fixed modernization cost, causing the welfare gains from trade to fall relative to the untaxed baseline. Only a sufficiently strong tax forces producers to reallocate enough land into crops to fully justify the fixed investment. A strong policy consequently draws high-productivity pasture farms into modern agriculture, lifting technology adoption toward the social optimum such that less deforestation increases the welfare gains from trade.

9.1 The Economy

The economy is populated by a mass N of households that supply one unit of labor inelastically. Production is organized across two sectors, agriculture denoted by a and non-agriculture n, indexed by $s \in \{a, n\}$. In each sector the domestic good Y_s is combined with an imported good I_s through a CES aggregator to form the composite good Q_s which is used for all final consumption and intermediate inputs.

$$Q_s = \left[\alpha_s^{\frac{1}{\sigma}} (Y_s)^{\frac{\sigma-1}{\sigma}} + (1-\alpha_s)^{\frac{1}{\sigma}} (I_s)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}. \tag{13}$$

Here σ is the elasticity of substitution between domestic and foreign goods, and α_s is a share parameter reflecting home bias. The optimal demands for foreign and domestic goods are

$$Y_s = \alpha_s \Big(\frac{P_{d,s}}{P_s}\Big)^{-\sigma} Q_s, \qquad I_s = (1-\alpha_s)\Big(\frac{\mathcal{H}P^*_{f,s}}{P_s}\Big)^{-\sigma} Q_s, \tag{14}$$

where P_s is the CES price index of the composite good, combining the domestic producer price $P_{d,s}$, the exogenous world price of imports $P^*_{f,s}$, and the nominal exchange rate $\mathcal{H}$.

$$P_s = \left[\alpha_s P_{d,s}^{1-\sigma} + (1-\alpha_s)(\mathcal{H}P^*_{f,s})^{1-\sigma}\right]^{\frac{1}{1-\sigma}}. \tag{15}$$

We assume a representative household combines the two composite goods into final consumption C through a Cobb-Douglas aggregator,

$$C = C_a^{\phi_a} C_n^{\phi_n} \qquad \text{where} \quad \phi_a + \phi_n = 1, \tag{16}$$

where C_s is demand for the composite good Q_s. The ideal consumption price index is $P_c = P_a^{\phi_a} P_n^{\phi_n}/(\phi_a^{\phi_a}\phi_n^{\phi_n})$ and expenditure satisfies $P_s C_s = \phi_s P_c C$. The non-agricultural composite Q_n also serves as the intermediate input in both sectors.

9.2 Domestic Production

Within each sector, a unit mass of potential entrants can create an establishment to produce differentiated varieties j. A non-agricultural establishment supplies a single variety y_{jn}, whereas each agricultural entrant produces a crop variety y_j^c and a pasture variety y_j^p (i.e., livestock). Intermediate varieties are then aggregated to form a domestic sectoral good using the following Dixit-Stiglitz aggregators with elasticity $\eta_s > 1$

$$Y_s = \begin{cases} \Big(\int y_{jn}^{\frac{\eta_n-1}{\eta_n}}\, dj\Big)^{\frac{\eta_n}{\eta_n-1}}, & s = n, \\ \Big(\int \big[\rho^{\frac{1}{\eta_a}} (y_j^c)^{\frac{\eta_a-1}{\eta_a}} + (1-\rho)^{\frac{1}{\eta_a}} (y_j^p)^{\frac{\eta_a-1}{\eta_a}}\big]\, dj\Big)^{\frac{\eta_a}{\eta_a-1}}, & s = a, \end{cases} \tag{17}$$

with the weight $\rho \in (0,1)$. The price index for this domestically produced non-agricultural good is $P_{d,n} = (\int p_{d,jn}^{1-\eta_n}\, dj)^{\frac{1}{1-\eta_n}}$ and $P_{d,a} = (\int [\rho\, (p^c_{d,j})^{1-\eta_a} + (1-\rho)\, (p^p_{d,j})^{1-\eta_a}]\, dj)^{\frac{1}{1-\eta_a}}$ in agriculture.

9.3 The Agricultural Sector

Within the agricultural sector, we assume that each potential agricultural firm owns a fixed plot of land of size h that cannot be sold or rented and can be split between growing crops l_c and pasture l_p under the binding owned-land aggregator[14]

$$\left[l_c^{\frac{\varepsilon_h - 1}{\varepsilon_h}} + l_p^{\frac{\varepsilon_h - 1}{\varepsilon_h}} \right]^{\frac{\varepsilon_h}{\varepsilon_h - 1}} \leq h \tag{18}$$

where ε_h governs how easily the farmer reallocates the plot between the two uses. Note that as $\varepsilon_h \to \infty$ the constraint collapses to $l_c + l_p \leq 1$, in which case crop land and pasture are perfectly substitutable within the plot. Here we depart from the assumption of perfectly substitutable land uses within the owned plot that we use in our parsimonious framework in Section 3. Specifically, we allow ε_h to be finite, so a plot split between the two uses leaves an area $v = h - l_c - l_p$ of the endowment vacant, which shrinks as the farmer concentrates the plot in one use. Our approach captures implicit gains from specialization, that can arise, for example from homogeneous land suitability within a plot or from use-specific managerial capacity and knowledge, which favor specialization (see e.g., Holmes & Lee 2012).[15] We plot the constraint for several values of ε_h in Appendix Figure F.1.

We then allow farmers to deforest an amount d to increase their total pasture area $\bar{l}_p$ at an exogenous unit (hectare) cost $c > 0$, which each potential entrant draws together with its productivity before making any entry decision, from an independent distribution $G(c)$ over the support $[\underline{c}, \infty)$.[16] The farm-specific cost c captures variation in land availability, property rights, and institutional constraints. Intuitively, a low draw for c captures an environment where adjacent uncultivated land is abundant and property rights are weakly enforced. Conversely, a high draw for c captures the constraints of densely settled regions where land is scarce and expansion is heavily regulated. Furthermore we assume that deforestation d is paid in units of the non-agricultural composite Q_n at price P_n, capturing the equipment and intermediate inputs required for land conversion (e.g., earthmoving equipment and contracted clearing services, transport, etc.).

The total land dedicated to pasture $\bar{l}_p$ for each farmer is then assumed to be a combination of the owned size l_p and deforested land d:

$$\bar{l}_p = \left[l_p^{\frac{\varepsilon_p - 1}{\varepsilon_p}} + d^{\frac{\varepsilon_p - 1}{\varepsilon_p}} \right]^{\frac{\varepsilon_p}{\varepsilon_p - 1}} \tag{19}$$

where ε_p is the elasticity of substitution between owned and cleared land. Similarly to the binding land constraint, we assume that established pasture within the owned plot l_p and newly deforested land d are not perfectly substitutable captured by the elasticity ε_p, reflecting that agricultural land expansion cannot seamlessly replace established pasture without diminishing returns.

[14] Here we abstract from an endogenous market for land. Note that in the 2017 Agricultural Census, owned land makes up on average around 85 percent of farm area while rented land is below 9 percent. We show a more detailed split by state in the Appendix Figure H.3. Excluding outliers such as São Paulo, Rio Grande do Sul, Paraná, and the Distrito Federal, owned land makes up almost 90 percent of total agricultural land with the share of rented land under 7 percent.

[15] Holmes & Lee (2012) document that farmers overwhelmingly plant fields within a plot in the same use, far beyond what independent choices across fields would produce. They attribute the specialization to similar land quality across fields within a plot and to gains from concentration itself, such as indivisible equipment and knowledge specific to a single use.

[16] We specify and calibrate a parametric form for G in Section 10.

Each potential entrant draws an ex-ante productivity z from a Pareto distribution $F(z)$ with tail parameter $\zeta > 0$. Having observed the pair (z, c), a potential entrant becomes active by paying a fixed entry cost κ_{ta} in units of labor. Each active farmer then produces a differentiated crop variety y_j^c and a pasture variety y_j^p using labor (n_c, n_p), intermediate inputs (x_c, x_p), and the respective land inputs $(l_c, \bar{l}_p)$ according to the Cobb-Douglas technologies

$$y_j^c = z\, A_i\, l_c^{\gamma_c}\, x_c^{\nu_c}\, n_c^{1-\nu_c-\gamma_c}, \qquad y_j^p = z\, \bar{l}_p^{\gamma_p}\, x_p^{\nu_p}\, n_p^{1-\nu_p-\gamma_p}, \tag{20}$$

where γ_c and γ_p denote the output elasticities of land in crop and pasture production, and ν_c and ν_p the corresponding elasticities of intermediate inputs.

A farmer can upgrade their crop-technology A_i from a traditional baseline A_t to a modern level $A_m > A_t$ by purchasing κ_{ma} units of the non-agricultural composite Q_n at price P_n.

Importantly, we assume that farmers face financial constraints to modernization. Inspired by Bigio & La'o (2020), we model these constraints as a within-period financing friction in which farmers must pay the fixed adoption cost upfront at the beginning of the period. As agricultural revenues are not realized until the end of the period, a modernizing farmer must take out an intraperiod loan to finance the upgrade. Specifically, the farmer borrows the required upgrading cost $P_n\kappa_{ma}$ from a banking sector at an exogenous within-period interest rate of $1 + r$. All interest payments accrued to the banking sector are then rebated lump sum to the household in the form of bank profits. Here we assume that farms are owned by the household such that farm and bank profits flow back to the household. Under this assumption, even a constrained social planner would prefer higher adoption rates such that, with the financing wedge in place, the social benefit of adoption exceeds the private benefit as adoption is distorted below its social optimum.

An active farmer then chooses prices (p_d^c, p_d^p), labor (n_c, n_p), intermediates (x_c, x_p), the owned-land split (l_c, l_p), and deforestation d to maximize joint operating profit,

$$\begin{aligned}
\pi_{i,a}^o(z,c) = \max_{p_d^c, p_d^p, n_c, n_p, x_c, x_p, l_c, l_p, d} \quad & \rho\, p_d^c \left(\frac{p_d^c}{P_{d,a}}\right)^{-\eta_a} Y_a + (1-\rho)\, p_d^p \left(\frac{p_d^p}{P_{d,a}}\right)^{-\eta_a} Y_a \\
& - P_n(x_c + x_p) - (n_c + n_p) - P_n c\, d - \tau\, d \\
\text{s.t.} \quad & z\, A_i\, l_c^{\gamma_c} x_c^{\nu_c} n_c^{1-\nu_c-\gamma_c} \geq \rho \left(\frac{p_d^c}{P_{d,a}}\right)^{-\eta_a} Y_a \\
& z\, \bar{l}_p^{\gamma_p} x_p^{\nu_p} n_p^{1-\nu_p-\gamma_p} \geq (1-\rho) \left(\frac{p_d^p}{P_{d,a}}\right)^{-\eta_a} Y_a \\
& \left[l_c^{\frac{\varepsilon_h - 1}{\varepsilon_h}} + l_p^{\frac{\varepsilon_h - 1}{\varepsilon_h}} \right]^{\frac{\varepsilon_h}{\varepsilon_h - 1}} \leq h \quad \text{and} \quad \bar{l}_p = \left[l_p^{\frac{\varepsilon_p - 1}{\varepsilon_p}} + d^{\frac{\varepsilon_p - 1}{\varepsilon_p}} \right]^{\frac{\varepsilon_p}{\varepsilon_p - 1}}
\end{aligned} \tag{21}$$

where the wage $w = 1$ is the numeraire. We assume that deforestation is subject to a perfectly enforced tax τ, and that all resulting revenues are rebated lump sum to the household. We think about τ as a strictly enforced environmental fine applied per hectare of newly cleared land. Since the government rebates all collected revenues directly to households, the tax operates purely as a financial penalty. The policy thus directly targets the deforestation margin and raises the hurdle for land expansion beyond the physical input cost.

The entry and modernization cutoffs are determined by evaluating operating profit. The thresholds solve the following indifference conditions:

$$\pi_t^o(z_{ta}(c), c) = \kappa_{ta}, \qquad \pi_m^o(z, c) - \pi_t^o(z, c) = (1 + r)\, P_n \kappa_{ma}. \tag{22}$$

The first condition defines the entry cutoff $z_{ta}(c)$. The second condition defines the productivity level z at which a farm is indifferent between the traditional and modern technology. Because the upgrade raises the return strictly to the crop activity and leaves the pasture technology unchanged, the value of upgrading moves directly with the size of the farm's crop operation. For a farm with low productivity, the crop operation is too small to generate a profit gain that covers the financed fixed cost $(1 + r)P_n\kappa_{ma}$, and the farm remains traditional. As productivity rises, the crop operation scales up, the gain from upgrading eventually exceeds the cost, and the farm enters the modern band at the root $z_{ma}(c)$.

Assumption 1. *(i)* $\eta_a > 1, \quad \varepsilon_p > 1,$

(ii) $\gamma_c \leq \gamma_p$,

(iii) $\gamma_p(\eta_a - 1) < \varepsilon_h - 1$,

(iv) $\frac{1}{\varepsilon_h} + \frac{\gamma_c}{\varepsilon_p}(\eta_a - 1) > 1.$

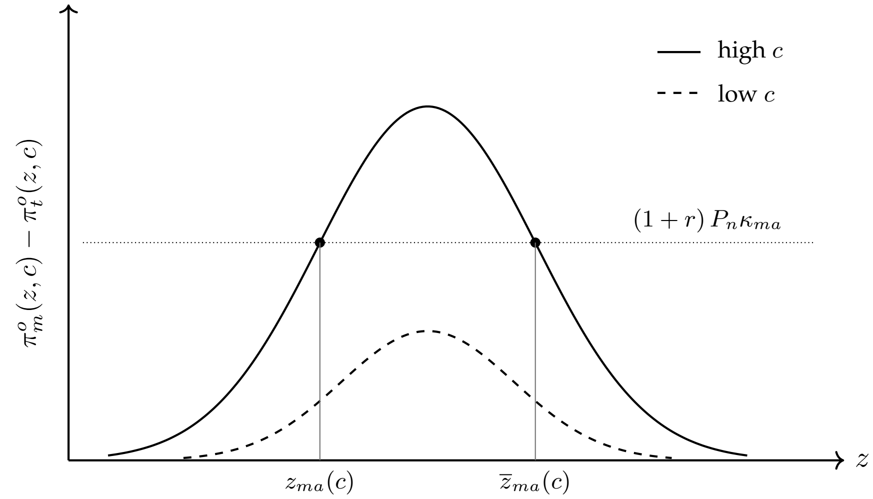
Throughout the remainder of the model we will occasionally invoke Assumption 1. This is the equivalent set of parametric conditions to those in our parsimonious framework from Section 3. In the limit $\varepsilon_h \to \infty$, the plot constraint collapses to $l_c + l_p \leq 1$, condition (iii) vanishes, and condition (iv) reduces to $\varepsilon_p < \gamma_c(\eta_a - 1)$, recovering the parametric restrictions of Proposition 1. With finite ε_h, the implicit gains from specialization within the owned plot provide an additional force toward pasture concentration for productive farms, relaxing the upper bound on ε_p to $\varepsilon_h\gamma_c(\eta_a - 1)/(\varepsilon_h - 1)$. Condition (iii) then ensures that ε_h is large enough for interior crop and pasture allocations.

Lemma 1 (Hump-shaped modernization gain)**.** *Under Assumption 1, the gain from modernizing $\pi_m^o(z, c) - \pi_t^o(z, c)$ is hump-shaped in z and vanishes as $z \to \infty$. The adoption set $\{z : \pi_m^o - \pi_t^o \geq (1 + r)P_n\kappa_{ma}\}$ is bounded at $\bar{z}_{ma}(c) < \infty$. The proof is in Appendix J.2.2.*

Next we show in Lemma 1 under Assumption 1, that the net gain from modernizing, $\pi_m^o - \pi_t^o$, does not rise monotonically with z, but is hump-shaped. This occurs because highly productive farms find it optimal to aggressively clear forest and shift their land allocation toward pasture. As the crop operation contracts for higher levels of z, the benefit of the crop-specific technology upgrade can fall below the financed adoption cost. This mechanism implies that the modernization indifference condition yields a second, upper root, $\bar{z}_{ma}(c)$. As a result, adoption is confined to an intermediate band $[z_{ma}(c), \bar{z}_{ma}(c)]$, where the most highly productive farms revert to the traditional technology. If $\frac{1}{\varepsilon_h} + \frac{\gamma_c}{\varepsilon_p}(\eta_a - 1) \leq 1$, the gain remains monotonically increasing in z and $\bar{z}_{ma}(c) = \infty$, such that the model collapses to a single-threshold rule.

In Figure 10 we plot the gain from modernizing against productivity z at different clearing cost draws, illustrating two cases where farms find it optimal not to adopt the modern technology good

Figure 10: Gain from Modernization and Adoption Thresholds



Notes: The figure plots the gain from modernizing, $\pi_m^o(z,c) - \pi_t^o(z,c)$, across productivity z for a high clearing cost (solid line) and a low clearing cost (dashed line) under the condition $\frac{1}{\varepsilon_h} + \frac{\gamma_c}{\varepsilon_p}(\eta_a - 1) > 1$. The horizontal dotted line represents the fixed cost $(1+r)P_n\kappa_{ma}$. For the high clearing cost, the gain intersects the fixed cost twice, defining the finite band $[\underline{z}_{ma}(c), \overline{z}_{ma}(c)]$. For the low clearing cost, the gain remains strictly below the fixed cost. If $\frac{1}{\varepsilon_h} + \frac{\gamma_c}{\varepsilon_p}(\eta_a - 1) \leq 1$, the gain would increase monotonically and $\overline{z}_{ma}(c) = \infty$.

even though they might be high z farmers. That is, for a sufficiently high clearing cost c, the gain crosses the financed fixed cost $(1+r)P_n\kappa_{ma}$ to form a finite adoption band $[\underline{z}_{ma}(c), \overline{z}_{ma}(c)]$. When productivity exceeds the upper threshold $\overline{z}_{ma}(c)$, the marginal value of pasture drives the farm to reallocate land away from crop production, which shrinks the relative scale of the crop operation and pushes the modernization gain below the fixed cost. Furthermore, if the clearing cost is low enough, the expected profit gain may remain strictly below the fixed cost across all productivity levels. Since a sufficiently low clearing cost induces a land expansion heavily dominated by pasture, the crop operation never reaches the minimum scale required to justify the fixed adoption cost, regardless of the farm's productivity draw. Hence, modernization fails to occur either because a farm's productivity exceeds the upper adoption threshold, or because a low clearing cost makes the upgrade strictly unprofitable at every productivity level.

The aggregate masses of active farms e_a and modern farms a_a are obtained by integrating these cutoffs over the clearing-cost distribution $G(c)$:

$$e_a = \int_0^\infty z_{ta}(c)^{-\zeta}\, G(dc), \qquad a_a = \int_0^\infty \left[\underline{z}_{ma}(c)^{-\zeta} - \overline{z}_{ma}(c)^{-\zeta}\right] G(dc). \tag{23}$$

In the following Proposition 2 we formalize how the farm's land allocation and technology adoption thresholds respond to shocks in partial equilibrium. Taking aggregate demand and prices as given, we derive comparative statics for an increase in the agricultural price $P_{d,a}$, an increase in the fixed modernization cost κ_{ma}, and stricter environmental enforcement τ.

Proposition 2. *Under Assumption 1, fix aggregate agricultural demand Y_a and all prices, and let $s_c \equiv l_c/(l_c + l_p + d)$.*

(1) (Extensification.) *A boom draws in entry, $\partial z_{ta}(c)/\partial P_{d,a} < 0$ for all c. If $\gamma_c < \gamma_p$ there is a finite clearing-cost threshold $c^\star > 0$ at which the land-use response splits by the cost of clearing. If $c < c^\star$ the farm shifts its owned plot from crops to pasture, expands cleared land, and lowers its crop share,*

$$\frac{\partial l_c(c)}{\partial P_{d,a}} < 0, \qquad \frac{\partial l_p(c)}{\partial P_{d,a}} > 0, \qquad \frac{\partial d(c)}{\partial P_{d,a}} > 0, \qquad \frac{\partial s_c(c)}{\partial P_{d,a}} < 0.$$

If $c > c^\star$ the farm instead shifts its owned plot toward crops,

$$\frac{\partial l_c(c)}{\partial P_{d,a}} > 0, \qquad \frac{\partial l_p(c)}{\partial P_{d,a}} < 0,$$

raising the crop share of the owned plot; the total-area crop share s_c carries no unconditional sign here, since cleared land may still expand near $c^\star$. If $\gamma_c = \gamma_p$ then $c^\star = \infty$ and the land-use response for $c < c^\star$ applies to all farms.

(2) (De-modernization.) *The modern region is bounded above by a finite upper threshold $\overline{z}_{ma}(c)$. A boom lowers this upper threshold, reverting the most productive modern farms to traditional methods, while entry at the floor leaves the modernization rate unchanged,*

$$\frac{\partial \overline{z}_{ma}(c)}{\partial P_{d,a}} < 0, \qquad \frac{\partial z_{ma}(c)}{\partial P_{d,a}} < 0, \qquad \frac{\partial (a_a/e_a)}{\partial P_{d,a}} = 0 \qquad \forall c.$$

(3) (Cost of modernizing.) *A rise in the modernization cost κ_{ma} leaves entry unchanged, raises the modernization floor, and sheds modern farms,*

$$\frac{\partial z_{ta}}{\partial \kappa_{ma}} = 0, \quad \frac{\partial z_{ma}}{\partial \kappa_{ma}} > 0, \quad \frac{\partial a_a}{\partial \kappa_{ma}} < 0, \quad \frac{\partial (a_a/e_a)}{\partial \kappa_{ma}} < 0.$$

Farms that revert to traditional methods shift from crops to pasture and clear more land; all other farms are unchanged,

$$\frac{\partial l_c}{\partial \kappa_{ma}} \leq 0, \qquad \frac{\partial l_p}{\partial \kappa_{ma}} \geq 0, \qquad \frac{\partial d}{\partial \kappa_{ma}} \geq 0, \qquad \frac{\partial s_c}{\partial \kappa_{ma}} \leq 0,$$

with the strict signs exactly for the farms that revert.

(4) (Environmental enforcement.) *A rise in the environmental policy τ pushes entry up and reduces cleared land,*

$$\frac{\partial z_{ta}}{\partial \tau} > 0, \qquad \frac{\partial d}{\partial \tau} < 0,$$

lowers the modernization floor z_{ma}, raises the upper threshold $\overline{z}_{ma}$, draws in modern farms, and tilts land toward crops,

$$\frac{\partial z_{ma}}{\partial \tau} < 0, \quad \frac{\partial \overline{z}_{ma}}{\partial \tau} > 0, \quad \frac{\partial a_a}{\partial \tau} > 0, \quad \frac{\partial (a_a/e_a)}{\partial \tau} > 0, \quad \frac{\partial s_c}{\partial \tau} > 0,$$

widening the modernization band from both ends.

The proof is in Appendix J.2.

9.4 The Non-Agricultural Sector

We assume a unit mass of potential non-agricultural firms that draw an ex-ante productivity z from the same Pareto distribution with shape parameter ζ as agricultural farms. If the firm chooses to produce and become active, the establishment supplies a differentiated variety and chooses between the same traditional and modern technology levels, A_t and A_m. After making its entry and technology adoption decisions, the establishment then draws an additional idiosyncratic ex-post productivity shock e^{ω}. This shock is drawn from a log-normal distribution, $\omega \sim \mathcal{N}\big(-\frac{\eta_n-1}{2}\delta^2,\, \delta^2\big)$, normalized such that $\mathbb{E}[e^{(\eta_n-1)\omega}] = 1$.

Production requires labor n and an intermediate input x. Facing standard constant elasticity of substitution (CES) demand, the establishment chooses its price p_d and input mix to maximize operating profit,

$$\begin{aligned} \pi^o_{i,n}(z,\omega) &= \max_{p_d,x,n}\; p_d\Big(\frac{p_d}{P_{d,n}}\Big)^{-\eta_n} Y_n - P_n\, x - n \\ &\text{s.t.} \quad z\, A_i\, e^{\omega}\, x^{\nu_n}\, n^{1-\nu_n} \geq \Big(\frac{p_d}{P_{d,n}}\Big)^{-\eta_n} Y_n. \end{aligned} \tag{24}$$

where ν_n denotes the output elasticity of the intermediate input. Because the production function exhibits constant returns to scale in variable inputs, the solution yields a standard constant markup over marginal cost. Operating profit is thus given by

$$\pi^o_{i,n}(z,\omega) = \frac{e^{(\eta_n-1)\omega}}{\eta_n}\Big(\frac{\eta_n-1}{\eta_n}\Big)^{\eta_n-1} \frac{(A_i z)^{\eta_n-1} P_{d,n}^{\eta_n} Y_n}{\big[P_n^{\nu_n}/\big(\nu_n^{\nu_n}(1-\nu_n)^{1-\nu_n}\big)\big]^{\eta_n-1}}. \tag{25}$$

The normalization of the ex-post shock leaves equilibrium aggregates unchanged and disperses establishment sizes within each technology class. Consequently, the entry and modernization decisions are entirely dictated by the firm's ex-ante productivity draw z, generating two productivity cutoffs $\{z_{tn}, z_{mn}\}$. A potential firm pays a fixed entry cost κ_{tn} in labor and is active for $z \geq z_{tn}$, running the traditional technology A_t. An active firm can then adopt the modern technology A_m by paying $P_n\kappa_{mn}$ units of the composite good Q_n, which is profitable for $z \geq z_{mn}$. These cutoffs are determined by the break-even conditions $\pi^o_{t,n}(z_{tn}) = \kappa_{tn}$ and $\pi^o_{m,n}(z_{mn}) - \pi^o_{t,n}(z_{mn}) = P_n\kappa_{mn}$. Because the ex-ante productivity z shares the same Pareto distribution, these cutoffs directly pin down the masses of active and modern firms as $e_n = z_{tn}^{-\zeta}$ and $a_n = z_{mn}^{-\zeta}$.

9.5 International Trade

Goods are traded with the rest of the world, and a flexible exchange rate clears the trade balance.

Exports. Foreign demand for the domestically produced good (exports, E_s) is a function of its price in foreign currency,

$$E_s = \Big(\frac{P_{d,s}}{\mathcal{H}}\Big)^{-\theta} \mathcal{D}^*_s, \tag{26}$$

where $\mathcal{H}$ is the nominal exchange rate, θ is the price elasticity of export demand, and $\mathcal{D}^*_s$ is a foreign demand shifter.

Trade Balance. The nominal exchange rate $\mathcal{H}$ adjusts so that the value of exports equals the value of imports in common currency. The equilibrium exchange rate solves:

$$\sum_s P_{d,s} E_s = \mathcal{H} \sum_s P^*_{f,s} I_s. \tag{27}$$

9.6 Market Clearing

The model is closed by market clearing for goods and labor and by the household budget constraint.

Goods Market. The non-agricultural composite serves consumption, intermediates, equipment, and clearing, and the agricultural composite serves consumption. Total supply of each domestic good equals domestic demand plus exports,

$$Y_n = \alpha_n \Big(\frac{P_{d,n}}{P_n}\Big)^{-\sigma} \Big[\phi_n \frac{P_c C}{P_n} + X_a + X_n + \kappa_{mn} a_n + \kappa_{ma} a_a + \mathcal{C}\Big] + E_n, \tag{28}$$

$$Y_a = \alpha_a \Big(\frac{P_{d,a}}{P_a}\Big)^{-\sigma} \phi_a \frac{P_c C}{P_a} + E_a, \tag{29}$$

where $N_n = \int_{z_{tn}}^{\infty} n(z)\, F(dz)$ and $X_n = \int_{z_{tn}}^{\infty} x(z)\, F(dz)$ in non-agriculture, while in agriculture the firm choices also depend on the clearing-cost draw and integrate over G,

$$N_a = \int_0^\infty \int_{z_{ta}(c)}^\infty n(z,c)\, F(dz)\, G(dc), \quad X_a = \int_0^\infty \int_{z_{ta}(c)}^\infty x(z,c)\, F(dz)\, G(dc), \quad D = \int_0^\infty \int_{z_{ta}(c)}^\infty d(z,c)\, F(dz)\, G(dc),$$

with $n = n_c + n_p$, $x = x_c + x_p$, D aggregate deforestation, and $\mathcal{C} = \int_0^\infty \int_{z_{ta}(c)}^\infty c\, d(z,c)\, F(dz)\, G(dc)$ the aggregate real clearing bill, with the per-hectare cost c varying by farm inside the integral.

Imports. Imports follow from CES demand for the imported component of each composite,

$$I_n = (1-\alpha_n)\Big(\frac{\mathcal{H} P^*_{f,n}}{P_n}\Big)^{-\sigma} Q_n, \qquad I_a = (1-\alpha_a)\Big(\frac{\mathcal{H} P^*_{f,a}}{P_a}\Big)^{-\sigma} Q_a, \tag{30}$$

where $Q_n = \phi_n P_c C/P_n + X_a + X_n + \kappa_{mn} a_n + \kappa_{ma} a_a + \mathcal{C}$ and $Q_a = \phi_a P_c C/P_a$ are the composite quantities demanded.

Labor Market. The total supply of labor N is allocated to production and entry,

$$N = \sum_s N_s + \sum_s e_s \kappa_{ts}. \tag{31}$$

Household Budget. The household supplies labor firms, receives all firm and bank profits, and collects the lump-sum transfer of the environmental tax revenue.

$$P_c C = N + \Pi_a + \Pi_n + \tau D, \tag{32}$$

where

$$\Pi_a = \int_0^\infty \int_{z_{ta}(c)}^\infty \pi^o_{i(z,c),a}(z,c)\, F(dz)\, G(dc) \;-\; \kappa_{ta}\, e_a \;-\; P_n \kappa_{ma}\, a_a, \tag{33}$$

$$\Pi_n = \int_{z_{tn}}^{\infty} \mathbb{E}_\omega\big[\pi^o_{i(z),n}(z,\omega)\big]\, F(dz) \;-\; \kappa_{tn}\, e_n \;-\; P_n \kappa_{mn}\, a_n. \tag{34}$$

The proceeds from the environmental enforcement charge τD are rebated to the household as a lump-sum transfer.[17]

9.7 Welfare and Environmental Policy

In this section we evaluate how the environmental policy τ shapes aggregate welfare. Because household utility is strictly increasing in aggregate consumption, we measure welfare using real consumption C. To understand why an environmental policy can improve aggregate welfare, we first isolate how the financial friction distorts the farmer's technology choice. At the modernization cutoffs $\{z_{ma}(c), \overline{z}_{ma}(c)\}$, a farm is indifferent between traditional and modern technologies. The private net profit gain equates to the financed adoption cost, $\pi^o_m(z,c) - \pi^o_t(z,c) = (1+r)P_n\kappa_{ma}$. The physical adoption cost to our economy is exactly $P_n\kappa_{ma}$ and the interest payment $rP_n\kappa_{ma}$ is a transfer from the farmer to the household. The social return to modernization therefore exceeds the private return by the financing wedge $rP_n\kappa_{ma}$. Because farmers do not internalize this return, the economy features under-adoption. By taxing land expansion, the environmental policy τ restricts aggregate deforestation D and tilts the relative returns toward technology adoption. As established in Proposition 2(4), a higher τ widens the modern band and increases the total mass of modern farms $\partial a_a/\partial\tau > 0$.

To formalize this trade-off, we evaluate the welfare impact in partial equilibrium to isolate the policy's direct effect on agricultural choices. In the comparative statics below we hold fixed the price indices (P_a, P_n, P_c) and the demand level Y_a. We restrict attention to interior land allocations under Assumption 1, so that the modernization band is finite and $\partial a_a/\partial\tau > 0$. The commodity boom is a rise in the agricultural producer price $P_{d,a}$.

Proposition 3 (Value of the policy)**.** *Under Assumption 1, in partial equilibrium, an environmental policy changes welfare by*

$$\frac{\partial C}{\partial \tau} = \frac{1}{P_c}\Big[\, r\, P_n\kappa_{ma}\,\frac{\partial a_a}{\partial\tau} \;+\; \tau\,\frac{\mathrm{d}D}{\mathrm{d}\tau}\,\Big]. \tag{35}$$

The first term captures the welfare gain from correcting the financial friction. This gain is the uninternalized return $rP_n\kappa_{ma}$ multiplied by the mass of new adopters $\partial a_a/\partial\tau > 0$. The second term captures the welfare loss of distorting the land expansion margin below its private optimum. This loss is the total derivative $\mathrm{d}D/\mathrm{d}\tau \le 0$, which is of order τ and vanishes at $\tau = 0$.

(i) If $r = 0$, only the second term remains and welfare decreases in τ.

(ii) If $r > 0$, then $\partial C/\partial\tau|_{\tau=0} > 0$ and a marginal environmental policy raises welfare. The two forces exactly offset when $rP_n\kappa_{ma}\,\partial a_a/\partial\tau = -\tau\,\mathrm{d}D/\mathrm{d}\tau$.

[17]Because the interest on the adoption loan is rebated to the household, the financed cost $(1+r)P_n\kappa_{ma}$ paid by the farmer and the interest income $rP_n\kappa_{ma}a_a$ received by the household consolidate, so Π_a subtracts only the physical cost $P_n\kappa_{ma}a_a$.

Proposition 4 (Policy and the gains from trade)**.** *Under Assumption 1, in partial equilibrium, let the boom raise the producer price from $P^0_{d,a}$ to $P^1_{d,a}$ and apply the policy $0 \to \tau$ after the boom occurs. If $r > 0$ and we restrict τ such that $\tau \in (0, \tau^*]$, where τ^* is the optimal policy evaluated at $P^1_{d,a}$, then*

$$\left[C(P^1_{d,a}, \tau) - C(P^0_{d,a}, 0)\right] - \left[C(P^1_{d,a}, 0) - C(P^0_{d,a}, 0)\right] = \int_0^{\tau} \frac{\partial C}{\partial \tau'}\Big|_{P^1_{d,a}} d\tau' > 0.$$

By correcting the adoption friction, the environmental policy increases the realized gains from trade.

10 Calibration

We calibrate the model parameters in two steps. First, we assign values to an external block of parameters based on standard values from the literature and aggregate data (see Table G.3 in Appendix G). Second, we jointly calibrate an internal block of parameters such that the model-implied moments match their empirical counterparts (see Table G.1 in Appendix G).

The internally calibrated block consists of fifteen parameters: the agricultural production weight ρ, the land-use elasticities $(\varepsilon_h, \varepsilon_p)$, a mean, dispersion, and lower bound $(\bar{c}, \varepsilon, \underline{c})$ of the clearing cost distribution $G(c)$, the modern technology level A_m, the fixed costs $(\kappa_{tn}, \kappa_{mn}, \kappa_{ta}, \kappa_{ma})$, the Pareto tail parameter ζ, the idiosyncratic shock dispersion δ, and the foreign demand shifters $(\mathcal{D}^*_a, \mathcal{D}^*_n)$. We compute the model moments as aggregate expectations over the joint distribution of productivity $F(dz)$ and clearing costs $G(dc)$.

10.1 Externally Calibrated Parameters

We assign values to an external block of parameters using aggregate data and the literature. Using the OECD Input-Output tables, we measure the sectoral consumption shares $\phi_a = 0.04$ and $\phi_n = 0.96$ as each sector's proportion of total household final consumption, and we calculate the home bias parameters $\alpha_a = 0.96$ and $\alpha_n = 0.93$ as the domestic share of total sectoral expenditure. The non-agricultural intermediate input expenditure share $\nu_n = 0.54$ is derived from the labor share reported in the OECD System of National Accounts. For the agricultural production function, we set the intermediate input shares $\nu_c = 0.44$ and $\nu_p = 0.17$ and the land elasticity shares $\gamma_c = 0.50$ and $\gamma_p = 0.53$ to match the estimates from Pellegrina (2022). We set the elasticity of substitution between domestic and foreign goods $\sigma = 2$ following Feenstra et al. (2018), and the export elasticity $\theta = 5$ following Qiu et al. (2026). The elasticities of substitution between varieties are set to $\eta_a = 4.1$ in agriculture and $\eta_n = 3.2$ in non-agriculture, implying sectoral markups consistent with Heresi (2024). It is well known that borrowing costs in Brazil are exceptionally high. We set the intra-period interest rate to $r = 0.45$, matching the credit-weighted spread paid by firms in Cavalcanti et al. (2023), so that a farmer repays 45 percent in interest on the loan financing the technology upgrade. Finally, we normalize the total labor endowment N, the baseline traditional technology A_t, the common plot size h, and the foreign import prices $P^*_{f,a}$ and $P^*_{f,n}$ to one, and set the baseline environmental policy τ to zero.

10.2 Employment Size Distributions

We calibrate the parameters $\{A_m, \kappa_{tn}, \kappa_{mn}, \kappa_{ta}, \kappa_{ma}, \zeta, \delta\}$ by fitting model predictions for the firm employment distribution to microdata jointly across all sectors. We build on the insight of Buera et al. (2021) and Buera & Trachter (2024) that the cross sectional distribution of employment identifies entry and adoption costs. The discrete decisions to enter the market and adopt modern technologies determine the share of employment mass across different firm sizes. We identify the fixed entry and adoption costs, along with the modern productivity premium and ex post shock dispersions, by matching empirical employment shares across discrete size bins.

For the non agricultural sectors, we compute employment shares across 21 bins per sector using Brazil's Relação Anual de Informações Sociais (RAIS), a registry covering strictly formal workers. In agriculture however, RAIS would substantially understate the true farm size distribution because agricultural labor is highly informal. Therefore, we estimate the agricultural entry and adoption cost using the Censo Agropecuário 2017, which captures informal labor and reports establishment counts across six size buckets.

In the non-agricultural sector, employment is proportional to operating profit. Conditional on the ex-ante productivity z and the ex-post shock ω, the labor bill is

$$n_{i,n}(z,\omega) = e^{(\eta_n-1)\omega}\,\frac{(1-\nu_n)(\eta_n-1)}{\eta_n}\left(\frac{\eta_n-1}{\eta_n}\right)^{\eta_n-1}\frac{(A_i z)^{\eta_n-1}P_{d,n}^{\eta_n}Y_n}{\left[P_n^{\nu_n}/\left(\nu_n^{\nu_n}(1-\nu_n)^{1-\nu_n}\right)\right]^{\eta_n-1}}. \tag{36}$$

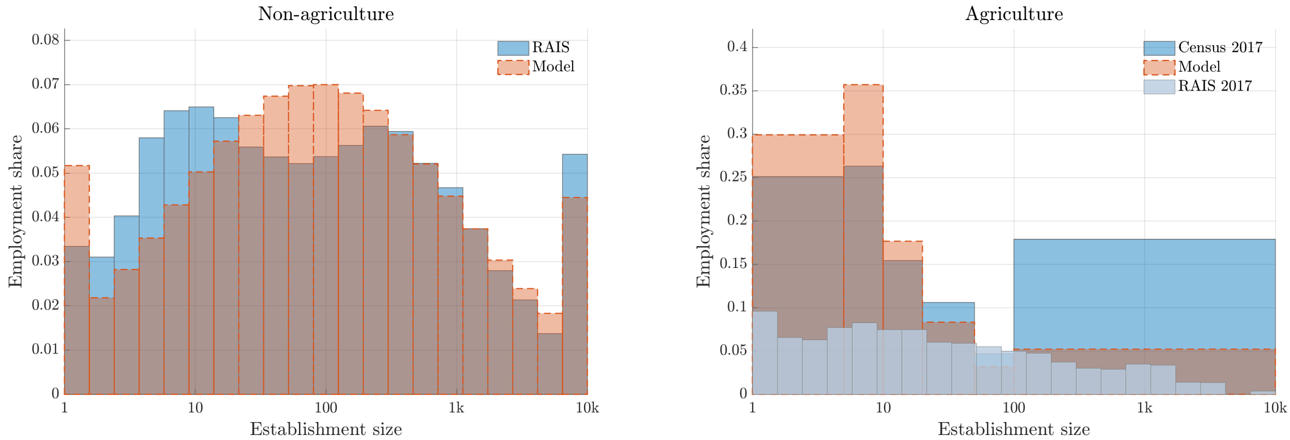
In the agricultural sector, the employment distribution is derived from the sum of crop and pasture labor inputs:

$$n_a(z,c) = \tfrac{\eta_a-1}{\eta_a}\left[(1-\nu_c-\gamma_c)\,p_d^c\,y^c + (1-\nu_p-\gamma_p)\,p_d^p\,y^p\right]. \tag{37}$$

We match the theoretical firm size distributions from both sectors directly to the empirical data to jointly identify the parameters $\{A_m, \kappa_{tn}, \kappa_{mn}, \kappa_{ta}, \kappa_{ma}, \zeta, \delta\}$. Here the sector specific fixed entry costs κ_{tn} and κ_{ta} determine the minimum size of an active firm and anchor the lower bounds of the respective employment distributions. Since the underlying productivity draws govern the proportion of larger firms, the common Pareto tail parameter ζ is identified from the steady decline in employment shares across upper bins in both sectors. Note that adopting the modern technology increases a firm's labor demand, which implies the size difference between traditional and modern firms pins down the common modern premium A_m along with the sector specific adoption costs κ_{mn} and κ_{ma}. Finally, the ex post shock ω disperses employment sizes within each non agricultural technology class to yield a smooth theoretical distribution, so we identify the dispersion parameter δ by matching the model to the RAIS registry.

In Figure 11 we compare the resulting employment shares in the model and in the data. Outside agriculture our model captures the wide middle of the size distribution together with the concentration of employment in very large establishments. In agriculture, the model is able to match the concentration of workers on small farms.

Figure 11: Employment size distributions in the model and in the data



Notes. Sectoral employment shares by establishment size (measured in workers, logarithmic scale). Left panel (Non-agriculture): Compares the fitted model (red) with the RAIS (blue) across 21 bins in 2003. Right panel (Agriculture): Compares the model (red) with the 2017 Censo Agropecuário across six bins and RAIS data from 2017.

10.3 The Land-Use Elasticities: Identifying ε_h and ε_p

The substitution elasticities $(\varepsilon_h, \varepsilon_p)$ govern the farm's reallocation of land between crops, pasture, and new deforestation. We calibrate these parameters by matching the model's crop-share responses to a foreign-demand shock in the Legal Amazon and the rest of Brazil to the peak responses ($\tau = 4$) from our empirical specification (Table 1). A well-known challenge in mapping cross-sectional estimates to structural models is that regressions with time fixed effects absorb any general equilibrium response common to all units, recovering only relative effects across differentially exposed units (e.g., Nakamura & Steinsson 2014, Guren et al. 2021, Wolf 2023). In our setting, the year fixed effects in our baseline regression in equation (10) absorb any common adjustment in agricultural prices that the foreign-demand shock induces in general equilibrium. To construct model counterparts that are consistent with our estimation, we perturb $\mathcal{D}_a^*$ and re-solve for the new general equilibrium, then subtract the national average crop-share response from each region's response. Since the general-equilibrium price adjustment is common to every farm regardless of its clearing cost draw, the national average captures the same common component that the year fixed effects absorb in the empirical regression.[18]

By construction, our model abstracts from explicit spatial variation, however the clearing cost draw c introduces heterogeneity in the profitability of forest conversion and thus captures underlying differences in land abundance between the Legal Amazon and the rest of Brazil. Because abundant land in the Legal Amazon drives down forest conversion costs, agricultural production within the Amazon favors land intensive cattle ranching over crop cultivation. A lower cost draw therefore translates directly into a lower crop share $s_c \equiv l_c/(l_c+l_p+d)$, which we use as an observable geographic

[18] Note that $\mathcal{D}_a^*$ enters the model only through the goods market clearing condition for Y_a, so a change in $\mathcal{D}_a^*$ is isomorphic to a change in $P_{d,a}$. The variation that survives demeaning is therefore the cross-sectional heterogeneity in the response of $l_c/(h+d)$ to the common change in $P_{d,a}$ across the joint distribution of (z, c), which is governed by $(\varepsilon_h, \varepsilon_p)$.

footprint (based on microdata from the CAR matched with land use data from Mapbiomas) to estimate each farmer's probability of being located within the Amazon. More specifically, we estimate the conditional probability that a farm is located in the Legal Amazon given its crop share via a logistic regression,

$$\hat{\beta}(s_{c,i}) \;\equiv\; \widehat{\Pr}\big(j(i) \in \text{Legal Amazon} \,\big|\, s_{c,i}\big) \in [0,1], \tag{38}$$

and assign each simulated farm a propensity $\beta(z,c) \equiv \hat{\beta}\big(s_c(z,c)\big)$. We estimate $\beta(z,c)$ with a cubic logistic regression on 6.7 million farms using our data from the CAR and Mapbiomas.[19] Table G.2 in Appendix G reports the coefficients and in Figure 12 we plot the fitted curve. Our estimate suggests that farms without crop land are located within the Legal Amazon with a probability of roughly one third, which falls below five percent once crops cover a third of the farm. Hence, farms devoted mainly to pasture are far more likely to be located within the Legal Amazon.[20]

For any farm-level outcome $X(z,c)$, we then compute the regional conditional expectations by weighting the joint distribution of productivity and clearing costs by the estimated propensity $\hat{\beta}(s_{c,i})$, integrating over active farms with $z \geq z_{ta}(c)$:

$$\begin{aligned}
\mathbb{E}\big[X \mid \text{Amazon}\big] &= \frac{\displaystyle\int_{\underline{c}}^{\infty}\int_{z_{ta}(c)}^{\infty} X\,\beta\,F(dz)\,G(dc)}{\displaystyle\int_{\underline{c}}^{\infty}\int_{z_{ta}(c)}^{\infty} \beta\,F(dz)\,G(dc)}, \\
\mathbb{E}\big[X \mid \text{non-Amazon}\big] &= \frac{\displaystyle\int_{\underline{c}}^{\infty}\int_{z_{ta}(c)}^{\infty} X\,(1-\beta)\,F(dz)\,G(dc)}{\displaystyle\int_{\underline{c}}^{\infty}\int_{z_{ta}(c)}^{\infty} (1-\beta)\,F(dz)\,G(dc)}.
\end{aligned} \tag{39}$$

We identify $(\varepsilon_h, \varepsilon_p)$ by targeting the crop-share responses

$$\frac{\partial\,\mathbb{E}\big[l_c/(h+d)\,\big|\,\text{Amazon}\big]}{\partial \ln \mathcal{D}_a^*} \quad \text{and} \quad \frac{\partial\,\mathbb{E}\big[l_c/(h+d)\,\big|\,\text{non-Amazon}\big]}{\partial \ln \mathcal{D}_a^*}, \tag{40}$$

each expressed relative to the national average $\partial\,\mathbb{E}[l_c/(h+d)]/\partial \ln \mathcal{D}_a^*$ (demeaned). We target -1.84 percentage points for the Legal Amazon and $+0.37$ for the rest of Brazil, matching the peak responses at horizon $\tau = 4$ from the separate Amazon and non-Amazon specifications. Panel B of Table 1 shows the model fit. In our calibration we pin down $\varepsilon_h = 3.10$ and $\varepsilon_p = 1.95$. The remaining demeaned responses (pasture, forest, and adoption expenditure) as well as the response to subsidized credit are untargeted moments reported in Table 2.

[19] Specifically, we estimate $\hat{\beta}(s_{c,i}) = \Lambda\big(\beta_0 + \beta_1 s_{c,i} + \beta_2 s_{c,i}^2 + \beta_3 s_{c,i}^3\big)$ by maximum likelihood on $N = 6{,}729{,}685$ farming properties in the 2003 CAR cross section matched with land use data from MapBiomas, where $\Lambda(\cdot)$ denotes the logistic function and $s_{c,i} \equiv l_{c,i}/(l_{c,i} + l_{p,i} + d_i)$ is the crop share of farm i. Table G.2 reports the coefficients.

[20] In Appendix I we estimate an alternative specification $\hat{\beta}(s_c, x)$ that also conditions on farming area, so that large pasture farms can reach Amazon probabilities above 70 percent. Recalibrating the model with this richer bridge leaves all parameters and quantitative conclusions virtually unchanged.

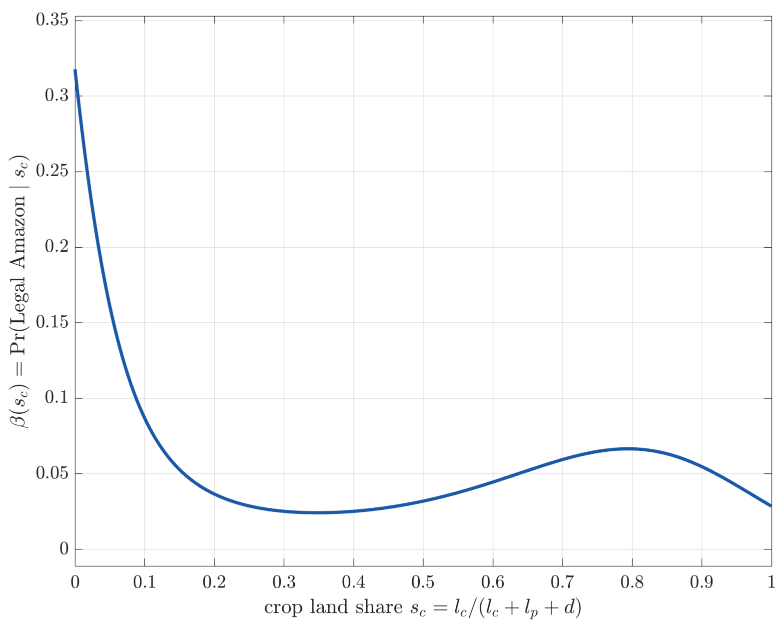
Table 1: Land-use responses to foreign demand and the model fit

Panel A. Empirical responses (area share, pp)			
	Foreign demand		Credit access
Dependent variable	Legal Amazon	Non-Amazon	ATT
Crop (soybean)	-1.84^{***}	0.37^{**}	5.50^{***}
	(0.29)	(0.17)	(1.25)
Pasture	1.78^{***}	-0.21^{**}	-4.16^{***}
	(0.32)	(0.09)	(1.00)
Forest	-0.41^{***}	0.05	0.18
	(0.13)	(0.03)	(0.24)

Panel B. Targeted moments (demeaned, pp)		
	Data	Model
Crop-share response, Legal Amazon	$[-2.41,\ -1.27]$	-1.84
Crop-share response, Non-Amazon	$[+0.04,\ +0.70]$	$+0.19$

Notes. Panel A reports the peak response at horizon $\tau = 4$ from equation (10), estimated separately for the Legal Amazon and the rest of Brazil. The credit column reports the average treatment effect at $\tau = 4$ from our difference-in-differences specification. Standard errors in parentheses. ***, **, and * denote significance at the 1, 5, and 10% levels. Panel B reports 95 percent confidence intervals from Panel A alongside the demeaned model counterpart, in percentage points (pp).

Figure 12: The probability of Amazon location as a function of the crop land share



Notes. The curve plots the fitted probability that a farm lies in the Legal Amazon conditional on its crop share of farm area, estimated by a cubic logistic regression on 6.7 million farms in the CAR matched with data from Mapbiomas for 2003.

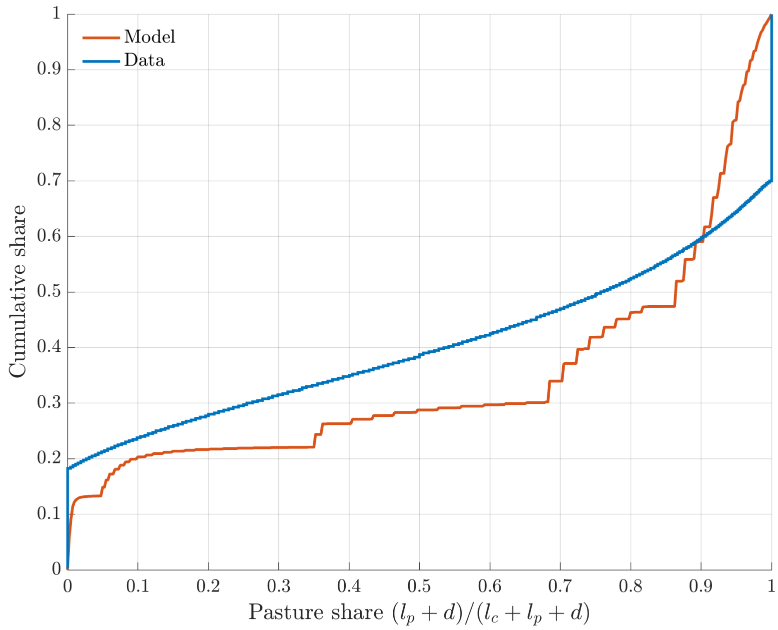
10.4 Remaining Internally Calibrated Parameters

The remaining internal parameters are jointly calibrated such that the model's predictions match empirical moments. The agricultural production weight ρ targets the crop share of agricultural production value from FAO data, and the foreign demand shifters $\mathcal{D}_a^*$ and $\mathcal{D}_n^*$ target the sectoral export-to-consumption ratios from the OECD Input-Output tables.

The distribution of clearing costs is the primary driver of land use in our model. Because converted land is used as pasture, the pasture share of a farm acts as the observable footprint of its clearing cost. Farms with cheap access to forest operate large pastures, whereas farms facing high clearing costs concentrate on crops cultivated on their endowed owned land h. The empirical distribution of pasture across farms is therefore directly informative about the underlying distribution of clearing costs.

To match the model to the empirically observed pasture share distribution, we assume that clearing costs follow a shifted log-normal distribution, $\ln(c - \underline{c}) \sim \mathcal{N}(\ln \bar{c}, \varepsilon^2)$, defined by the parameters $(\bar{c}, \varepsilon, \underline{c})$. The log-normal functional form captures the two salient features of the land-use cross-section: a large mass of farms with moderate costs and a long right tail of farms for which clearing is effectively prohibitive. The floor $\underline{c} > 0$ bounds costs away from zero, reflecting that even the most accessible forest requires machinery and fuel to convert. The lower bound $\underline{c}$ is also quantitatively necessary because without $\underline{c}$, farms drawing a very low clearing cost would deforest unrealistically large areas.

Figure 13: Cumulative distribution of pasture shares across farms



Notes. The figure plots the cumulative distribution of the pasture share of farm area in the model and in the 2003 CAR registry. The model distribution aggregates farm level shares over the calibrated joint distribution of productivity and clearing costs. The mean and the standard deviation of this distribution are targeted in the calibration, while the rest of the shape is left free.

The parameters $(\bar{c}, \varepsilon, \underline{c})$ are identified by three moments: We pin down $\bar{c}$ and ε with the mean and standard deviation of the pasture share of farm area in the CAR matched with Mapbiomas data from

2003. Uniformly higher costs shift every farm away from pasture and lower the average share, while cost dispersion translates into the dispersion of pasture shares across farms. However, moments from the distribution of pasture shares cannot identify the cost floor $\underline{c}$ because the pasture share is bounded at one. Once a farm allocates land entirely to pasture, a lower clearing cost draw simply expands total cleared land without changing the share. Since farms with the lowest costs operate at this corner solution and account for the majority of total cleared land, the floor $\underline{c}$ determines their total cleared area. We therefore identify $\underline{c}$ by matching the model's implied aggregate pasture-to-crop land ratio to aggregate data from MapBiomas. The data reflect the disproportionate weight of large, pasture-intensive farms: the average farm has a pasture share of roughly 0.60, whereas pasture accounts for approximately 0.80 of aggregate farm area.

All parameters are estimated jointly, with model moments computed as aggregate expectations over the joint distribution of productivity and clearing costs. The mean pasture share pins down $\bar{c}$, its dispersion pins down ε, and the aggregate land ratio pins down $\underline{c}$. Figure 13 shows the resulting fit.

11 The Calibrated Economy and External Validity

Our model predicts highly productive farmers optimally forgo modern technology when facing low land-clearing costs. To see this more clearly we map the entry and technology decision of every farm in our calibrated model across their productivity z and clearing cost c draw in Figure 14.

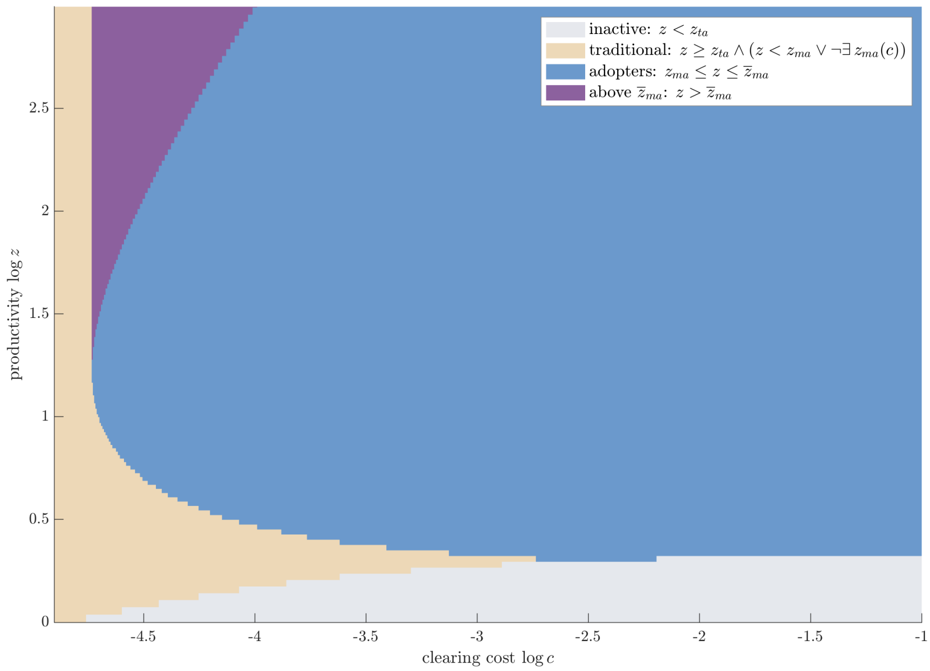
Note that the x-axis in Figure 14 maps directly to our estimated probability of being located within the Legal Amazon through $\beta(z, c)$. Hence, for the sake of intuition, we can think of a lower clearing cost as equivalent to a higher probability of being in the Legal Amazon with a higher optimal share of pasture.

Farms that draw a low clearing cost (and thus are assigned a higher probability of being in the Amazon) face a lower entry threshold $z_{ta}(c)$ because access to inexpensive land raises expected profits, allowing less productive operations to cover the fixed entry cost. Thus, the least productive active farms in our economy are those with the highest probability of being in the Amazon. Overall, we show below that the probability weighted share of farms within the Amazon (around 9%) matches closely the observed data.

Once active, farms adopt the modern technology inside the band, $z_{ma}(c) \leq z \leq \overline{z}_{ma}(c)$, and stay traditional either if $z < z_{ma}(c)$, or when clearing costs are so low that $z_{ma}(c)$ simply does not exist (formally, we write $\neg\exists\, z_{ma}(c)$ in the Figure's legend). For farms drawing low clearing costs, the marginal value of pasture drives a reallocation away from crops such that $\overline{z}_{ma}(c)$ becomes finite. Farms drawing a productivity level z that exceeds $\overline{z}_{ma}(c)$ therefore forgo technology and operate under the traditional technology. Since we think of a lower clearing cost being equivalent to a higher probability of being located within the Legal Amazon, the model implies a lower aggregate adoption rate within the Amazon and we show below that our calibration also closely matches the relative adoption rate as an untargeted empirical moment.

Our model is able to match a range of key untargeted moments that we summarize in Table 2 which

Figure 14: Technology choice across productivity and clearing draws



Notes. We partition the space of productivity z and clearing cost c, both on logarithmic axes, into the regions implied by the calibrated thresholds. Farms below $z_{ta}(c)$ stay out of agriculture. Active farms adopt the modern technology inside the band $[z_{ma}(c), \overline{z}_{ma}(c)]$ and operate the traditional technology elsewhere, including where $z_{ma}(c)$ does not exist and no farm modernizes. Where $\overline{z}_{ma}(c)$ does not exist the band is unbounded from above. In equilibrium 28 percent of active farms are modern.

capture the difference in land use and technology adoption between the Legal Amazon and the rest of Brazil, and is also able to rationalize our empirical findings that we do not specifically target.

First, we examine in Panel A how our model divides land use between the Amazon and the rest of Brazil. Our calibration targets the distribution of pasture shares across individual farms, yet our framework accurately predicts the aggregate regional land division. Pasture represents the dominant farmland use across both regions, but the concentration is exceptionally high within the Amazon. Specifically, our model assigns 93 percent of Amazonian farmland to pasture, aligning closely with the 89 percent observed in the data. Furthermore, outside the Amazon, we match the empirical pasture share almost exactly. [21]

Second, we shift our focus in Panel A from intra-regional land allocation to the aggregate size of the Amazonian agricultural economy. Our model predicts that 9 percent of all farms locate within the Amazon and generate 13 percent of total agricultural value added, closely matching the observed data from the CAR and IBGE.[22] Furthermore, our model is able to track the concentrated nature of Amazonian agriculture, where fewer farms operate at a significantly larger size ($h + d$) and generate

[21]We measure regional land use from MapBiomas land cover in 2003, aggregated within and outside the Legal Amazon.

[22]We take agricultural value added by state in 2003 from IBGE's municipal GDP accounts (PIB dos Municípios, SIDRA table 5938) and aggregate the nine Legal Amazon states, with Maranhão entering in proportion to its statutory municipalities' share of agricultural value added. Farm counts are from the Rural Environmental Registry (CAR).

higher revenues, closely matching the observed farm revenues and size ratios in the data.

Third, we examine the level and regional heterogeneity of technology adoption in Panel A. To estimate an empirical counterpart for the model-implied share of farms operating modern technology, we proxy modern operations using the observed national shares of establishments owning machinery, applying pesticides, and applying fertilizer from the 2006 Censo Agropecuário. For the non-agricultural sector, we exploit data from PINTEC and the World Bank Enterprise Survey to proxy the modern share through metrics like process innovation and fixed asset purchases.[23] Our model predicts 28 percent of farms and 40 percent of non-agricultural firms operate modern technology, which are within the ranges spanned by our empirical proxies. Importantly, our model implies the share of Amazonian farms operating modern technology is 0.41 times the non-Amazonian share, closely aligning with the respective census ratios of 0.42 for pesticides, 0.39 for machinery, and 0.20 for fertilizer. Hence, the low technological modernization within the Legal Amazon—a central feature of our model—emerges strictly as an endogenous prediction of the model that closely aligns with the observed data.

Fourth, although our calibration targets only the demeaned crop-share responses, the model closely reproduces our farm-level empirical estimates of the reallocation from crops into pasture and the decline in adoption expenditure in the Legal Amazon. Outside the Amazon, land use and technology spending are largely unresponsive to the demand shock in both the model and the data, a pattern the model attributes to heterogeneity in the cost of clearing forest. One caveat in our model is that deforestation is always an available margin to expand production, however in the data many farms may not have remaining forest on their plot, such that the model overpredicts the magnitude of the forest response relative to our empirical estimates. Note that we measure deforestation only within the registered farm boundary, so any clearing that farmers undertake outside their property is not captured in our empirical estimate on forest loss.

Fifth, in Panel C we examine how the model responds to subsidized credit. We set r to zero at fixed $(P_{d,a}, P_{d,n}, Y_a, Y_n, \mathcal{H})$ in partial equilibrium (PE), removing the financing spread on the modernization loan, while holding prices fixed to account for the time fixed effects in our difference-in-differences specification that absorb any common price adjustment between treated and control farms. Cheaper financing moves farms out of pasture and into crops, and the model's crop share response lies inside the confidence interval of our difference-in-differences estimate. The same shock raises the forest share, though the model overpredicts the magnitude relative to the data.

Lastly, recall that we depart from a perfectly substitutable owned-plot allocation between pasture l_p and crop cultivation l_c such that there exists non-farmed vacant land $v = h - l_c - l_p > 0$ if a farmer

[23] For farms, we use the 2006 Censo Agropecuário, extracting state-level counts of establishments owning machinery (including tractors, trucks, pickups, and trailers), applying pesticides, and applying fertilizer. Nationally, 10 percent of the 5.2 million establishments own machinery, 27 percent use pesticides, and 36 percent use fertilizer. To construct the regional ratio, we aggregate establishment counts across the nine Legal Amazon states and compare the totals against the remaining states, yielding an Amazon adoption rate of 0.39 times the non-Amazon share for machinery, 0.42 for pesticides, and 0.20 for fertilizer. For the non-agricultural sector, we capture the modern share using survey measures of firm sophistication. In PINTEC 2017, 39,329 of the 116,962 surveyed firms (34 percent) report implementing product or process innovations. In the World Bank Enterprise Survey for Brazil 2009, the survey-weighted share of firms offering formal worker training is 0.42, operating a website is 0.56, and purchasing fixed assets is 0.61.

Table 2: Untargeted moments

	Model counterpart	Model	Data	Source
Panel A. Cross-sectional moments				
Land use				
Pasture share of farmland, Legal Amazon	$\frac{\int_{\underline{c}}^{\infty}\int_{z_{ta}(c)}^{\infty}(l_p+d)\,\beta\,F(dz)\,G(dc)}{\int_{\underline{c}}^{\infty}\int_{z_{ta}(c)}^{\infty}(l_c+l_p+d)\,\beta\,F(dz)\,G(dc)}$	0.93	0.89	MapBiomas
Pasture share of farmland, non-Amazon	$\frac{\int_{\underline{c}}^{\infty}\int_{z_{ta}(c)}^{\infty}(l_p+d)\,(1-\beta)\,F(dz)\,G(dc)}{\int_{\underline{c}}^{\infty}\int_{z_{ta}(c)}^{\infty}(l_c+l_p+d)\,(1-\beta)\,F(dz)\,G(dc)}$	0.79	0.76	MapBiomas
The Amazon in the aggregate				
Share of farms in the Amazon	$\mathbb{E}[\,\beta\,]$	0.09	0.13	CAR
Amazon share of agricultural value added	$\frac{\int_{\underline{c}}^{\infty}\int_{z_{ta}(c)}^{\infty}(p_d^c y_j^c + p_d^p y_j^p)\,\beta\,F(dz)\,G(dc)}{\int_{\underline{c}}^{\infty}\int_{z_{ta}(c)}^{\infty}(p_d^c y_j^c + p_d^p y_j^p)\,F(dz)\,G(dc)}$	0.13	0.17	IBGE
Mean farm revenue, Amazon / non-Amazon	$\frac{\mathbb{E}[\,p_d^c y_j^c + p_d^p y_j^p \mid \text{Amazon}\,]}{\mathbb{E}[\,p_d^c y_j^c + p_d^p y_j^p \mid \text{non-Amazon}\,]}$	1.43	1.28	IBGE, CAR
Mean farm size, Amazon / non-Amazon	$\frac{\mathbb{E}[\,l_c+l_p+d \mid \text{Amazon}\,]}{\mathbb{E}[\,l_c+l_p+d \mid \text{non-Amazon}\,]}$	1.67	1.94	CAR
Technology adoption				
Share of modern farms	a_a/e_a	0.28	0.10–0.36	Censo Agropecuário
Share of modern nonagricultural firms	a_n/e_n	0.40	0.34–0.61	PINTEC, Enterprise Survey
Adoption rate, Amazon / non-Amazon	$\frac{\Pr\left[\,z \in [z_{ma}(c), \overline{z}_{ma}(c)] \mid \text{Amazon}\,\right]}{\Pr\left[\,z \in [z_{ma}(c), \overline{z}_{ma}(c)] \mid \text{non-Amazon}\,\right]}$	0.41	0.20–0.42	Censo Agropecuário
Panel B. Response to foreign demand (demeaned)				
Legal Amazon				
Pasture share (pp)	$\partial\,\mathbb{E}[\,(l_p{+}d)/(h{+}d) \mid \text{Amazon}\,]/\partial \ln \mathcal{D}_a^*$	+1.84	[+1.15, +2.41]	Own estimate
Forest share (pp)	$-\,\partial\,\mathbb{E}[\,d/(h{+}d) \mid \text{Amazon}\,]/\partial \ln \mathcal{D}_a^*$	−2.14	[−0.67, −0.15]	Own estimate
Adoption expenditure (%)	$\partial\,\mathbb{E}[\,\log(1{+}\mathbf{1}\{i{=}m\}\,P_n\kappa_{ma}) \mid \text{Amazon}\,]/\partial \ln \mathcal{D}_a^*$	−0.24	[−0.45, −0.16]	Own estimate
Non-Amazon				
Pasture share (pp)	$\partial\,\mathbb{E}[\,(l_p{+}d)/(h{+}d) \mid \text{non-Amazon}\,]/\partial \ln \mathcal{D}_a^*$	−0.19	[−0.39, −0.03]	Own estimate
Forest share (pp)	$-\,\partial\,\mathbb{E}[\,d/(h{+}d) \mid \text{non-Amazon}\,]/\partial \ln \mathcal{D}_a^*$	+0.22	[−0.01, +0.11]	Own estimate
Adoption expenditure (%)	$\partial\,\mathbb{E}[\,\log(1{+}\mathbf{1}\{i{=}m\}\,P_n\kappa_{ma}) \mid \text{non-Amazon}\,]/\partial \ln \mathcal{D}_a^*$	+0.02	[−0.05, +0.08]	Own estimate
Panel C. Response to subsidized credit (PE)				
Crop share (pp)	$\Delta\,\mathbb{E}[\,s_c\,]\big\rvert_{r\to 0}$	+12.5	[+4.20, +14.54]	Own estimate
Forest share (pp)	$-\,\Delta\,\mathbb{E}[\,d/(h+d)\,]\big\rvert_{r\to 0}$	+1.60	[−2.04, +1.14]	Own estimate

Notes. For any farm level outcome $X(z,c)$ we write $\mathbb{E}[X] = \int_{\underline{c}}^{\infty}\int_{z_{ta}(c)}^{\infty} X\,F(dz)\,G(dc) \,/\, \int_{\underline{c}}^{\infty}\int_{z_{ta}(c)}^{\infty} F(dz)\,G(dc)$, and Amazon and non-Amazon following equation (39). In Panel A, we use MapBiomas land cover in 2003, aggregated within and outside the Legal Amazon. Farm counts and farm areas are from the 6.7 million farms in the 2003 Rural Environmental Registry (CAR), and agricultural value added by state from IBGE's municipal GDP accounts (PIB dos Municípios). To measure technology adoption, we exploit data from the 2006 Censo Agropecuário to calculate shares in pesticide use (0.27), tractor ownership (0.10), and fertilizer use (0.36), and ratios between the Amazon and non-Amazon (0.42, 0.39, and 0.20). For nonagricultural firms, we use innovation activity in PINTEC 2017 (0.34) and worker training (0.42), website use (0.56), and fixed asset purchases (0.61) in the World Bank Enterprise Survey Brazil 2009. In Panel B we report 95 percent confidence intervals of our own estimates ($\tau = 4$ for pasture and forest, $\tau = 7$ for *technology goods and services*), demeaned by the national average. In Panel C we report 95 percent confidence intervals of our own estimates ($\tau = 7$). To replicate the empirical estimates for access to subsidized credit, we set $r = 0$ on the modernization loan at fixed $(P_{d,a}, P_{d,n}, Y_a, Y_n, \mathcal{H})$ in partial equilibrium (PE).

does not specialize into one activity. This is a reduced-form modeling approach to capture implicit gains from specialization that the literature documents. In our calibrated model $\varepsilon_h = 3.1$ which implies that the share of vacant land inside the owned plot v/h is roughly 28% if the farmer splits the owned-land equally into crops and pasture. In Appendix Figure F.2 we investigate this implication for more than 6 million properties in the CAR and show that in the median our model prediction about the size of vacant land within properties aligns with the data (a dimension we do not target).[24]

12 The Commodity Boom

We now use the calibrated model to trace how an agricultural commodity boom changes the allocation of land and technology across the farm distribution. Therefore, we double foreign agricultural demand $\mathcal{D}_a^*$ and solve for the new equilibrium, allowing all prices, entry decisions, technology choices, and land allocations to adjust endogenously. While we derive the comparative statics in partial equilibrium in Proposition 2, we now show that the same forces govern the aggregate response in full general equilibrium. In Figure 15 we plot the resulting responses of land use and technology choice conditional on the clearing cost draw c.

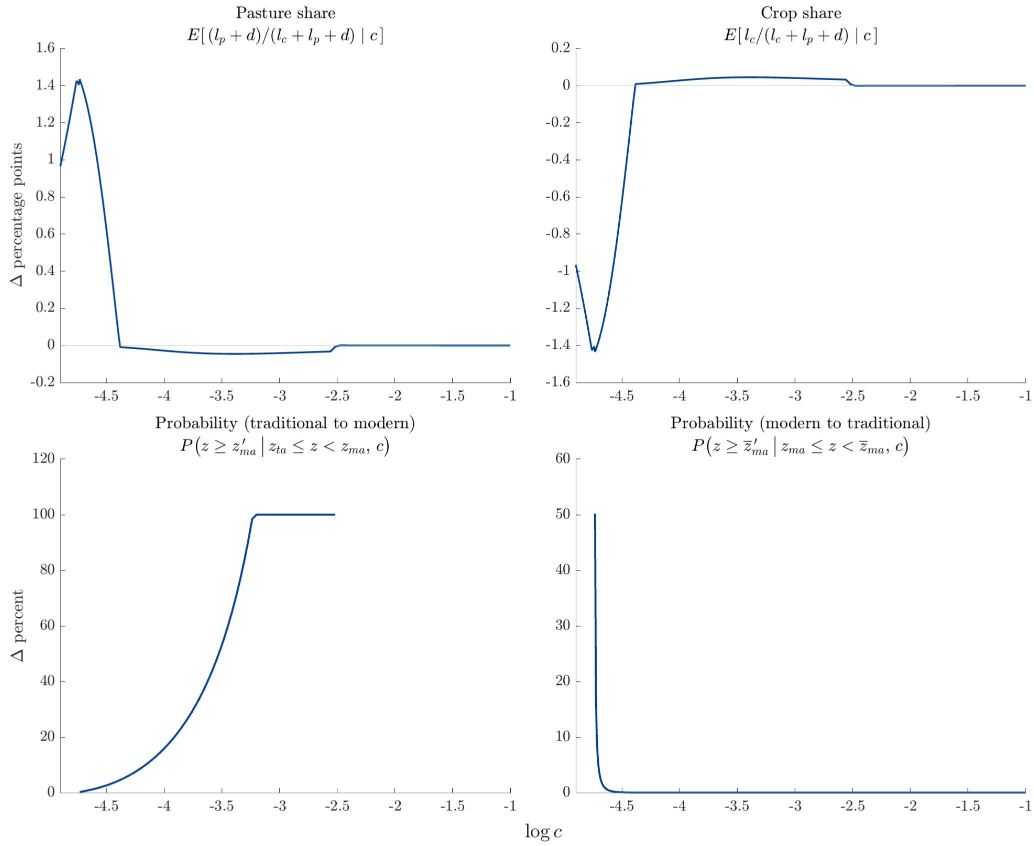
In the upper row of Figure 15 we plot the percentage point change of the average pasture and crop shares of farmed area conditional on the clearing cost, $\mathbb{E}[(l_p+d)/(l_c+l_p+d) \mid c]$ and $\mathbb{E}[l_c/(l_c+l_p+d) \mid c]$, across $\log c$. An increase in foreign demand $\mathcal{D}_a^*$ shifts aggregate demand outward and raises equilibrium prices. For farms with a low clearing cost c, the most profitable path to expansion is to clear new forest, which substitutes imperfectly for established pasture and thereby raises the marginal value of operating owned pasture alongside the newly cleared land. Consequently, farms with low c clear new forest and shift owned plots away from crops and into pasture, raising the average pasture share by up to 1.4 percentage points while the average crop share falls by the same amount. By contrast, for farms with a high clearing cost c, the land allocation is relatively unresponsive, because farms facing expensive clearing convert to forest. On the owned plot, the boom raises the return to crop land and the return to pasture land almost in proportion, as the two output elasticities of land are nearly identical, $\gamma_c = 0.50$ against $\gamma_p = 0.53$ (as per estimates from Pellegrina (2022)). Given a high c, with relative returns essentially unchanged, farms scale up production without reallocating land within the owned plot.

The lower two panels plot the probability of switching technologies for a given clearing cost draw. The left panel shows the probability that a farm that is traditional before the shock, $z_{ta} \leq z < z_{ma}$, transitions to the modern technology after the shock, $P(z \geq z'_{ma} \mid z_{ta} \leq z < z_{ma}, c)$, where z'_{ma} denotes the post-shock adoption threshold. The increase in foreign agricultural demand lowers all three thresholds, $z_{ta}(c)$, $z_{ma}(c)$, and $\overline{z}_{ma}(c)$. Here the probability of farms switching from traditional to modern increases with the clearing cost draw until farmers modernize with a probability of one. For clearing cost draws above around $\log c = -2.5$, the entry and adoption thresholds coincide $z_{ta}(c) = z_{ma}(c)$.

The right lower panel shows the probability that a farm that is modern before the shock, $z_{ma} \leq z < \overline{z}_{ma}$, reverts to the traditional technology, $P(z \geq \overline{z}'_{ma} \mid z_{ma} \leq z < \overline{z}_{ma}, c)$, where $\overline{z}'_{ma}$ denotes the post-shock upper threshold. Farms drawing very low clearing cost (around $\log c = -4.7$) then transition

[24] We take this result with a grain of salt since vacant land may obviously also arise from land characteristics such as unsuitable terrain, degraded land or even legal forest reserve requirements which our model abstracts from. Nevertheless, our model is able to capture the distribution of both farmed and non-farmed areas within properties that we observe in Brazil.

Figure 15: Farm responses to a doubling of foreign agricultural demand



Notes. Each panel plots the response to a doubling of foreign agricultural demand $\mathcal{D}_a^*$, as a function of the log clearing cost. The upper panels show the percentage point change of the average pasture and crop share of farmed area. The lower panels show the percentage changes in transition probabilities: the probability that a farm traditional before the shock transitions to the modern technology, $P(z \geq z'_{ma} \mid z_{ta} \leq z < z_{ma}, c)$, and the probability that a farm modern before the shock reverts to the traditional technology, $P(z \geq \bar{z}'_{ma} \mid z_{ma} \leq z < \bar{z}_{ma}, c)$, where z'_{ma} and $\bar{z}'_{ma}$ denote the post-shock modernization thresholds.

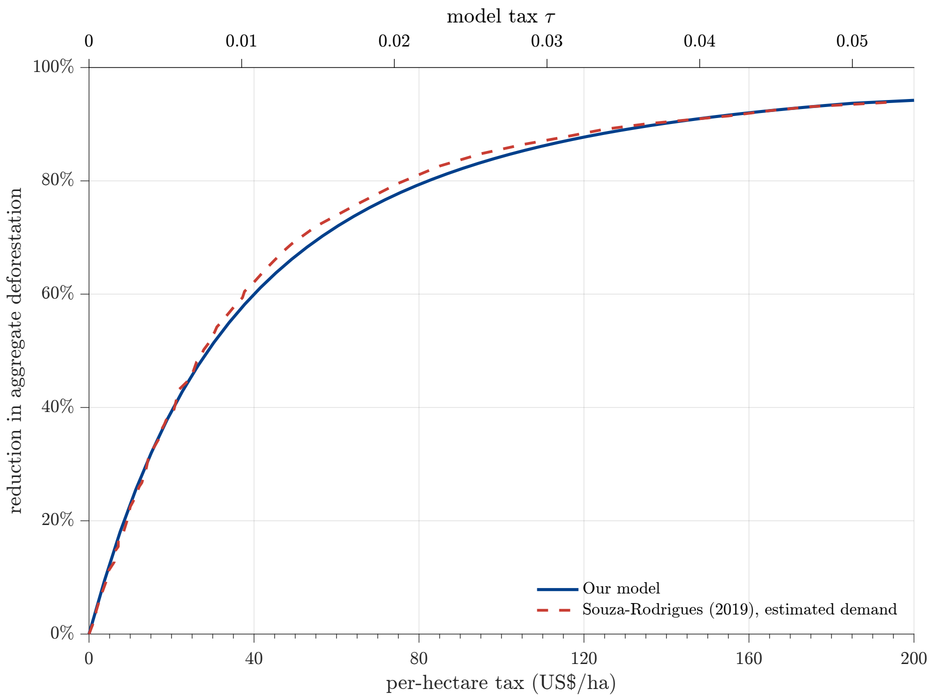
from modern to traditional with a probability of around 50%. Given our probability mapping β, farms in the Amazon reduce their spending into the modern technology as they reallocate land away from the capital-intensive crop activity and into land-intensive pasture.

13 Environmental Policy & Welfare gains from Trade

In this section, we show that an environmental policy can raise welfare by incentivizing farms to adopt more technology through the reallocation into capital-intensive crops and away from land-intensive pasture. Since a farmer must borrow the fixed cost of the technology upgrade and repay it at a credit spread, while the interest payment is a transfer to the household, the private cost of adopting exceeds the social cost and farmers consequently adopt below the efficient level. An environmental policy that raises the cost of clearing land hence restricts deforestation and simultaneously draws high-productivity pasture farms into crop cultivation, thereby raising technology adoption toward the efficient level. In the following counterfactual exercise, we increase the environmental policy τ from zero and solve for a new general equilibrium at each level of the policy. The policy raises the cost of clearing land beyond the market cost $P_n c$, and we assume that all proceeds are rebated lump sum to the household. We then show that correcting the adoption margin increases the welfare gains from trade, such that an increase in foreign demand results in more consumption under less deforestation.

13.1 Effect of the Deforestation Tax

Figure 16: The environmental policy and the estimate of Souza-Rodrigues (2019)



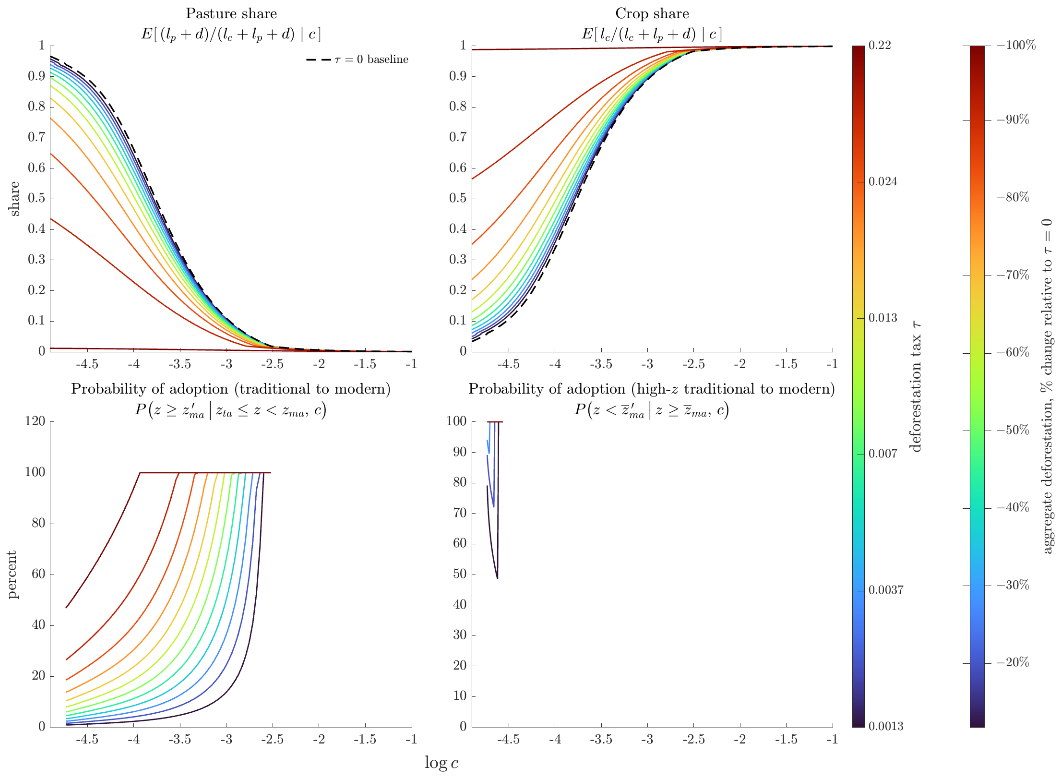
Notes. The solid curve is the reduction in aggregate deforestation in our calibrated model against the deforestation tax, shown in 2006 dollars on the lower axis and in the model's units τ on the upper axis. The dashed curve is the estimated demand for agricultural land from Souza-Rodrigues (2019) (Figure 4) and expressed as a reduction from its untaxed level.

We first evaluate how our deforestation tax τ affects aggregate deforestation. Therefore we use data on agricultural value added per hectare, which implies that one unit of τ is roughly US$3,700 per hectare in 2006 dollars.[25] In Figure 16 we then plot the percentage reduction in aggregate deforestation D resulting from an increase in τ in our calibrated model. Surprisingly our estimate aligns closely

[25]We pin down a value of τ through the ratio of agricultural value added per hectare in the data to agricultural value added per hectare in the model. In the model this is sales net of intermediate inputs over the land in use, $[(p_d^c y^c + p_d^p y^p) - P_n(x_c + x_p)]/(l_c + l_p + d) = 0.0677$, aggregated across active farms. In the data it is Brazilian agricultural value added over agricultural land, about R$464 per hectare in 2003. One unit of τ is therefore worth $464/0.0677 \approx$ R$6,859 per hectare in 2003 reais, which we restate in 2006 reais with the Brazilian consumer price index and convert at the 2006 exchange rate to about US$3,700 per hectare. Data on value added comes from the IBGE Contas Regionais, SIDRA table 5938 and land from MapBiomas Brazil Collection 10.

with Souza-Rodrigues (2019), who finds that a perfectly enforced tax of US$42.5 per hectare raises private forest cover in the Amazon from 44 to 80 percent, a fall of roughly 64 percent in the land under agriculture. At that same tax our model eliminates 62 percent of deforestation. A tax of about US$29 per hectare lowers aggregate deforestation by half, US$138 per hectare by ninety percent, and US$496 per hectare by ninety-nine percent.

Figure 17: Farm responses to the environmental policy



Notes. Each panel plots the level of a conditional moment of the farm distribution as a function of the log clearing cost. Each curve corresponds to one level of the deforestation tax, and the dashed curve in the top row is the untaxed baseline. The left color bar reports the tax and the right color bar the respective reduction in aggregate deforestation. The top row shows the average pasture and crop shares of farm area. The bottom row shows, in percent, two transition probabilities: the probability that a farm traditional at $\tau = 0$ transitions to the modern technology, $P(z \geq z'_{ma} \mid z_{ta} \leq z < z_{ma}, c)$, and the probability that a farm above the modernization band transitions to the modern technology, $P(z < \overline{z}'_{ma} \mid z \geq \overline{z}_{ma}, c)$, where z'_{ma} and $\overline{z}'_{ma}$ denote the values of the two modernization thresholds under the deforestation tax $\tau > 0$.

Next, we evaluate how the deforestation tax changes land use and the technology choice across the distribution of clearing costs in Figure 17, where we plot the average pasture and crop shares, $\mathbb{E}[(l_p + d)/(l_c + l_p + d) \mid c]$ and $\mathbb{E}[l_c/(l_c + l_p + d) \mid c]$, the probability that a traditional farm in the baseline economy with $\tau = 0$ transitions to the modern technology, $P(z \geq z'_{ma} \mid z_{ta} \leq z < z_{ma}, c)$ after being taxed $\tau > 0$, and equivalently, the probability that a farm above the modernization band at baseline, $z \geq \overline{z}_{ma}$, transitions to the modern technology, $P(z < \overline{z}'_{ma} \mid z \geq \overline{z}_{ma}, c)$ after the tax, where z'_{ma} and $\overline{z}'_{ma}$ denote the modernization thresholds under the deforestation tax. Each curve is drawn at a given tax τ and the dashed curve in the upper panels is the untaxed baseline.

The tax raises the cost of clearing forest and reshapes land use and technology choices for farms with a low clearing cost. The deforestation tax effectively taxes pasture expansion such that owned land reallocates into crop cultivation. Under a perfectly enforced no-deforestation policy, the pasture share falls to nearly zero at every clearing cost. A farm with a high clearing cost already allocates almost all owned land to crop cultivation and thus does not respond to the tax. Since the returns to modernizing accrue to crops, the tax then pushes traditional farms across the adoption threshold such that the probability of switching from the traditional to the modern technology (i.e., $P(z \geq z'_{ma} \mid z_{ta} \leq z < z_{ma}, c)$) rises. Equivalently, the deforestation tax causes a rise in $\overline{z}'_{ma}(c)$, such that even the highly productive (high z) pasture farms that draw the lowest clearing costs optimally start to modernize.

Note that the responses are not proportional. A tax large enough to remove 90 percent of aggregate deforestation still leaves a farm with the lowest clearing cost draw allocating almost half of farm area in pasture, with a probability of switching from traditional to modern of only one third. The largest changes in land use and technology arrive with the final tenth of forest saved, which pushes the pasture share on the cheapest land from almost one half to essentially zero and raises the probability of adoption on the cheapest land to one half.

13.2 The Gains from Trade

We first define the welfare gains from agricultural trade, $\mathcal{G}(\tau)$, as the response of aggregate consumption to foreign demand, evaluated at tax τ:

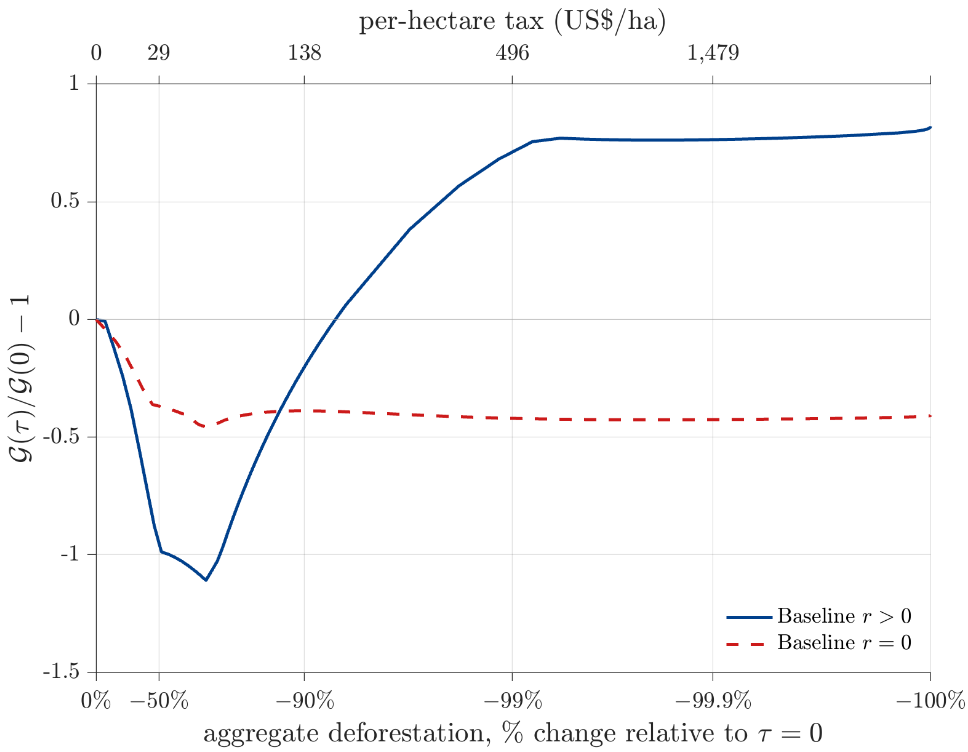
$$\mathcal{G}(\tau) = \left.\frac{\Delta \ln C}{\Delta \ln \mathcal{D}_a^*}\right|_{\tau} \tag{41}$$

In other words, we compute $\mathcal{G}(\tau)$ by doubling $\mathcal{D}_a^*$ at each level of the tax and solving for the new equilibrium. $\mathcal{G}(\tau)$ then captures the welfare improvement when foreign demand for agricultural goods rises, accounting for adjustments of prices, entry, technology, and land use. In Figure 18 we plot the change relative to the untaxed baseline economy where $\tau = 0$, given by $\left[\mathcal{G}(\tau)/\mathcal{G}(0) - 1\right]$, against the reduction in aggregate deforestation D generated by τ.

In Figure 18 we show that a perfectly enforced no-deforestation tax of around US$700 per hectare increases the welfare gains from trade by more than 75 percent. Importantly, welfare gains from trade only materialize for a sufficiently strong tax above US$170 per hectare which reduces deforestation by more than 90%.

A moderate tax below US$170 per hectare instead lowers the gains from trade, because the policy distorts the land allocation of farms before inducing a switch to the modern technology towards their efficient level. Transitioning to the modern technology requires a fixed upfront investment, so the expected profit gain from crop cultivation must exceed the financed investment to make modernization viable. A moderate tax thus reduces land expansion but fails to raise the relative return of crops enough to incentivize modernization, and the welfare gains from trade fall by more than one hundred percent relative to the untaxed level. Only a sufficiently strong tax forces farms to reallocate enough owned land away from pasture and into capital-intensive crops to justify the fixed cost of adopting the modern technology. At those tax levels, farms previously meeting foreign demand by clearing new land instead meet the same demand by adopting the modern technology and reallocating land from land-intensive pasture to capital-intensive crop cultivation.

Figure 18: Environmental policy and the gains from trade



Notes. The vertical axis shows the change, relative to the $\tau = 0$ equilibrium, of the gains from trade $\mathcal{G}(\tau)$ defined in equation (41), computed by doubling $\mathcal{D}_a^*$ at every level of τ. The lower horizontal axis converts each tax into the change in aggregate deforestation on a logarithmic scale, and the upper horizontal axis reports the corresponding per-hectare tax in 2006 US dollars. The tax under the calibrated baseline economy is in blue and under no financial frictions $r = 0$ in red.

In our model, a farmer must borrow the fixed cost of the technology upgrade and repay it at a credit spread, while the interest payment is a transfer to the household and farmers consequently adopt below the social optimum. This means, once the deforestation tax is sufficiently strong to incentivize adoption, the welfare gains from trade exceed the untaxed level which materializes around a tax of US$170 per hectare (equivalent to a 93 percent reduction in deforestation). To see this, we plot in Figure 18 a counterfactual economy where the credit spread is zero $r = 0$, such that all farmers adopt at their efficient level at $\tau = 0$. A deforestation tax then merely distorts land allocation such that the welfare gains from trade strictly fall across all tax levels.

14 Conclusion

In this paper, we ask how agricultural trade shapes farmers' decisions over technology adoption and land use. We combine confidential records on the near universe of cross-bank electronic transfers with geo-referenced farm boundaries and satellite imagery, which allows us to observe both land use changes and input purchases at the farm level during the commodity boom in Brazil. Using a shift-share shock based on destination-country import demand, we find that farmers in the Amazon meet higher foreign demand by reallocating land from capital-intensive soybean toward land-intensive pasture. The reallocation shrinks the area under modern cultivation, such that spending on machinery and technology falls in the years that follow, and forest is cleared to make room for the additional pasture. Outside the Legal Amazon, we find that exporters pass through significantly less foreign demand to farmers, and we consequently find limited changes in land allocation or technology spending following a foreign demand shock. In a complementary difference-in-differences design, we then show that access to rural credit pushes farmers in the opposite direction, raising spending on modern inputs and shifting land out of pasture and into soybean, encouraging modernization.

We show that our empirical observations emerge naturally from the asymmetric use of land and technology across agricultural activities. Drawing on natural sciences, we assume that crops and pasture differ in their underlying use of technology and land such that the returns to adopting modern technology accrue solely to crop cultivation, while only pasture can replace forest, as freshly deforested land is largely unsuitable for immediate crop cultivation. Land is also inherently heterogeneous in our framework, as a farmer can neither substitute freely between pasture and crops nor between the established owned plot and newly deforested pasture. Cleared forest substitutes only in part for owned pasture, such that an additional deforested hectare raises the value of the owned pasture operated alongside it. This owned pasture then competes with the land a farmer would otherwise dedicate to crops. A farmer who deforests consequently specializes into pasture, and for a sufficiently low clearing cost, the crop operation remains too small to justify the fixed adoption cost, regardless of the farm's productivity draw. We embed this mechanism into a general equilibrium model in which every farmer draws a productivity and a cost of converting forest into pasture. The clearing cost draw captures variation in land scarcity and property rights, and allows us to abstract from explicit spatial units. In equilibrium, even highly productive farmers can find it optimal to stay traditional and expand pasture, in line with what we observe in the data. We then calibrate the model to Brazilian data from the beginning of the commodity boom in 2003, drawing on granular micro-data, aggregate statistics, and our own empirical estimates. The calibrated model rationalizes our empirical findings on land allocation and technology spending, and it reproduces untargeted differences in land use, farm scale, and modernization between the Legal Amazon and the rest of Brazil.

In a counterfactual exercise, we show that a sufficiently strong deforestation tax can increase the welfare gains from agricultural trade without assuming carbon externalities, biodiversity losses, or climatic changes that affect firm productivity or household utility. In our model, a farmer finances the fixed cost of the upgrade with a loan and repays it at a credit spread. The interest on this loan is a transfer to households, such that the private cost of adopting exceeds its social cost and farmers adopt below their efficient level. A deforestation tax adds to the cost of pasture expansion and thereby pushes owned land back into crop cultivation, which raises the return on technology investments. Highly productive pasture farms inside the Amazon are consequently drawn into modern agriculture, and technology adoption moves toward the efficient level. In our calibration, the welfare gains from trade rise by more than 75 percent under a perfectly enforced no-deforestation tax of around US$700 per hectare. Importantly, we show that welfare gains from trade only materialize after a reduction in aggregate deforestation of more than 90% (at a tax above US$170 per hectare). A moderate tax merely distorts the land allocation, as crop returns do not rise enough to cover the fixed modernization cost. Only a sufficiently strong policy moves enough land into crop cultivation to fully justify the investment.

A Appendix: Shift-Share Diagnostics

This appendix reports diagnostics for the shift-share instrument constructed in Section 5, following the framework of Borusyak et al. (2025).

Table A.1 summarizes the properties of the demand shock for the combined instrument as well as for the separate beef and soy instruments. The combined instrument draws on 143 destination countries. The effective number of independent demand shocks, computed as the inverse Herfindahl index of the importance weights $(\sum_c S_c^2)^{-1}$, is 8.79, indicating that the identifying variation is not driven by a single destination. The largest importance weight falls on China at 29.1%, reflecting its dominant role in global demand for both beef and soybeans. The next four destinations by importance weight are Russia (8.5%), Hong Kong (7.7%), Iran (7.0%), and the Netherlands (5.7%). For the separate instruments, the effective number of shocks is 8.66 for beef and 4.37 for soy. The soy instrument is more concentrated, with China alone accounting for 45.8% of the importance weight, reflecting the outsized role of Chinese soybean imports during the sample period.

The concentration of importance weights on China raises the question of whether our results are driven by a single demand shifter. We address this concern directly through the leave-one-destination-out exercise reported in Appendix E, which shows that dropping China (or any other top-5 destination) and reweighting the remaining shares does not materially change our estimates. The identifying variation in Chinese demand for beef and soybeans is plausibly driven by structural income growth and dietary transition rather than by conditions in Brazilian agriculture, supporting the exogeneity of this dominant shift.

Table A.1: Shift-Share Diagnostics

	Combined	**Beef**	**Soy**
Number of destination countries (C)	143	129	76
Effective number of shifts, $(\sum S_c^2)^{-1}$	8.79	8.66	4.37
Largest importance weight (S_c)	0.291	0.204	0.458
Δz_{it} Mean	−0.016	0.004	−0.005
Δz_{it} Std. Dev.	0.422	0.253	0.337
$\Delta \ln g_{ct}^k$ Mean (weighted by S_c)	0.102	−0.006	0.065
$\Delta \ln g_{ct}^k$ Std. Dev. (weighted by S_c)	0.483	0.861	0.503
Observations	283,356	123,867	170,328
Unique farmers	108,942	44,660	64,051
Top 5 Destinations (by S_c)			
1	China (29.1%)	Hong Kong (20.4%)	China (45.8%)
2	Russia (8.5%)	Russia (19.6%)	Netherlands (8.1%)
3	Hong Kong (7.7%)	Iran (13.7%)	France (5.1%)
4	Iran (7.0%)	Egypt (10.0%)	Thailand (5.0%)
5	Netherlands (5.7%)	Venezuela (3.3%)	Spain (4.1%)

Notes: The table reports shift-share diagnostics following Borusyak et al. (2025). Importance weights S_c measure the contribution of each destination to the overall variance of the instrument, computed as the weighted average of farmer-level exposure to each destination. The effective number of shifts is the inverse Herfindahl index of these weights. The combined instrument pools beef and soy as described in equation (9). The separate beef and soy instruments use product-specific shares s_{ic}^k. Sample restricted to the regression sample (non-missing leads, lags, and controls).

Table A.2 reports the distribution of initial years t_0 at which farmer-exporter weights w_{ix} are fixed. About a quarter of farmers enter the sample in 2010, with the remainder distributed across subsequent years. Table A.3 reports the corresponding distribution of t_0' for exporter-destination shares s_{xc}^k. For beef, 97 out of 418 exporter-destination pairs (23%) have their shares fixed in 2010, with the rest spread across later years. For soy, the distribution is more uniform, with 261 out of 1,153 pairs (23%) fixed in 2010. We verify in Appendix E that restricting the sample to farmers entering in 2010 does not change our results.

Table A.2: Distribution of Initial Years: Farmer-Exporter Weights (w_{ix})

Year	Farmers	Percent	Cumulative
2010	33,110	25.8	25.8
2011	11,805	9.2	35.0
2012	10,372	8.1	43.1
2013	9,528	7.4	50.5
2014	9,122	7.1	57.6
2015	8,318	6.5	64.1
2016	8,666	6.8	70.9
2017	9,792	7.6	78.5
2018	6,076	4.7	83.3
2019	12,313	9.6	92.9
2020	9,170	7.2	100.0
Total	128,272	100.0	

Notes: The table reports the distribution of the initial year t_0 at which each farmer's exporter weights w_{ix} are fixed. t_0 is defined as the first year farmer i appears in the payment data.

Table A.3: Distribution of Initial Years: Exporter-Destination Shares (s^k_{xc})

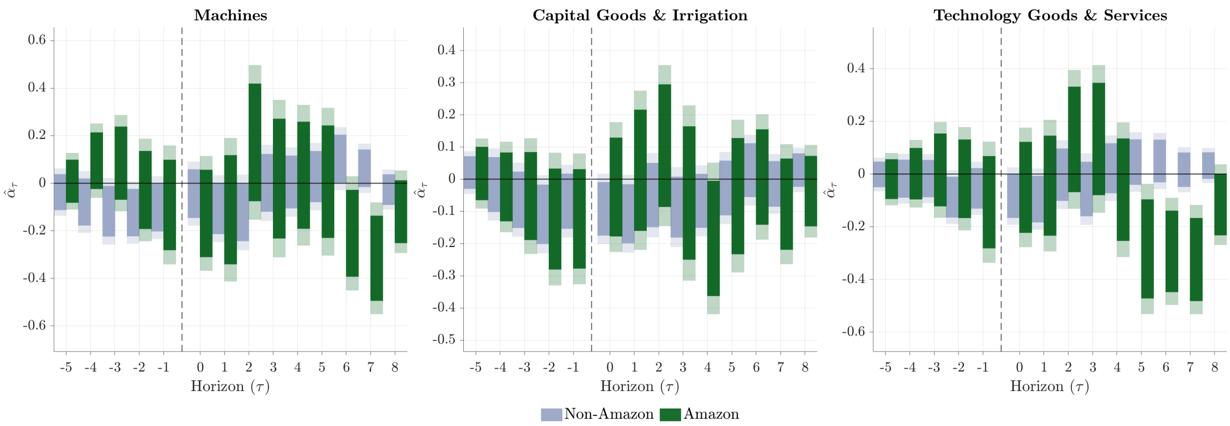
	Beef		**Soy**	
Year	Exporters	Percent	Exporters	Percent
2010	97	23.2	261	22.6
2011	38	9.1	127	11.0
2012	29	6.9	93	8.1
2013	24	5.7	115	10.0
2014	18	4.3	92	8.0
2015	31	7.4	98	8.5
2016	23	5.5	75	6.5
2017	40	9.6	94	8.2
2018			54	4.7
2019	74	17.7	77	6.7
2020	44	10.5	67	5.8
Total	418	100.0	1,153	100.0

Notes: The table reports the distribution of the initial year t'_0 at which each exporter's destination shares s^k_{xc} are fixed, separately for beef and soy. t'_0 is defined as the first year the exporter appears in the TRASE data. Beef data excludes 2018 (not available in TRASE).

B Appendix: Alternative Outcome Transformations

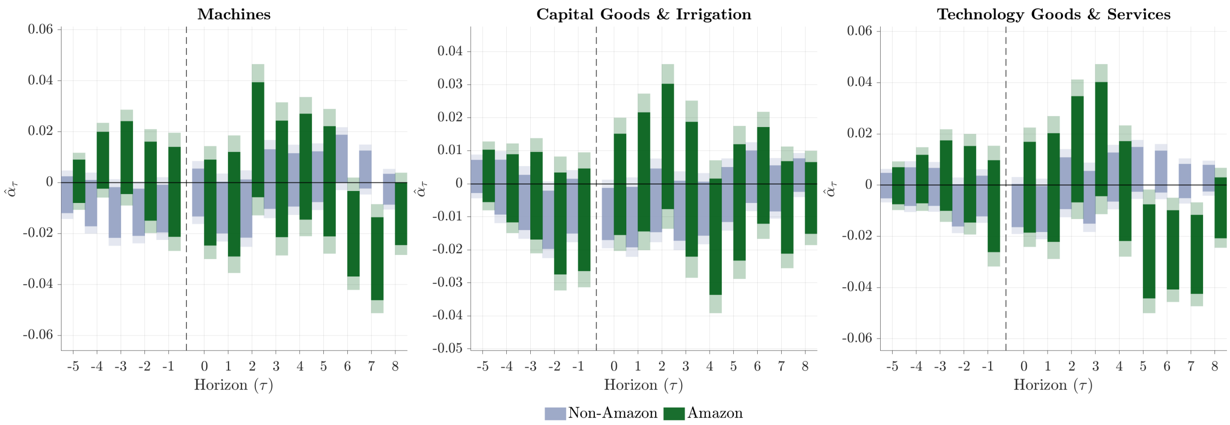
The main text reports spending outcomes using $\log(1 + Y_{it})$ as the baseline transformation. This appendix verifies that our spending results are robust to alternative approaches to handling the mass of zeros in farm-level investment. Figure B.1 reports event study estimates using the inverse hyperbolic sine transformation $\ln(Y_{it} + \sqrt{Y_{it}^2 + 1})$, which has similar properties to $\log(1 + Y_{it})$ but is exactly invariant to the units of measurement (Burbidge et al. 1988). Figure B.2 uses an extensive margin dummy $\mathbb{1}[Y_{it} > 0]$, which isolates the effect of the demand shock on the probability that a farmer makes any purchase in a given category, abstracting from the intensive margin. Figure B.3 uses $\log(1 + N_{it})$ where N_{it} is the number of distinct suppliers in each category, measuring changes in the breadth of the sourcing base rather than the value of payments. In all cases, we estimate equation (10) with the same controls and sample splits as in the baseline.

Figure B.1: Inverse Hyperbolic Sine Transformation



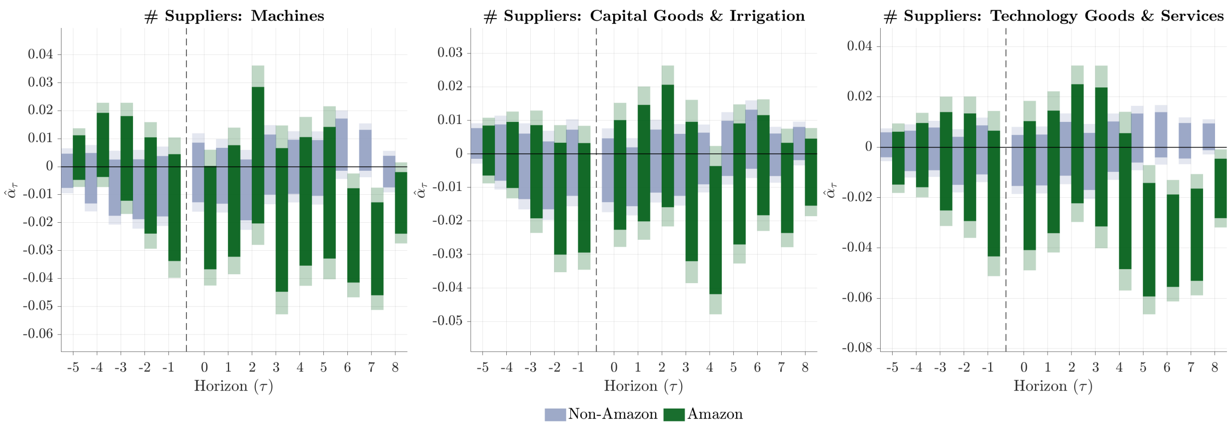
Notes: Each panel plots estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$, separately for farmers in the Legal Amazon (green) and outside it (blue). The outcome is the inverse hyperbolic sine $\ln(Y_{it} + \sqrt{Y_{it}^2 + 1})$ of total annual spending in each category. Solid bars denote 95% confidence intervals and shaded areas denote 99% confidence intervals, based on standard errors clustered at the farmer level.

Figure B.2: Extensive Margin Dummies



Notes: Each panel plots estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$, separately for farmers in the Legal Amazon (green) and outside it (blue). The outcome is an extensive margin dummy $\mathbb{1}[Y_{it} > 0]$, isolating the effect on the probability that a farmer makes any purchase in a given category. Solid bars denote 95% confidence intervals and shaded areas denote 99% confidence intervals, based on standard errors clustered at the farmer level.

Figure B.3: Supplier Counts

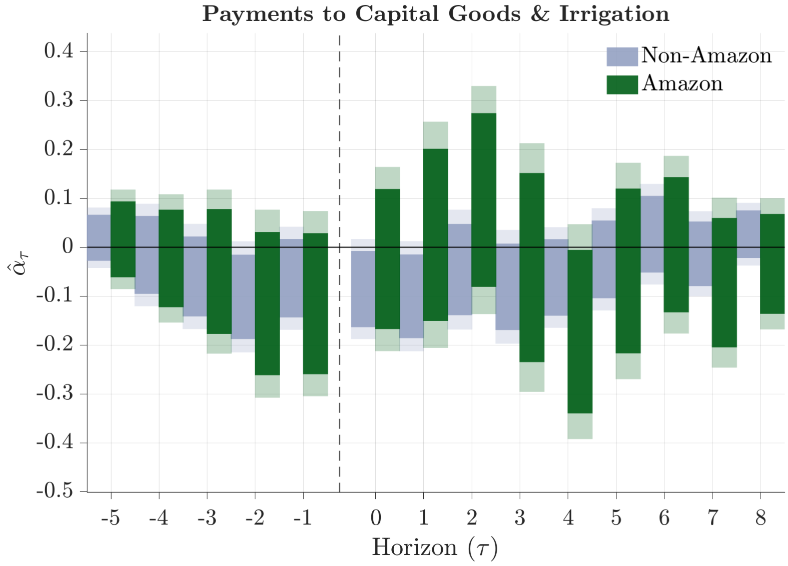


Notes: Each panel plots estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$, separately for farmers in the Legal Amazon (green) and outside it (blue). The outcome is $\log(1 + N_{it})$ where N_{it} is the number of suppliers in each category, measuring changes in the breadth of the sourcing base. Solid bars denote 95% confidence intervals and shaded areas denote 99% confidence intervals, based on standard errors clustered at the farmer level.

C Appendix: Capital Goods & Irrigation

The main text reports baseline spending results for Machines and Technology Goods & Services. Figure C.1 reports the corresponding event study for Capital Goods & Irrigation, the third spending category described in Section 5. The specification is identical to that in Figure 6.

Figure C.1: Agricultural Export Demand and Capital Goods & Irrigation Spending



Notes: The figure plots the estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$, separately for farmers in the Legal Amazon (green) and outside it (blue). The outcome is $\log(1 + Y_{it})$ where Y_{it} is total annual spending on suppliers classified under Capital Goods & Irrigation (see Appendix Table E.1). Solid bars denote 95% confidence intervals and shaded areas denote 99% confidence intervals, based on standard errors clustered at the farmer level.

D Appendix: Separate Beef and Soy Shocks

The baseline specification in Section 5 pools beef and soy into a single demand shock Δz_{it}. This appendix decomposes the instrument into separate product-specific shocks to verify that both commodities contribute to the patterns documented in the main text. For each product $k \in \{\text{beef}, \text{soy}\}$, we construct the product-specific shock

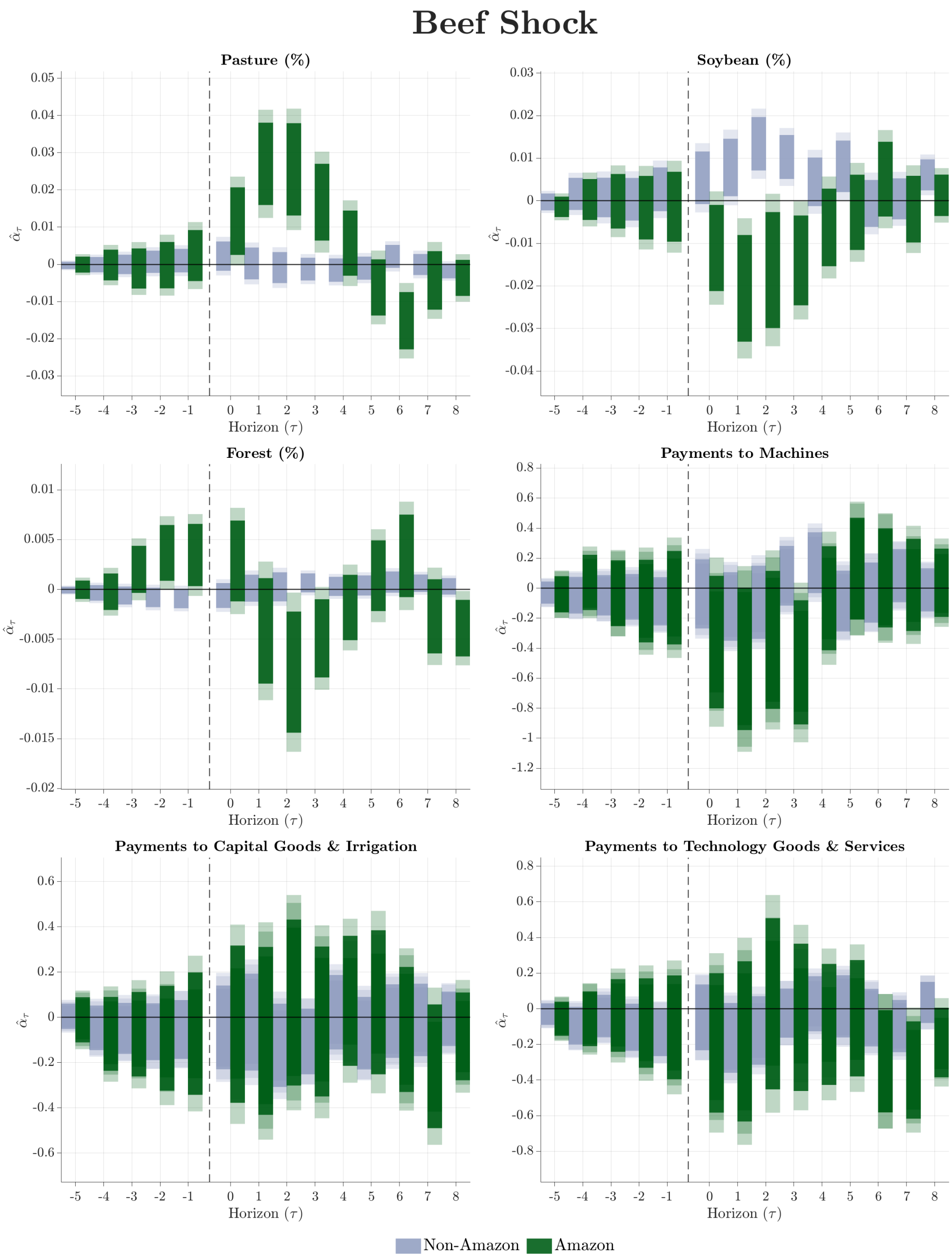
$$\Delta z_{it}^k = \sum_c s_{ic}^k \cdot \Delta \ln g_{ct}^k \tag{42}$$

where the farmer-destination shares s_{ic}^k are the same as those used in the baseline (equation 8), but now each product enters as a separate regressor rather than being summed. We estimate equation (10) replacing Δz_{it} with Δz_{it}^k and restricting the sample to farmers linked to at least one exporter of product k. The absorbed controls include farmer and year fixed effects, the sum of product-specific shares $\bar{s}_i^k = \sum_c s_{ic}^k$ interacted with year fixed effects (to control for incomplete shares), and municipality-level log land-use controls for land-use outcomes.[26]

Figures D.1 and D.2 report the results. The beef shock generates a pattern consistent with the baseline, with pasture expansion, soybean contraction, and spending declines concentrated in the Amazon. The soy shock generates qualitatively similar effects, confirming that our baseline findings are not driven by the composition of the pooled instrument.

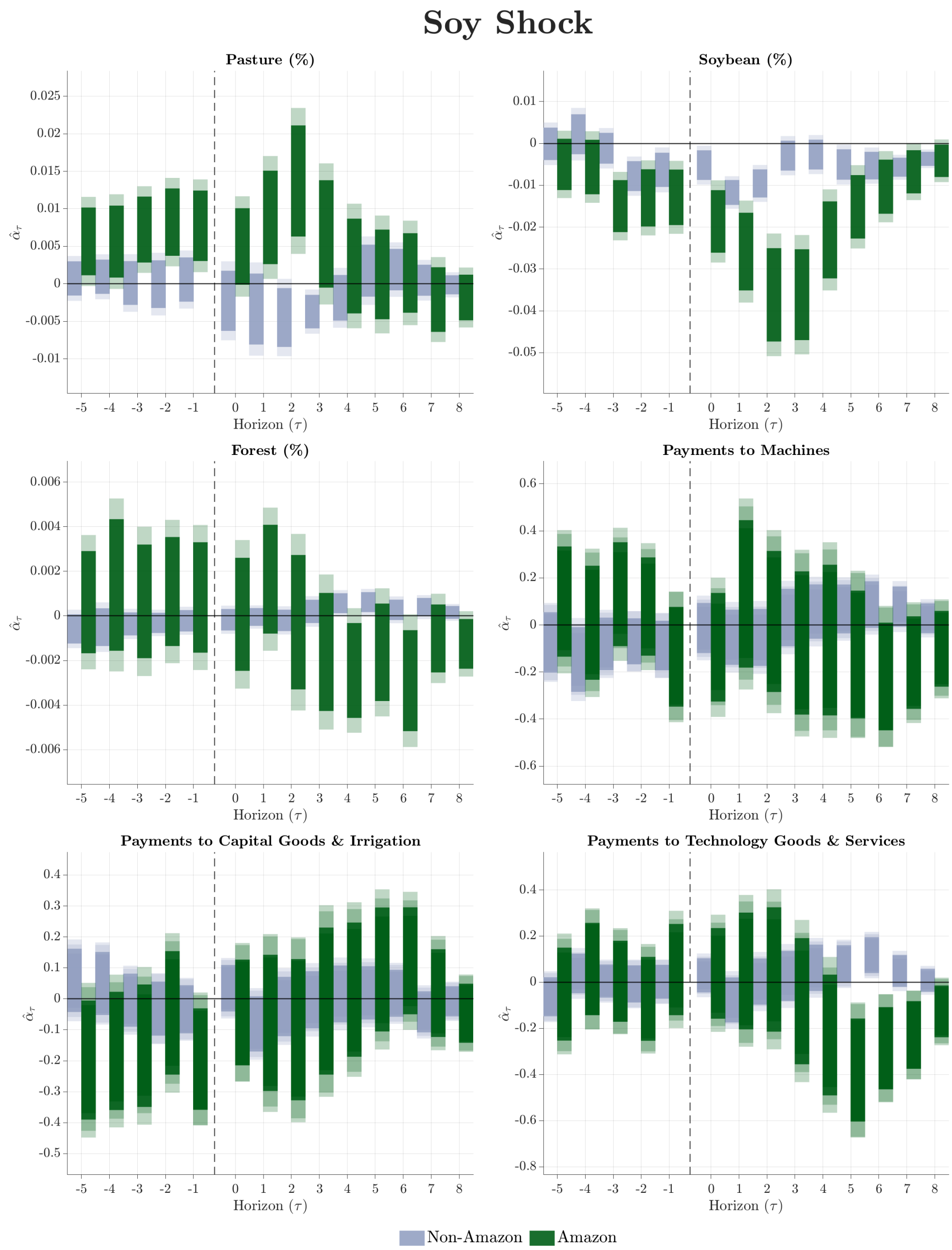
[26] Municipality-level controls are the log of total municipal area under forest, pasture, and soybean. These absorb local trends in land use that may be correlated with the product-specific shock. We also estimate variants with and without initial product-specific land shares interacted with year fixed effects (Pasture share at t_0 for beef, Soybean share at t_0 for soy).

Figure D.1: Separate Beef Shock



Notes: The figure plots the estimated coefficients from the separate beef shock specification for $\tau = -5, \ldots, +8$, separately for farmers in the Legal Amazon (green) and outside it (blue). The sample is restricted to farmers linked to at least one beef exporter. Panels report effects on land-use shares (pasture, soybean, forest) and $\log(1 + Y_{it})$ spending outcomes (machines, capital goods, technology). Solid bars denote 95% confidence intervals and shaded areas denote 99% confidence intervals, based on standard errors clustered at the farmer level. Controls include municipality-level log land use for land-use outcomes.

Figure D.2: Separate Soy Shock



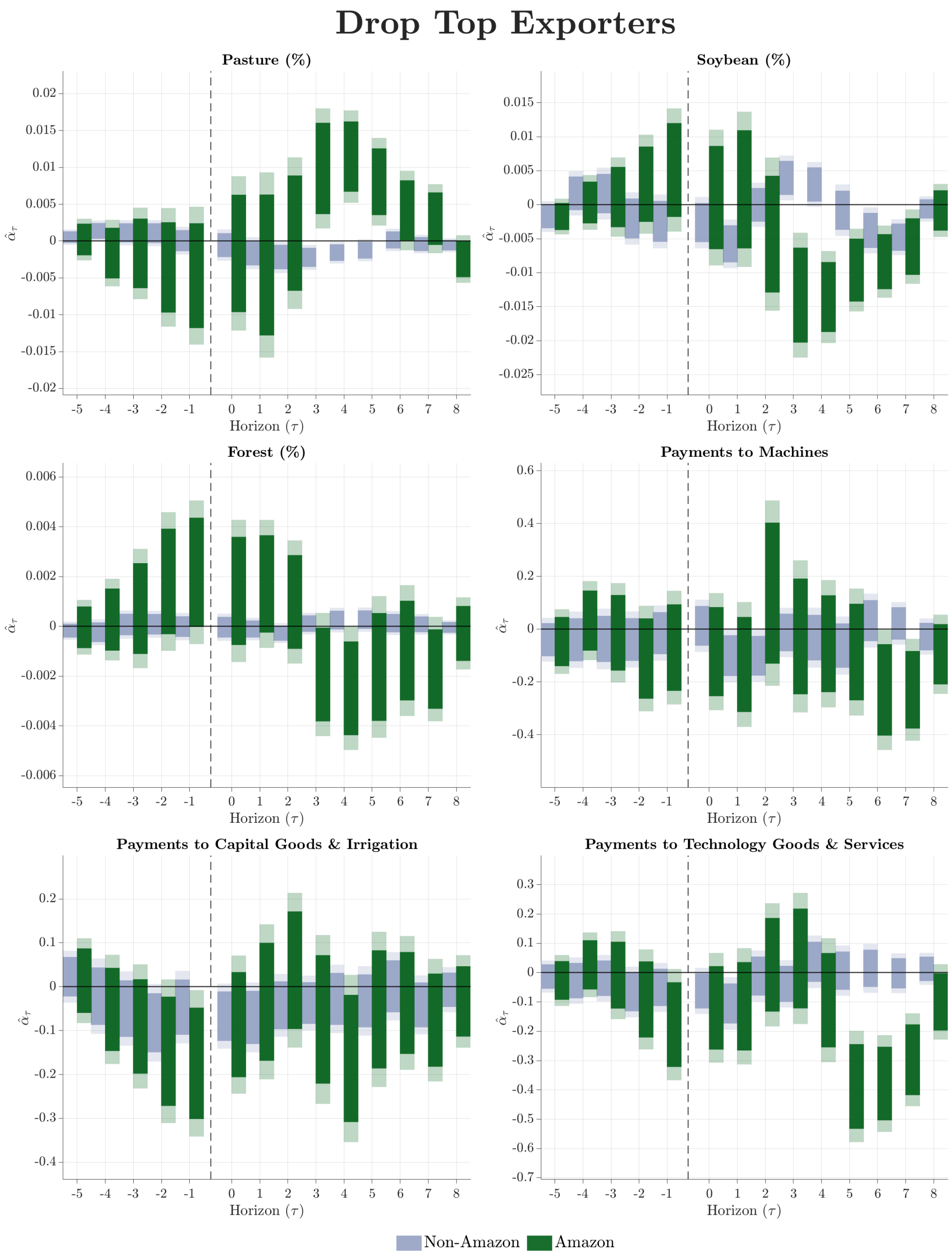
Notes: The figure plots estimated coefficients from the separate soy shock specification for $\tau = -5, \ldots, +8$, separately for farmers in the Legal Amazon (green) and outside it (blue). The sample is restricted to farmers linked to at least one soy exporter. Panels report effects on land-use shares (pasture, soybean, forest) and $\log(1 + Y_{it})$ spending outcomes (machines, capital goods, technology). Solid bars denote 95% confidence intervals and shaded areas denote 99% confidence intervals, based on standard errors clustered at the farmer level. Controls include municipality-level log land use for land-use outcomes.

E Appendix: Sample and Specification Robustness

This appendix examines the robustness of our baseline results to a range of sample restrictions, alternative specifications, and instrument modifications. In each exercise, we re-estimate equation (10) and report event study coefficients for the six main outcomes (pasture share, soybean share, forest share, machines, capital goods, and technology spending), separately for the Amazon and non-Amazon subsamples. Each figure replicates Figure 6 under the modified specification.

Dropping Top Exporters. We drop the three largest exporters by average yearly FOB value and recompute farmer-exporter weights w_{ix} and the demand shock Δz_{it} from scratch using the remaining exporters. This exercise tests whether our results are sensitive to the concentration of the export sector in a small number of firms such as JBS, Cargill, and Bunge. Figure E.1 reports the results.

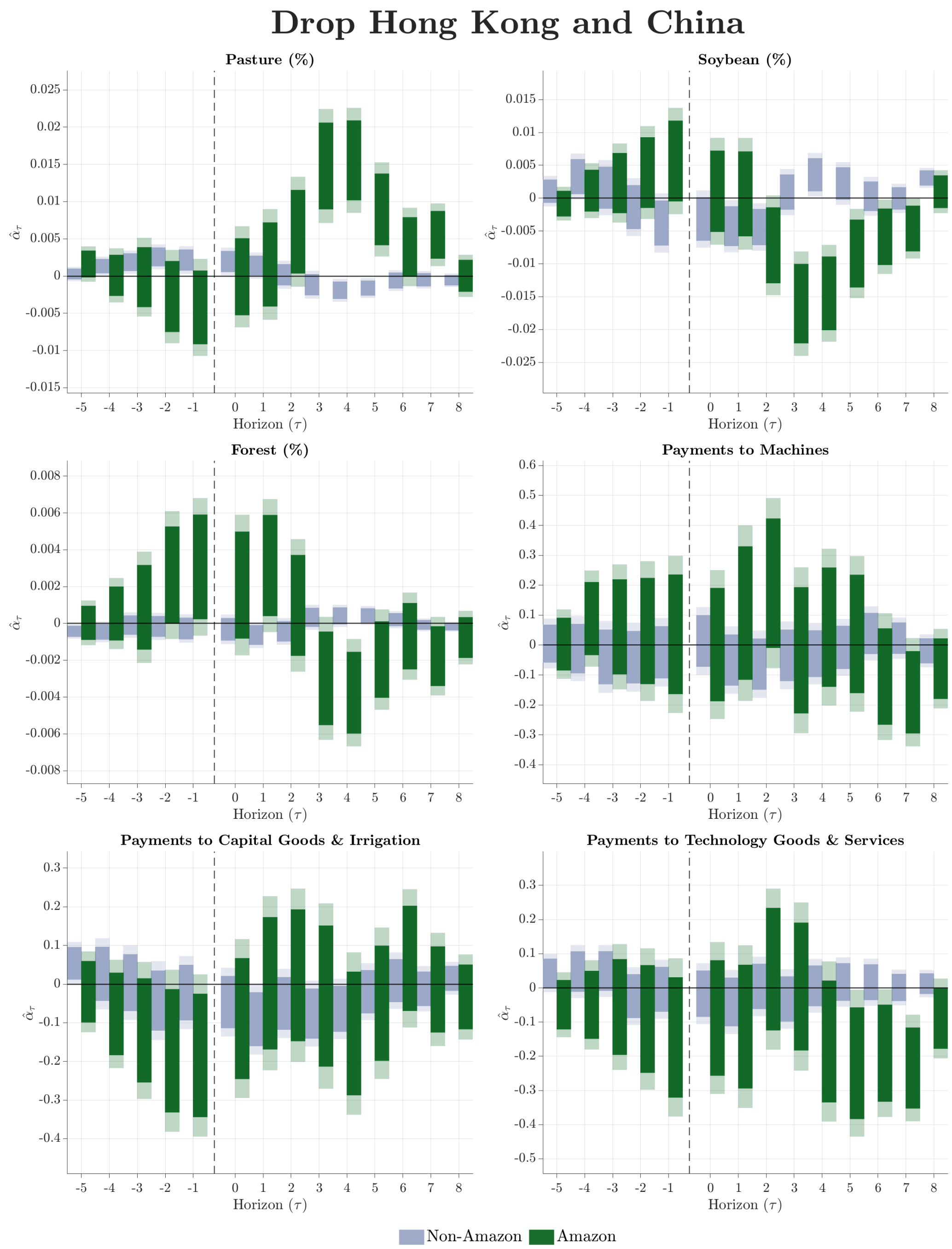
Figure E.1: Robustness: Dropping Top 3 Exporters



Notes: The figure plots estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$ after dropping the three largest exporters by average yearly FOB and recomputing farmer-exporter weights and demand shocks from scratch. Green bars denote Amazon, blue bars denote non-Amazon. Panels report effects on land-use shares and $\log(1 + Y_{it})$ spending. Solid bars denote 95% confidence intervals, shaded areas 99%, clustered at the farmer level.

Dropping Key Destinations. We drop Hong Kong from beef destinations and China from soy destinations, then jointly reweight the remaining shares s^k_{ic} across both products so that $\sum_{k,c} s^k_{ic} = 1$ for each farmer. Because shares sum to one after reweighting, the incomplete share control $\bar{s}_i \times \delta_t$ is absorbed by the year fixed effects and is dropped from the specification. Figure E.2 reports the results.

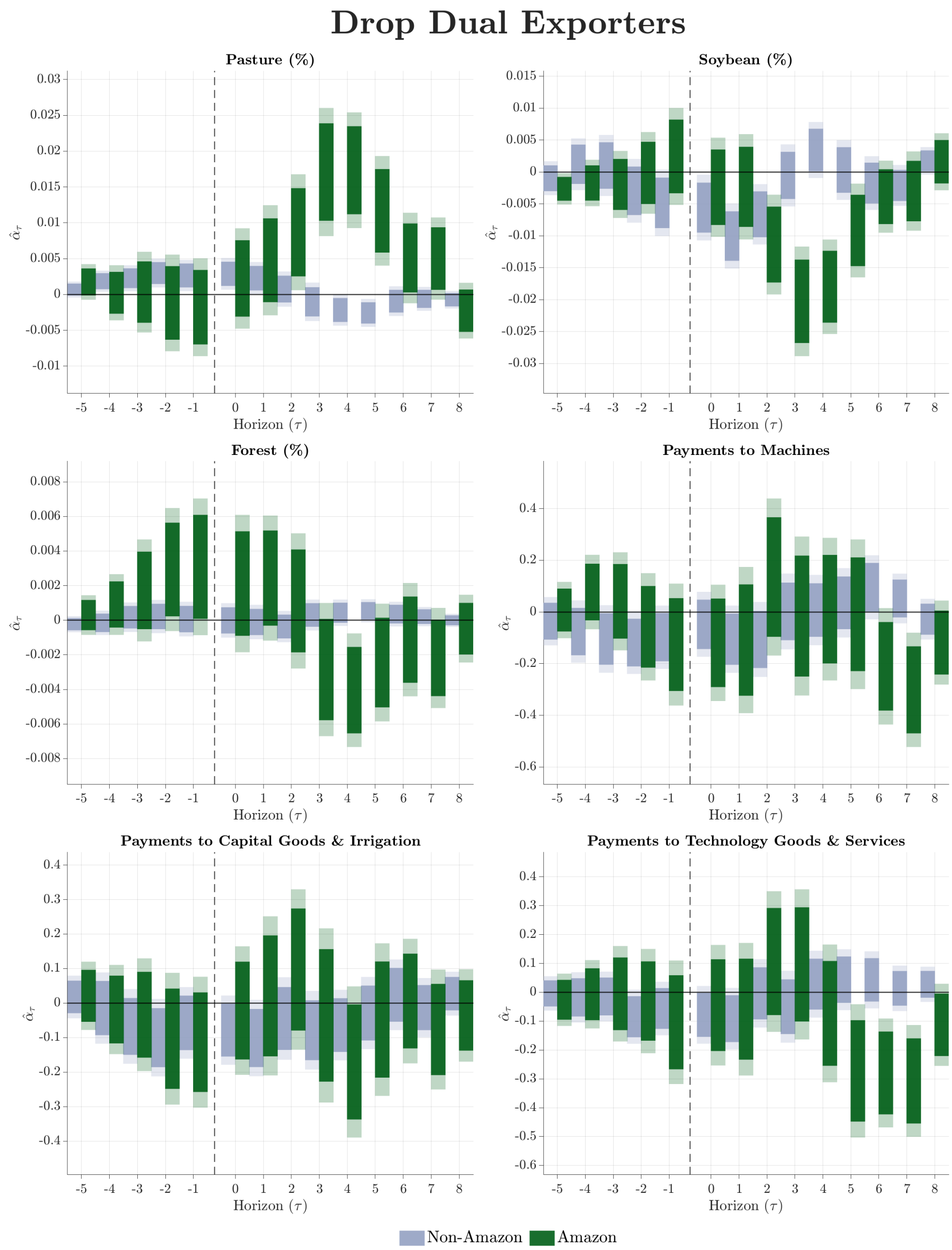
Figure E.2: Robustness: Dropping Key Destinations



Notes: The figure plots estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$ after dropping Hong Kong from beef and China from soy destination shares and reweighting remaining shares to sum to one. Green bars denote Amazon, blue bars denote non-Amazon. Panels report effects on land-use shares and $\log(1 + Y_{it})$ spending. Solid bars denote 95% confidence intervals, shaded areas 99%, clustered at the farmer level.

Dropping Dual Exporters. We remove all farmers connected to exporters that handle both beef and soy in any given year, recompute farmer-exporter weights w_{ix} from scratch using the remaining exporters, and jointly reweight destination shares so that $\sum_{k,c} s_{ic}^k = 1$. The incomplete share control is dropped after reweighting. In our sample, only 35 out of 4,066 exporter-year observations involve dual-commodity exporters (23 unique firms out of 1,325), so this restriction results in a modest sample reduction of 3.9 percent on the beef side and 1.9 percent on the soy side. Figure E.3 reports the results.

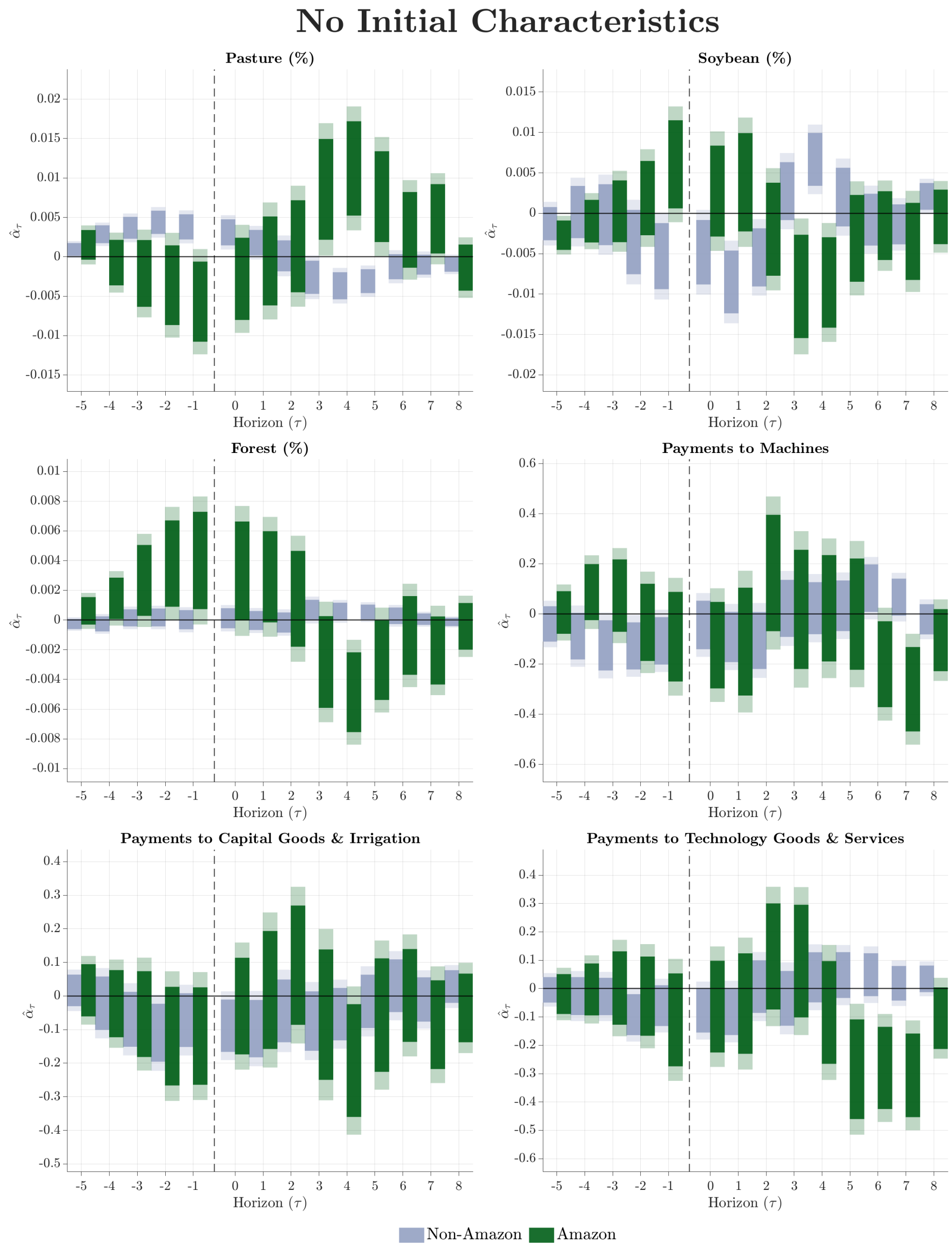
Figure E.3: Robustness: Dropping Dual Exporters



Notes: The figure plots estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$ after removing farmers connected to exporters that handle both beef and soy in any given year. Green bars denote Amazon, blue bars denote non-Amazon. Panels report effects on land-use shares and $\log(1 + Y_{it})$ spending. Solid bars denote 95% confidence intervals, shaded areas 99%, clustered at the farmer level.

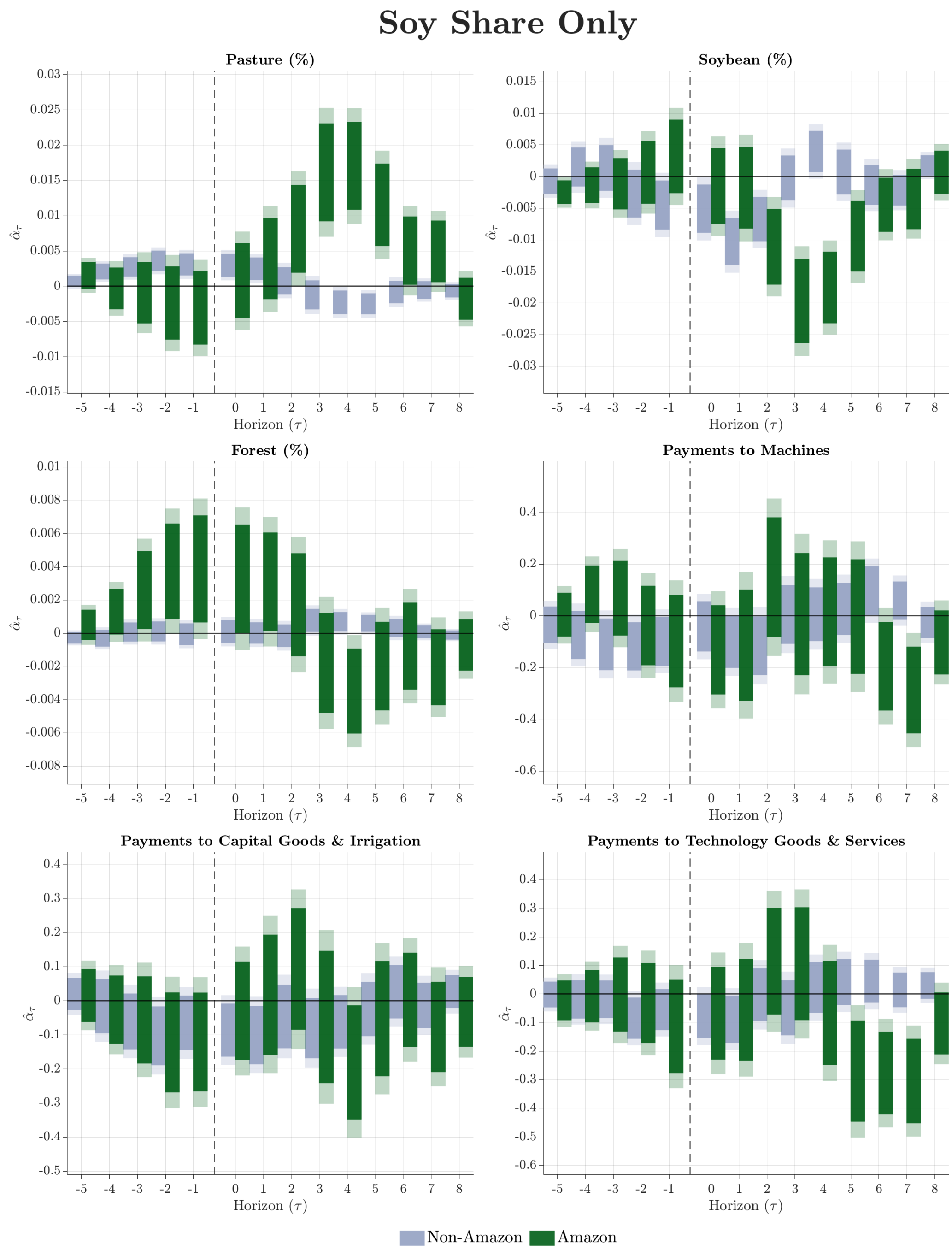
Alternative Initial Characteristics. Our baseline absorbs both the initial forest share and the initial soybean share of farmer i's property area, each separately interacted with year fixed effects, to control for differential trends correlated with initial land composition. Figures E.4–E.7 report four alternatives that deviate from this baseline: (a) no initial land-use characteristic, (b) initial soybean share only $\times\, \delta_t$, (c) initial pasture share only $\times\, \delta_t$, and (d) initial forest share only $\times\, \delta_t$.

Figure E.4: Robustness: No Initial Characteristic



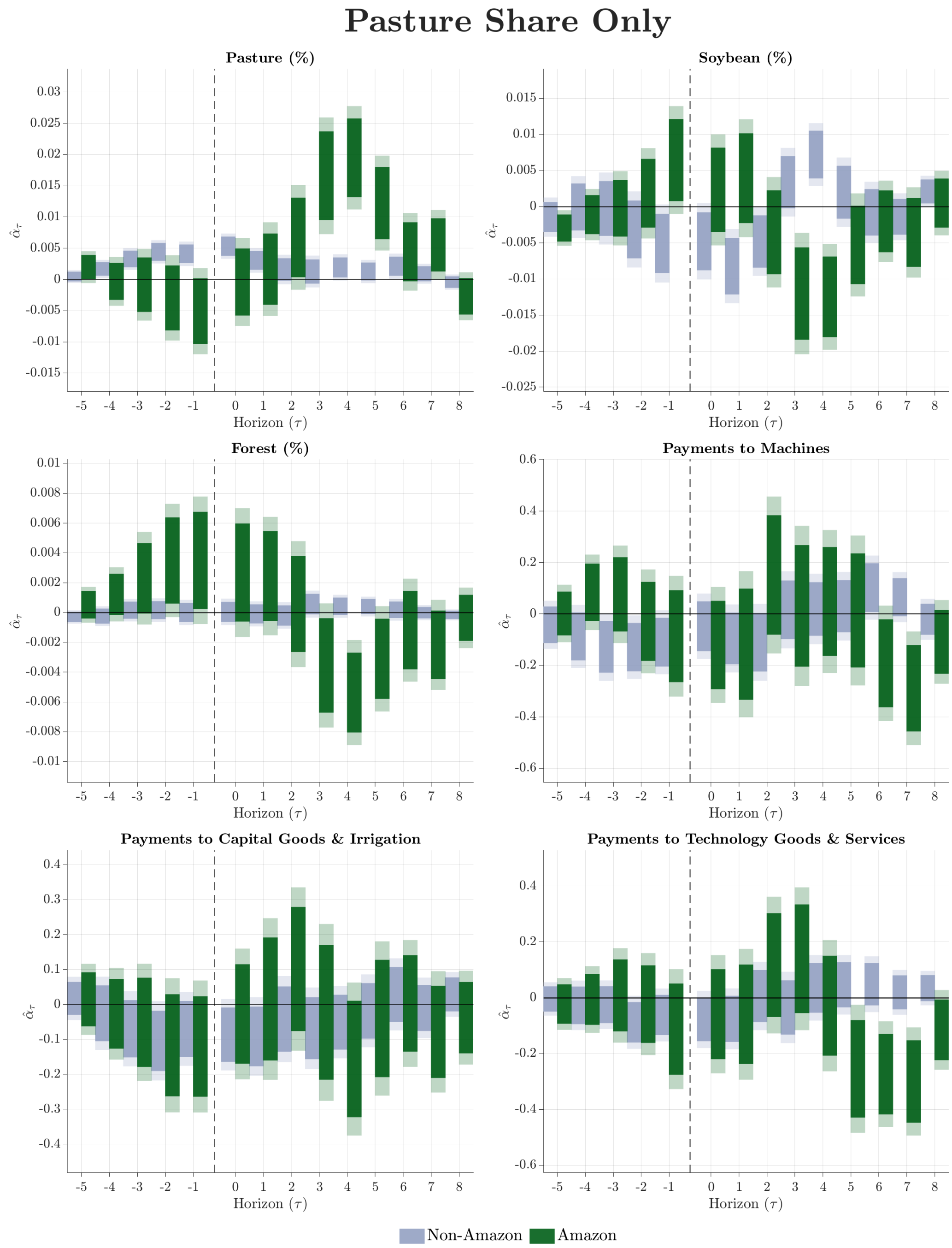
Notes: The figure plots estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$ without absorbing any initial land-use characteristic interacted with year fixed effects. Absorbed controls include farmer FE, year FE, coverage $\bar{s}_i \times$ year, and beef share $\times$ year. Green bars denote Amazon, blue bars denote non-Amazon. Panels report effects on land-use shares and $\log(1 + Y_{it})$ spending. Solid bars denote 95% confidence intervals, shaded areas 99%, clustered at the farmer level.

Figure E.5: Robustness: Initial Soybean Share Only



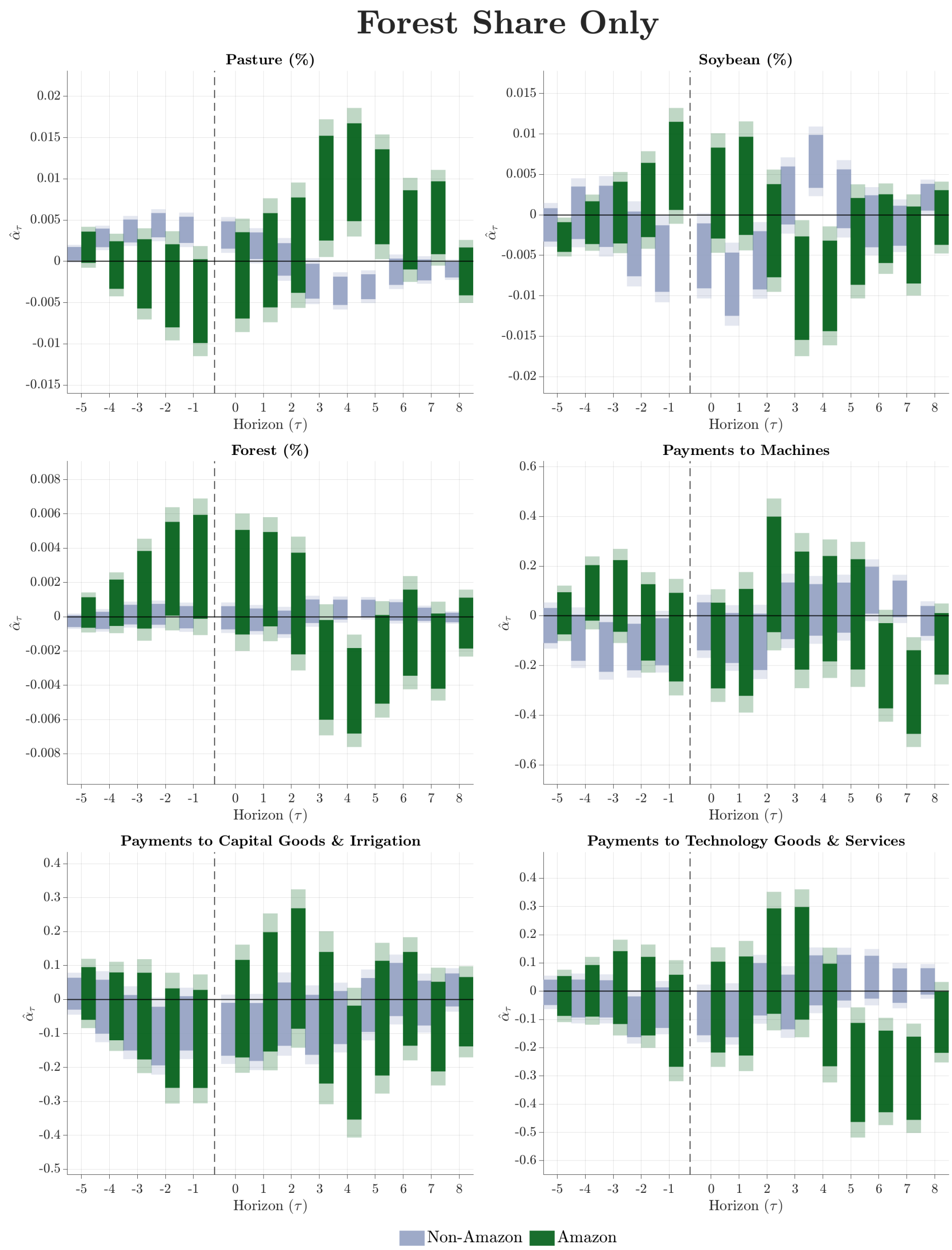
Notes: The figure plots estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$, dropping the initial forest share from the baseline absorb specification and retaining only initial soybean share interacted with year fixed effects. Absorbed controls include farmer FE, year FE, coverage $\bar{s}_i \times$ year, beef share $\times$ year, and Soybean share$_{t_0} \times$ year. Green bars denote Amazon, blue bars denote non-Amazon. Panels report effects on land-use shares and $\log(1 + Y_{it})$ spending. Solid bars denote 95% confidence intervals, shaded areas 99%, clustered at the farmer level.

Figure E.6: Robustness: Initial Pasture Share Only



Notes: The figure plots estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$, replacing the baseline initial forest and soybean shares with initial pasture share only interacted with year fixed effects. Absorbed controls include farmer FE, year FE, coverage $\bar{s}_i \times$ year, beef share $\times$ year, and Pasture share$_{t_0} \times$ year. Green bars denote Amazon, blue bars denote non-Amazon. Panels report effects on land-use shares and $\log(1 + Y_{it})$ spending. Solid bars denote 95% confidence intervals, shaded areas 99%, clustered at the farmer level.

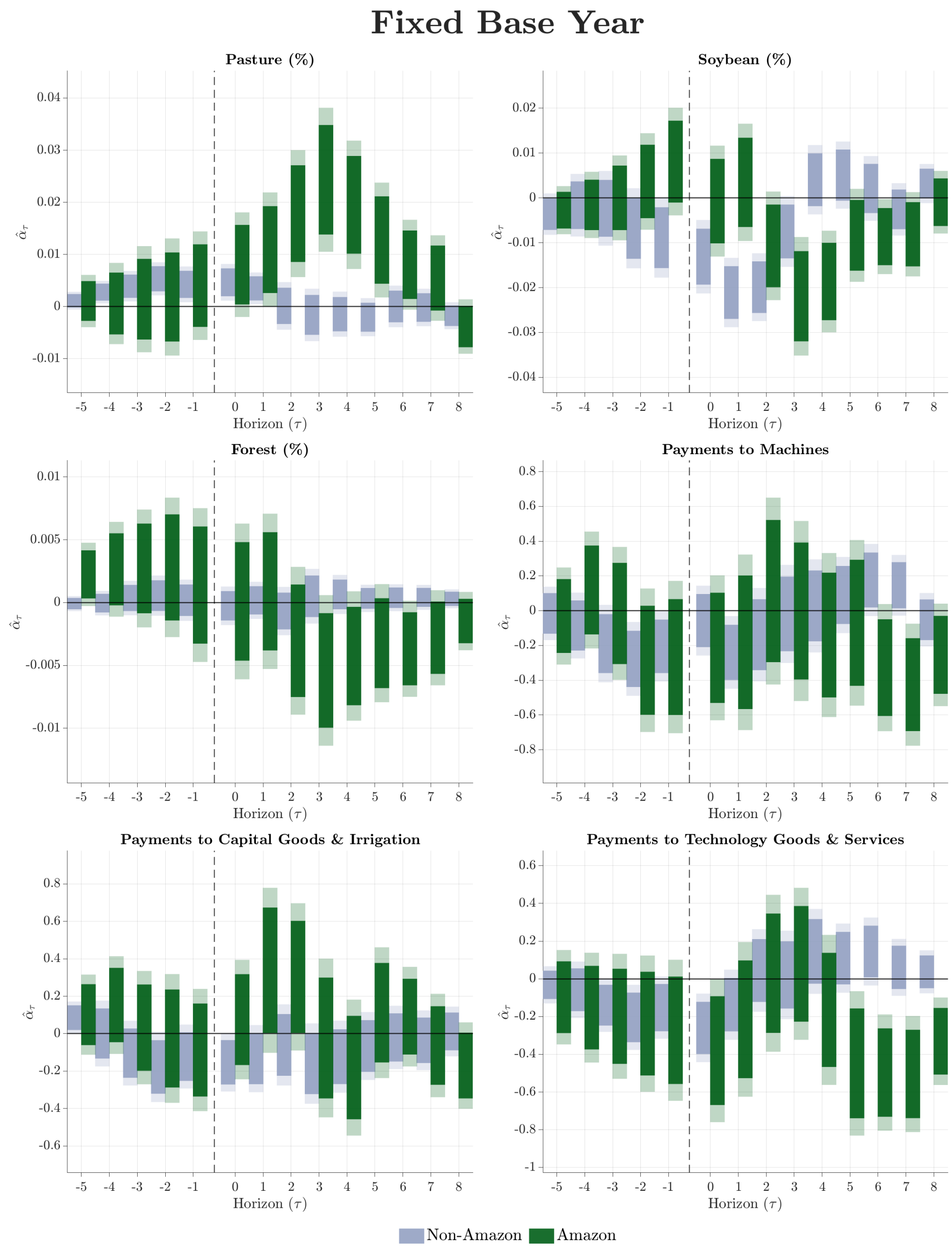
Figure E.7: Robustness: Initial Forest Share Only



Notes: The figure plots estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$, dropping the initial soybean share from the baseline absorb specification and retaining only initial forest share interacted with year fixed effects. Absorbed controls include farmer FE, year FE, coverage $\bar{s}_i \times$ year, beef share $\times$ year, and Forest share$_{t_0} \times$ year. Green bars denote Amazon, blue bars denote non-Amazon. Panels report effects on land-use shares and $\log(1 + Y_{it})$ spending. Solid bars denote 95% confidence intervals, shaded areas 99%, clustered at the farmer level.

Fixed Baseline Year. We restrict the sample to farmers whose first year in the data is 2010, so that $t_0 = 2010$ for all farmers and the initial weights w_{ix} and shares s^k_{ic} are measured at the same point in time. This eliminates variation arising from staggered entry and rolling baseline years. Figure E.8 reports the results.

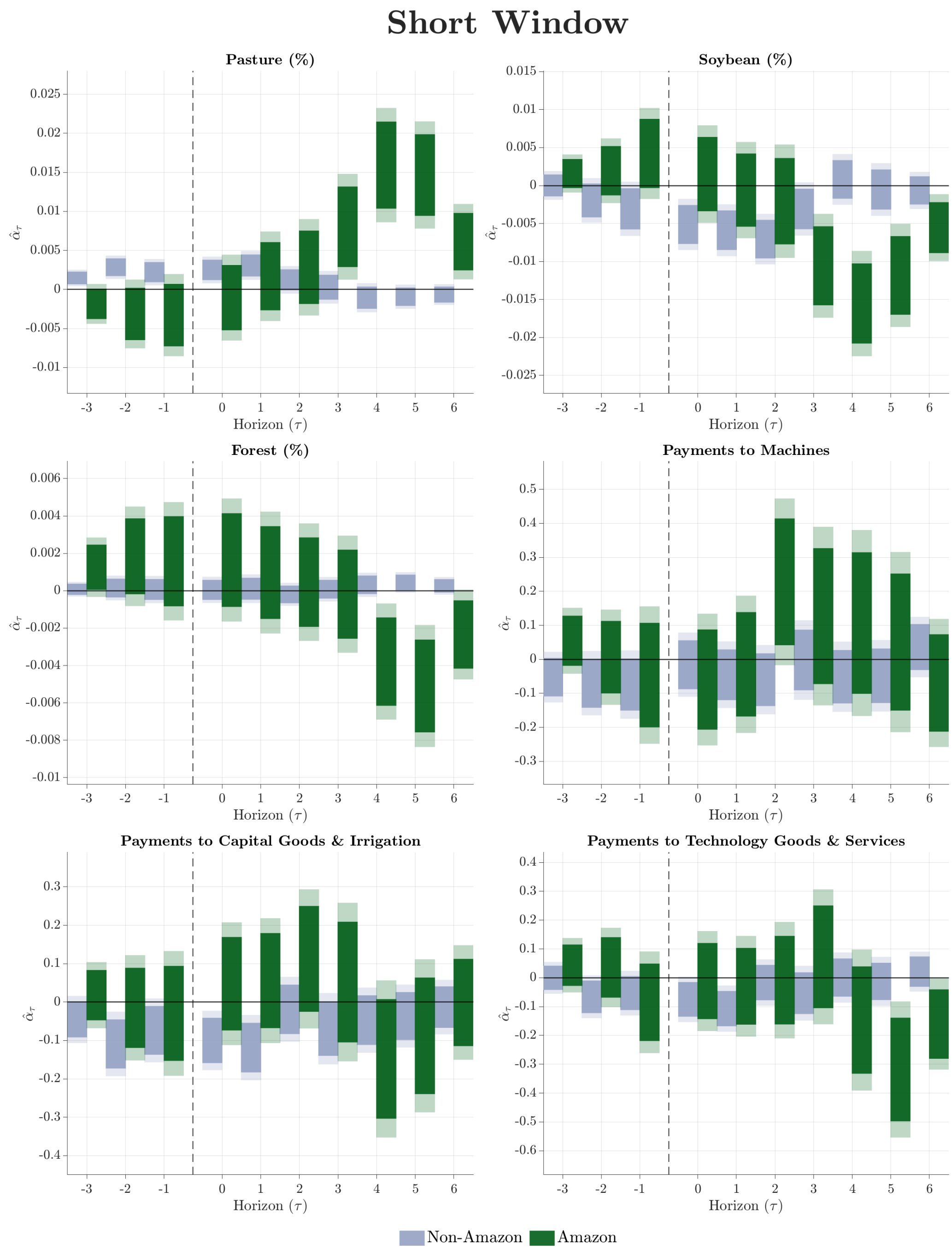
Figure E.8: Robustness: Fixed Baseline Year ($t_0 = 2010$)



Notes: The figure plots estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$, restricting the sample to farmers whose first year in the data is 2010 so that all initial weights and shares are measured at the same point in time. Green bars denote Amazon, blue bars denote non-Amazon. Panels report effects on land-use shares and $\log(1 + Y_{it})$ spending. Solid bars denote 95% confidence intervals, shaded areas 99%, clustered at the farmer level.

Shorter Event Window. We re-estimate equation (10) using an event window of $\tau = -3$ to $+6$ instead of $\tau = -5$ to $+8$. This tests whether our results are sensitive to the inclusion of distant leads and lags, which impose stronger sample balance requirements. Figure E.9 reports the results.

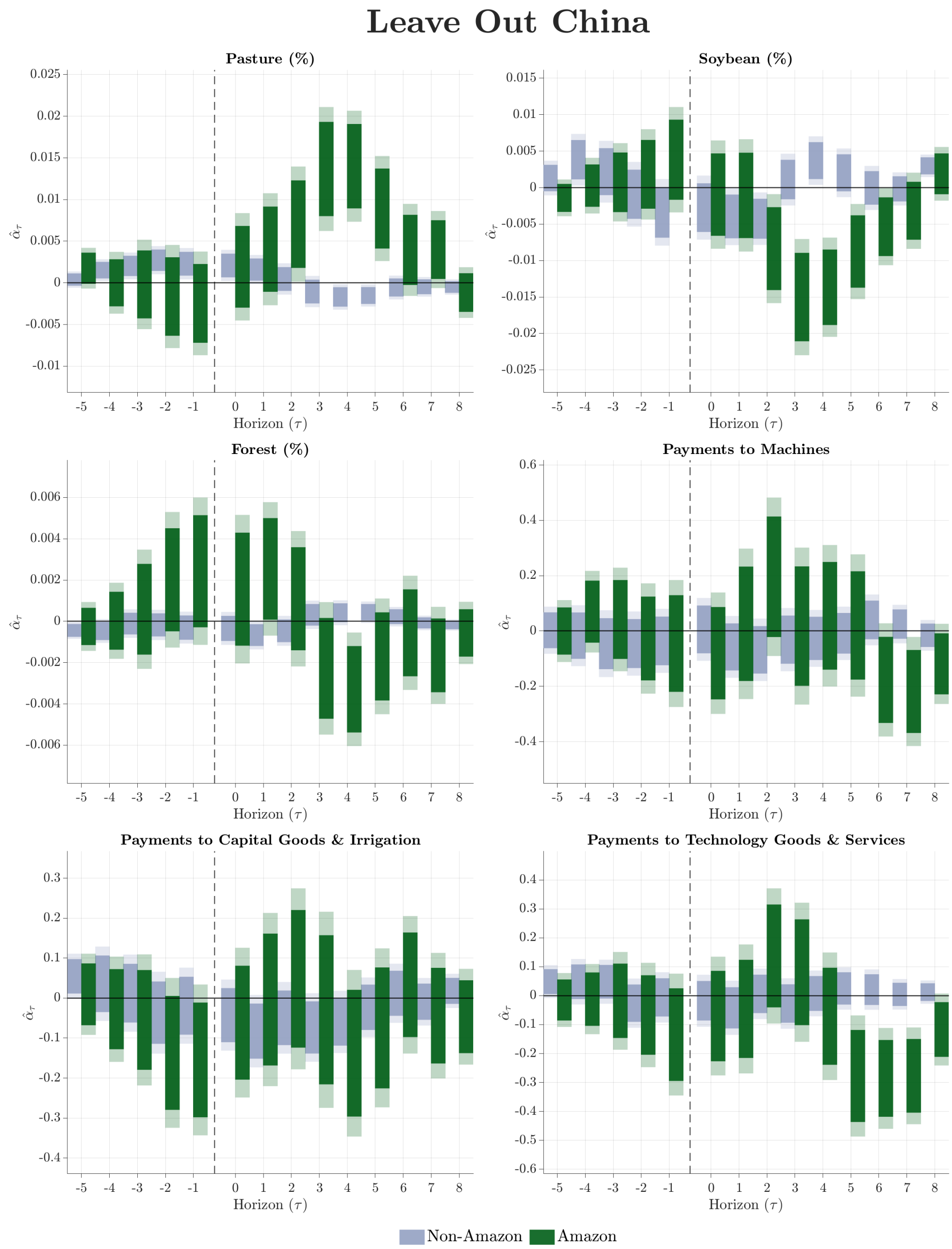
Figure E.9: Robustness: Shorter Event Window ($\tau = -3$ to $+6$)



Notes: The figure plots estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) with a shorter event window of $\tau = -3, \ldots, +6$ instead of the baseline $\tau = -5, \ldots, +8$. Green bars denote Amazon, blue bars denote non-Amazon. Panels report effects on land-use shares and $\log(1 + Y_{it})$ spending. Solid bars denote 95% confidence intervals, shaded areas 99%, clustered at the farmer level.

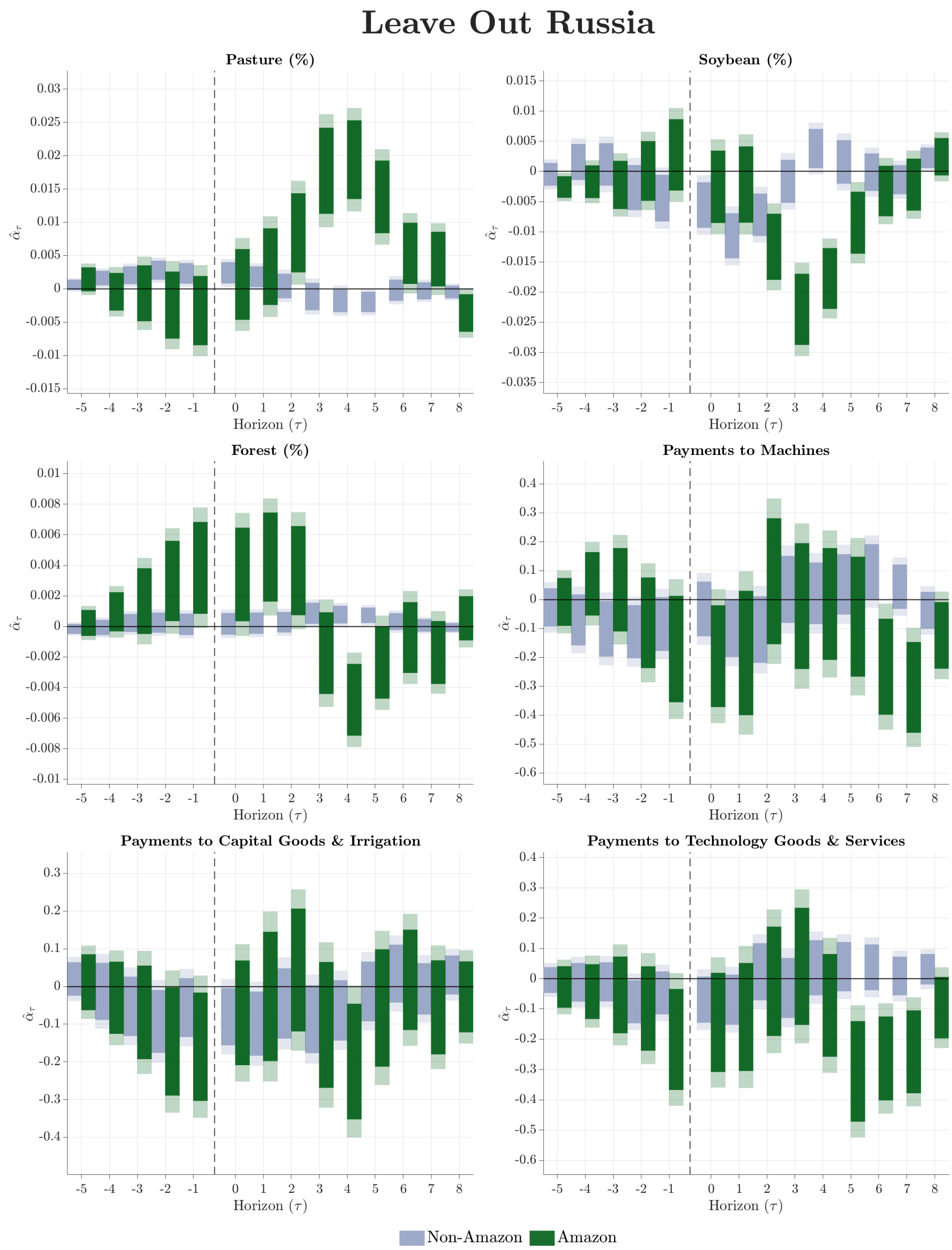
Leave-One-Destination-Out. We drop each of the five largest destinations by importance weight S_c one at a time (China, Russia, Hong Kong, Iran, the Netherlands), jointly reweight the remaining shares s_{ic}^k across both products so that $\sum_{k,c} s_{ic}^k = 1$, recompute Δz_{it}, and re-estimate. As in the destination-drop exercise above, the incomplete share control is dropped after reweighting. Figures E.10–E.14 report each variant. This exercise directly addresses the concern raised in Appendix A that the instrument may be driven by a small number of dominant demand shifters.

Figure E.10: Leave-One-Out: Dropping China



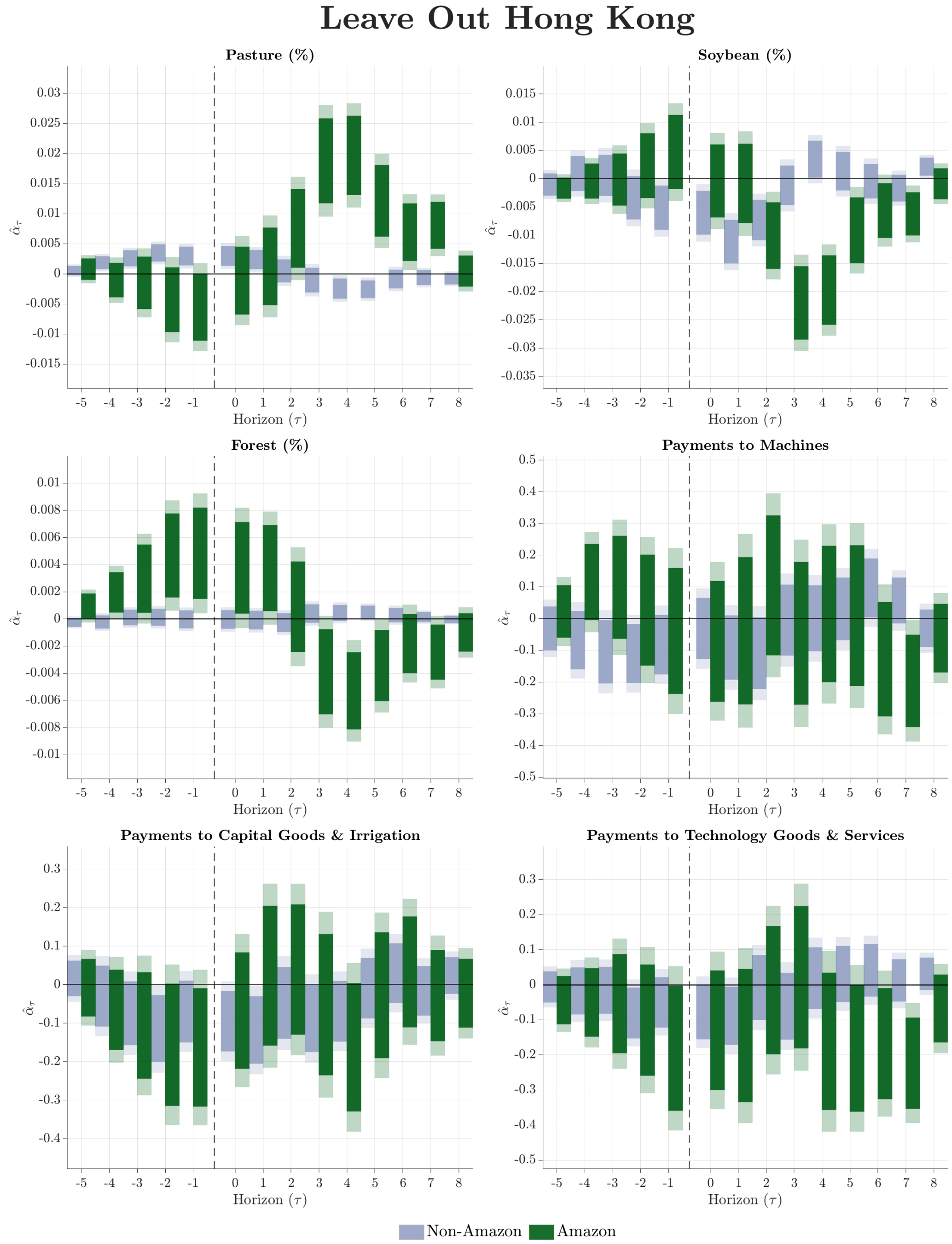
Notes: The figure plots estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$ after dropping China from all destination shares and reweighting remaining shares to sum to one. Green bars denote Amazon, blue bars denote non-Amazon. Panels report effects on land-use shares and $\log(1 + Y_{it})$ spending. Solid bars denote 95% confidence intervals, shaded areas 99%, clustered at the farmer level.

Figure E.11: Leave-One-Out: Dropping Russia



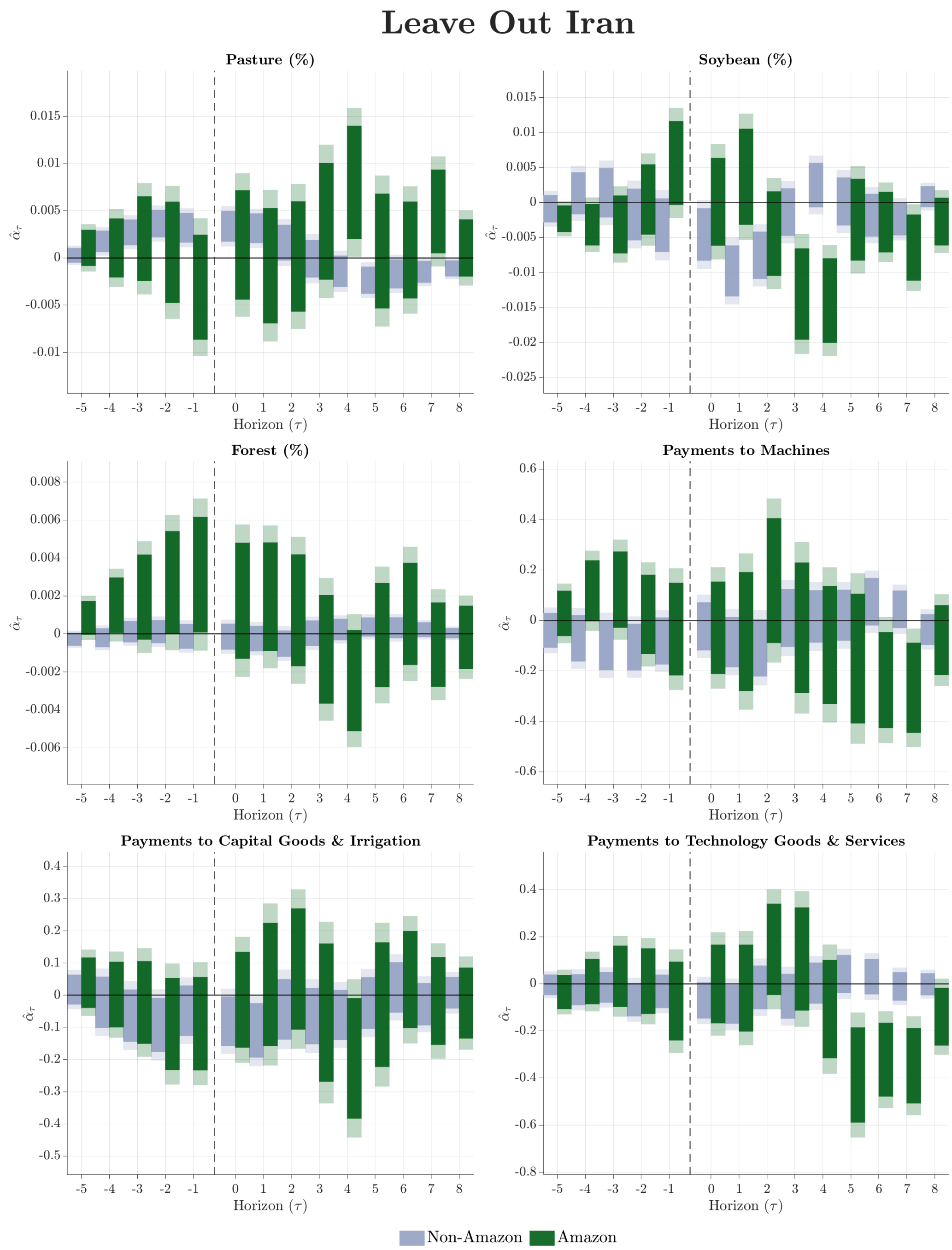
Notes: The figure plots estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$ after dropping Russia from all destination shares and reweighting remaining shares to sum to one. Green bars denote Amazon, blue bars denote non-Amazon. Panels report effects on land-use shares and $\log(1 + Y_{it})$ spending. Solid bars denote 95% confidence intervals, shaded areas 99%, clustered at the farmer level.

Figure E.12: Leave-One-Out: Dropping Hong Kong



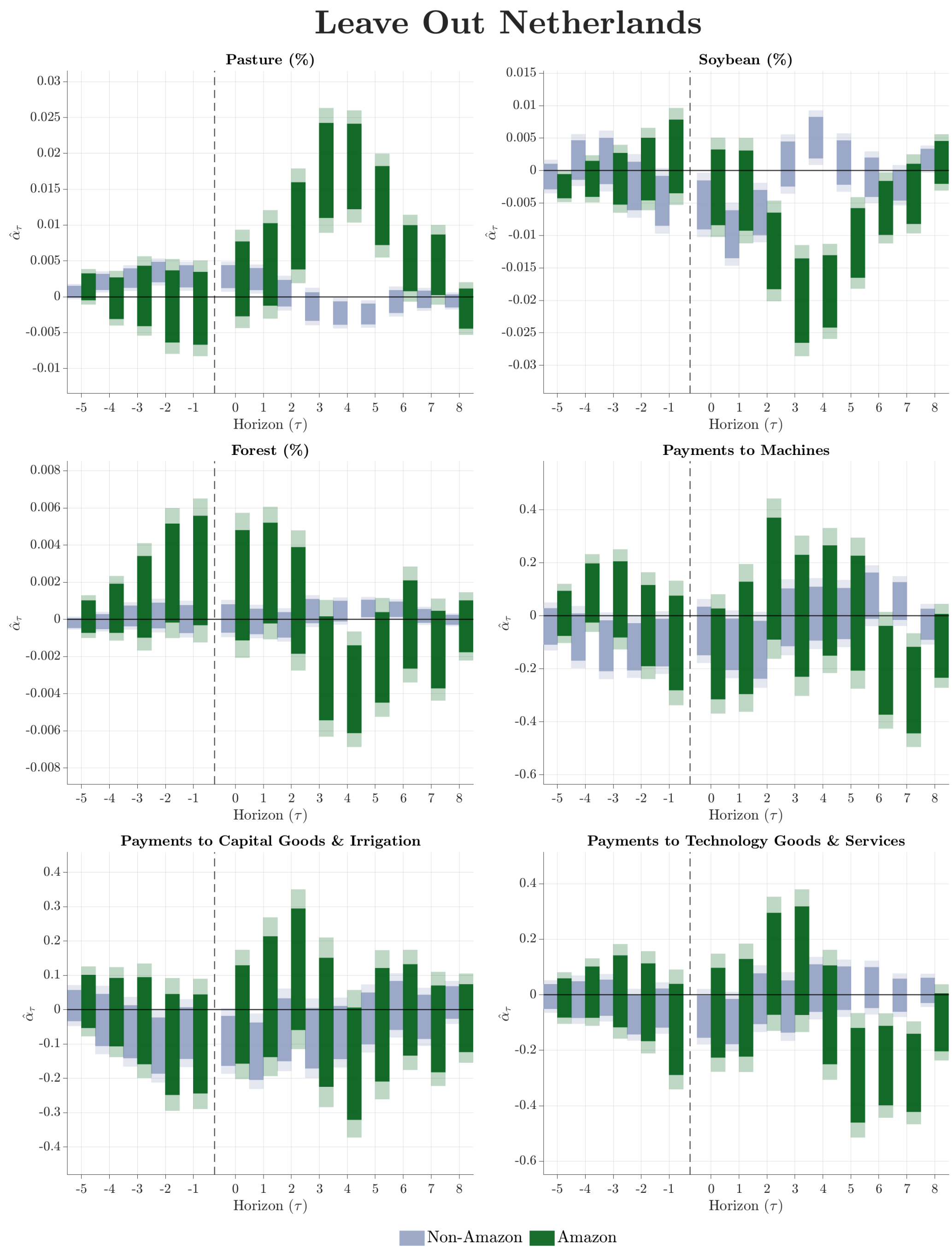
Notes: The figure plots estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$ after dropping Hong Kong from all destination shares and reweighting remaining shares to sum to one. Green bars denote Amazon, blue bars denote non-Amazon. Panels report effects on land-use shares and $\log(1 + Y_{it})$ spending. Solid bars denote 95% confidence intervals, shaded areas 99%, clustered at the farmer level.

Figure E.13: Leave-One-Out: Dropping Iran



Notes: The figure plots estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$ after dropping Iran from all destination shares and reweighting remaining shares to sum to one. Green bars denote Amazon, blue bars denote non-Amazon. Panels report effects on land-use shares and $\log(1 + Y_{it})$ spending. Solid bars denote 95% confidence intervals, shaded areas 99%, clustered at the farmer level.

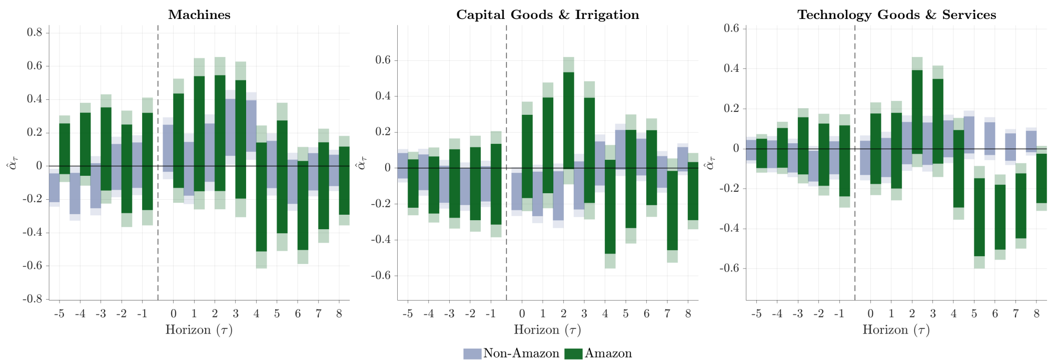
Figure E.14: Leave-One-Out: Dropping Netherlands



Notes: The figure plots estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$ after dropping the Netherlands from all destination shares and reweighting remaining shares to sum to one. Green bars denote Amazon, blue bars denote non-Amazon. Panels report effects on land-use shares and $\log(1 + Y_{it})$ spending. Solid bars denote 95% confidence intervals, shaded areas 99%, clustered at the farmer level.

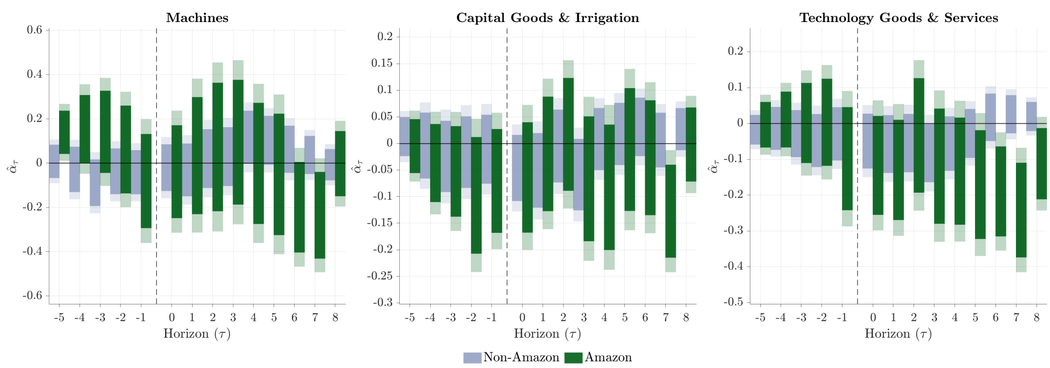
Classification Robustness. Our baseline classification of upstream suppliers into machines, capital goods and irrigation, and technology goods and services relies on specific cnae codes as documented in Table E.1. To assess the sensitivity of our results to these definitions, we estimate two alternative classifications. The *broader* classification expands each category to include additional supplier types that plausibly relate to farm investment. For machines, we add wholesale trade in agricultural machinery (cnae 4661-3), wholesale of earthmoving equipment (cnae 4662-1), vehicle commerce and repair (Division 45), and rental of agricultural equipment (cnae 7731-4). For capital goods, we add construction services (Divisions 41–43) and wholesale of electrical materials (cnae 4673-7). For technology, we add agricultural pest control (cnae 0161-0/01) and land preparation services (cnae 0161-0/03). The *narrower* classification restricts each category to the most tightly defined codes. For machines, we retain only agricultural tractors and machinery (cnae group 283), wholesale of agricultural machines (cnae 4661-3), and agricultural machinery repair (cnae 3314-7/11, 3314-7/12), dropping vehicles and earthmoving equipment. For capital goods, we retain only irrigation works (cnae 4222-7), irrigation equipment manufacturing (cnae 2832-1), hydraulic and pneumatic equipment (cnae 2812-7, 2813-5), and electrical machinery (Division 27). For technology, we retain only IT services (Divisions 62–63), R&D (Division 72), and agronomic consulting (cnae 7490-1/03), dropping engineering services and electronics manufacturing. Figures E.15 and E.16 report the results.

Figure E.15: Robustness: Broader Supplier Classification



Notes: The figure plots estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$ using the broader cnae-based classification of spending categories documented in Table E.2. Green bars denote Amazon, blue bars denote non-Amazon. Panels report effects on $\log(1 + Y_{it})$ spending in each category. Solid bars denote 95% confidence intervals, shaded areas 99%, clustered at the farmer level. Land-use outcomes are unaffected by supplier reclassification and are omitted.

Figure E.16: Robustness: Narrower Supplier Classification



Notes: The figure plots estimated coefficients $\{\hat{\alpha}_\tau\}$ from equation (10) for $\tau = -5, \ldots, +8$ using the narrower cnae-based classification of spending categories documented in Table E.3. Green bars denote Amazon, blue bars denote non-Amazon. Panels report effects on $\log(1 + Y_{it})$ spending in each category. Solid bars denote 95% confidence intervals, shaded areas 99%, clustered at the farmer level. Land-use outcomes are unaffected by supplier reclassification and are omitted.

Table E.1: Baseline Classification of Spending Categories by cnae Code

Category	cnae Code	Description (Portuguese)	English
Panel A: Machines			
	2831-3/00	Fab. de tratores agrícolas	Agricultural tractors
	2833-0/00	Fab. de máq. para agricultura e pecuária	Ag. machinery (excl. irrigation)
	2853-4/00	Fab. de tratores, exceto agrícolas	Non-agricultural tractors
	2854-2/00	Fab. de máq. para terraplenagem	Earthmoving equipment
	2910-7/01	Fab. de automóveis e utilitários	Automobiles and utility vehicles
	2920-4/01	Fab. de caminhões e ônibus	Trucks and buses
	2930-1/01	Fab. de cabines e reboques	Trailers and semi-trailers
	3314-7/11	Manut. de máq. para agricultura	Repair of ag. machinery
	3314-7/12	Manut. de tratores agrícolas	Repair of ag. tractors
Panel B: Capital Goods & Irrigation			
	4222-7/02	Obras de irrigação	Irrigation works
	Division 27	Máq., aparelhos e mat. elétricos	Electrical machinery and apparatus
	Division 30	Outros equip. de transporte	Other transport equipment
	Div. 28 remainder	Máq. e equipamentos (excl. Machines)	General/special machinery (residual)
	Div. 29 remainder	Veículos automotores (excl. Machines)	Motor vehicles and parts (residual)
Panel C: Technology Goods & Services			
	Division 26	Equip. de informática e eletrônicos	Computing and electronic equipment
	Division 62	Serviços de tecnologia da informação	IT services
	Division 63	Serviços de informação	Information services
	7112-0/00	Serviços de engenharia	Engineering services
	7119-7	Atividades técnicas de engenharia	Technical engineering activities
	7120-1/00	Testes e análises técnicas	Testing and technical analysis
	7490-1/03	Consultoria agronômica	Agronomic consulting
	Division 72	Pesquisa e desenvolvimento científico	Scientific R&D

Notes: The table reports the baseline mapping from cnae codes to the three spending categories used in the empirical analysis. "Division" refers to the 2-digit cnae division (all subclasses included). "Remainder" indicates all subclasses within the division not already classified under Machines. The cnae classification follows *Classificação Nacional de Atividades Econômicas* version 2.3. We verify robustness to broader and narrower definitions of these categories (see text).

Tables E.2 and E.3 document the broader and narrower classification variants used in Figures E.15 and E.16. The broader classification adds wholesale and rental of agricultural machinery, vehicle commerce and repair, construction services, and agricultural pest control. The narrower classification restricts machines to cnae group 283 (agricultural tractors and machinery only), drops most of the capital goods residual categories, and removes electronics manufacturing and engineering services from technology.

Table E.2: Broader Classification of Spending Categories by cnae Code

Category	cnae Code	English
Panel A: Machines (Broader)		
	All baseline codes	*See Table E.1*
	4661-3/00	Wholesale of agricultural machinery
	4662-1/00	Wholesale of earthmoving equipment
	Division 45	Vehicle commerce and repair
	7731-4/00	Rental of agricultural equipment
Panel B: Capital Goods & Irrigation (Broader)		
	All baseline codes	*See Table E.1*
	Divisions 41–43	Construction services
	4673-7/00	Wholesale of electrical materials
Panel C: Technology Goods & Services (Broader)		
	All baseline codes	*See Table E.1*

Notes: The table reports additional cnae codes included in the broader classification relative to the baseline (Table E.1). Technology Goods & Services is unchanged from baseline in the broader variant. Categories are mutually exclusive and assigned in the order Machines, Capital Goods, Technology.

Table E.3: Narrower Classification of Spending Categories by cnae Code

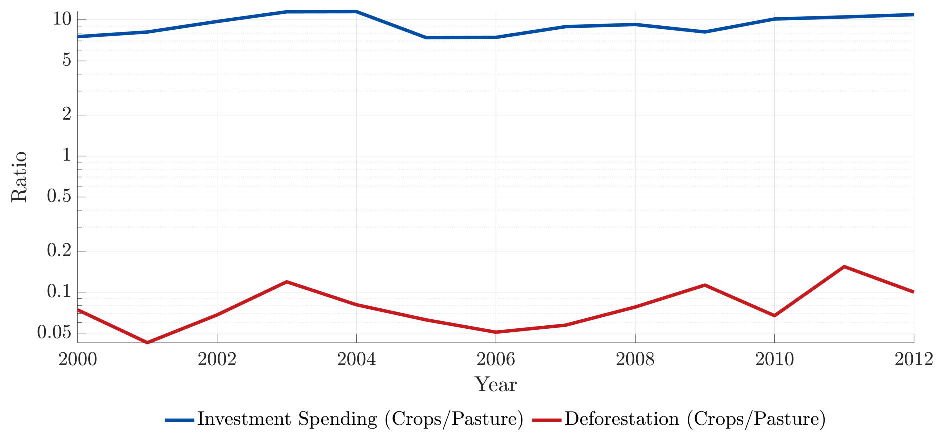
Category	cnae Code	English
Panel A: Machines (Narrower)		
	Group 283	Agricultural tractors and machinery
	4661-3/00	Wholesale of agricultural machinery
	3314-7/11	Repair of agricultural machinery
	3314-7/12	Repair of agricultural tractors
Panel B: Capital Goods & Irrigation (Narrower)		
	4222-7/02	Irrigation works
	2832-1/00	Irrigation equipment manufacturing
	2812-7/00	Hydraulic equipment
	2813-5/00	Pneumatic equipment
	Division 27	Electrical machinery and apparatus
Panel C: Technology Goods & Services (Narrower)		
	Divisions 62–63	IT and information services
	Division 72	Scientific R&D
	7490-1/03	Agronomic consulting

Notes: The table reports the restricted set of cnae codes included in the narrower classification. Relative to the baseline (Table E.1), Machines drops vehicles, earthmoving equipment, and non-agricultural tractors. Capital Goods drops Divisions 28–30 residuals and retains only irrigation and electrical equipment. Technology drops Division 26 (electronics manufacturing) and engineering services (cnae 7112, 7119, 7120).

F Appendix: Additional Descriptives

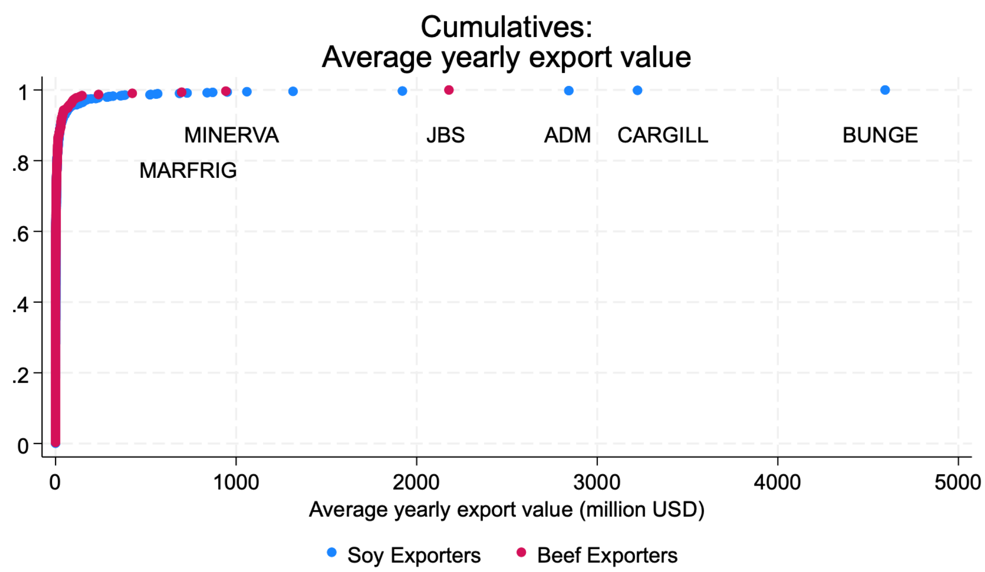
This appendix provides additional descriptive evidence on the sample used in the main analysis.

Figure F.1: Investment and Deforestation for Crops Relative to Pasture, 2000–2012



Notes. The blue line plots the ratio of crop to pasture investment spending and the red line plots the ratio of forest converted to cropland to forest converted to pasture, for Brazil over 2000 to 2012 (log scale). Investment spending is computed from the Investment Absorption Matrices (Miguez & Freitas 2021) in constant 2003 US dollars, excluding spending on crops, forestry products, and live animals. Forest conversion is measured as the area of native forest transitioning to cropland and to pasture between consecutive years in the MapBiomas land cover series, Collection 10.

Figure F.2: Cumulative Distribution of Average Yearly Export Value by Sector



Notes: The figure displays the cumulative distribution function (CDF) of average yearly export values (in million USD) for soy and beef exporting firms. For the agricultural sectors, the labels identify the top three exporters: Bunge, Cargill, and ADM for Soy; and JBS, Minerva, and Marfrig for Beef.

Table F.1: Summary Statistics: Amazon vs. Non-Amazon

	Legal Amazon				Outside Amazon			
	Mean	SD	Min	Max	Mean	SD	Min	Max
Trade & Transactions								
Export Value (mil. USD)	2,881	3,121	0	28,318	1,863	2,594	0	28,795
Payments to Suppliers (mil. USD)	0.40	5.70	0	664	0.73	15.98	0	2,972
# Suppliers per Farmer	10.2	48.9	1	5,125	13.1	132.8	1	12,157
# Exporters per Farmer	1.80	1.34	1	25	1.51	1.15	1	37
Beef Export Exposure	0.590	0.483	0	1	0.310	0.453	0	1
Soy Export Exposure	0.410	0.483	0	1	0.690	0.453	0	1
Land Use								
Forest Share	0.191	0.185	0	0.95	0.108	0.110	0	0.94
Pasture Share	0.464	0.346	0	1.00	0.211	0.276	0	1.00
Soybean Share	0.202	0.293	0	1.00	0.336	0.317	0	1.00
Total Area (ha)	3,065	8,356	0.09	345,555	1,758	4,513	0.09	171,162
# Properties per Farmer	2.74	3.04	1	56	3.59	4.08	1	130
Spending (Real USD / ha)								
Machines	14.2	1,075	0	234,057	165.9	21,939	0	8,983,155
Capital Goods & Irrigation	13.1	1,369	0	315,317	135.2	7,586	0	2,010,281
Technology Goods & Services	11.8	1,219	0	350,587	107.2	7,684	0	1,602,507
Instrument								
Share Coverage ($\bar{s}_i$)	1.000	0.003	0.260	1.000	1.000	0.009	0.001	1.000
Observations	99,499				252,541			
Unique Farmers	30,295				98,009			

Notes: The table reports summary statistics at the farmer-year level over 2010–2020. Farmers are classified as Legal Amazon if more than 50% of their CAR properties fall within the Legal Amazon boundary. *Export Value* is the total export value (in millions of USD) of all exporters the farmer is linked to through payment records. *Payments to Suppliers* is the total value of outgoing payments from the farmer to upstream input suppliers. *# Suppliers* and *# Exporters* count the unique upstream suppliers receiving payments from the farmer and the distinct export firms the farmer sells to, respectively. *Beef* and *Soy Export Exposure* are the shares of the farmer's total export exposure that flow through beef and soy exporters, respectively, and sum to one. Land use variables are computed by aggregating annual satellite imagery from MapBiomas (30×30 meter resolution) across all CAR properties associated with a given farmer, and dividing by the farmer's total registered property area. Each share therefore reflects the farmer's overall land allocation across all properties operated in a given year. Spending variables are total annual payments to suppliers in each category, converted to real USD using the BRL/USD exchange rate, and divided by total CAR property area in hectares. *Share Coverage* ($\bar{s}_i$) is the sum of farmer i's destination shares $\sum_{k,c} s^k_{ic}$; a value below one indicates that some of the farmer's export exposure flows through destinations for which FAO import data is unavailable, so that the demand shock is constructed from an incomplete set of destination weights. The mean is effectively one, but the minimum of 0.001 outside the Amazon shows that a small number of farmers have very low coverage, which we control for by absorbing $\bar{s}_i \times$ year fixed effects in the regression.

Table F.2 reports summary statistics split by whether the farmer is linked to beef or soy exporters. The two commodities display the expected differences in land use composition. Farms linked to beef exporters allocate on average 52% of their property area to pasture and 9% to soybean, while farms linked to soy exporters allocate 14% to pasture and 42% to soybean. Beef-linked farms are exposed to lower total export values (reflecting the smaller scale of beef exporters relative to soy) and are connected to fewer exporters on average. Spending per hectare on machines, capital goods, and technology services is broadly similar across the two groups in levels, though soy-linked farmers spend somewhat more on machinery.

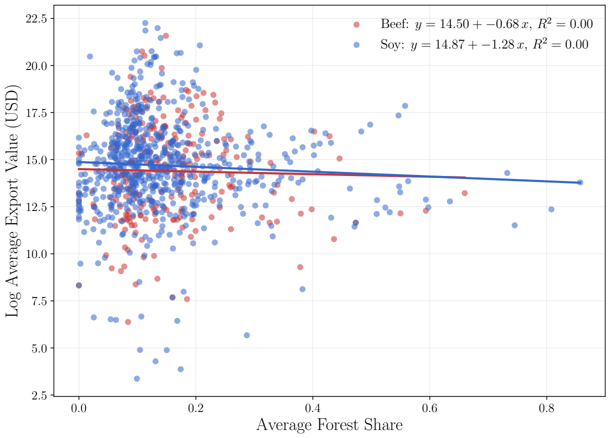
Table F.2: Summary Statistics: Beef vs. Soy

	Beef				Soy			
	Mean	SD	Min	Max	Mean	SD	Min	Max
Trade & Transactions								
Export Value (mil. USD)	1,217	1,288	0	6,576	2,487	3,122	0	28,795
Payments to Suppliers (mil. USD)	0.69	14.5	0	1,814	0.88	16.9	0	2,972
# Suppliers per Farmer	13.4	144.0	1	12,157	15.4	139.2	1	12,157
# Exporters per Farmer	1.33	0.66	1	10	1.60	1.27	1	35
Land Use								
Forest Share	0.143	0.156	0	0.95	0.125	0.129	0	0.95
Pasture Share	0.519	0.323	0	1.00	0.136	0.201	0	1.00
Soybean Share	0.090	0.197	0	1.00	0.424	0.304	0	1.00
Total Area (ha)	2,178	5,796	0.09	345,549	2,340	6,502	0.09	345,555
# Properties per Farmer	2.51	2.82	1	56	3.93	4.31	1	130
Spending (Real USD / ha)								
Machines	106.2	15,401	0	5,707,565	172.4	22,755	0	8,983,155
Capital Goods & Irrigation	139.4	8,807	0	2,010,281	140.9	7,874	0	2,010,281
Technology Goods & Services	95.7	7,278	0	1,419,847	116.4	7,998	0	1,602,507
Observations	142,389				234,455			
Unique Farmers	59,524				85,472			

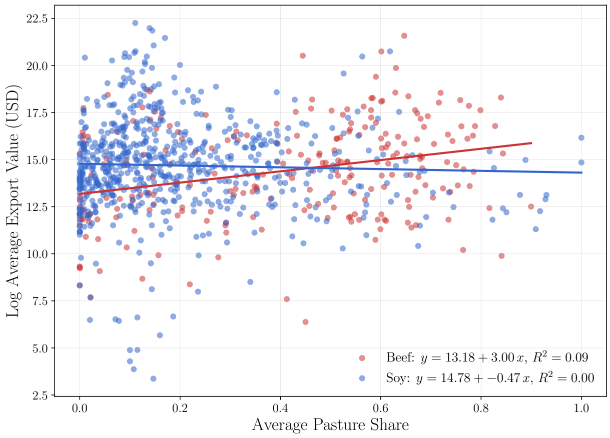
Notes: The table reports summary statistics at the farmer-year level over 2010–2020, split by whether the farmer is linked to at least one beef or soy exporter through payment records. Farmers linked to both appear in both columns. *Export Value* is the total export value (in millions of USD) of all exporters the farmer is linked to. Land use variables are computed by aggregating annual satellite imagery from MapBiomas across all CAR properties a farmer operates, divided by total registered property area. Spending variables are total annual payments to suppliers in each category, converted to real USD and divided by total property area in hectares.

Figure F.3 plots each exporter's average export value against the average land composition of its connected farmers. In contrast to the log-linear relationship between export value and the number of connected properties documented in the main text (Figure 4), there is no systematic correlation between exporter size and the average forest share, pasture share, or soybean share of connected farmers. The R^2 is essentially zero in all three panels. This pattern implies that larger exporters are not systematically connected to farmers with different land use characteristics, reducing the concern that exporter-level demand variation might be confounded with pre-existing differences in how farmers allocate their land.

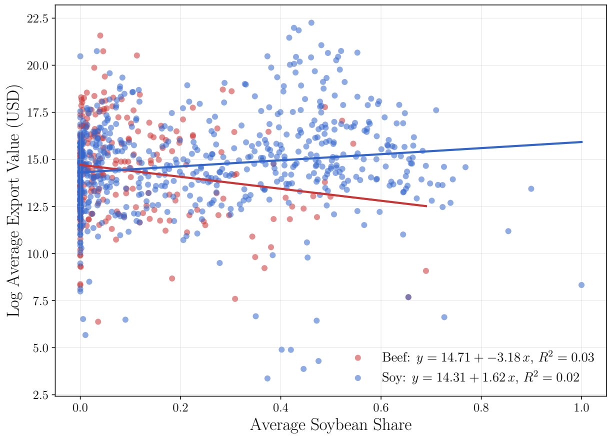
Figure F.3: Exporter Size and Average Land Composition of Connected Farmers



(a) Forest Share



(b) Pasture Share

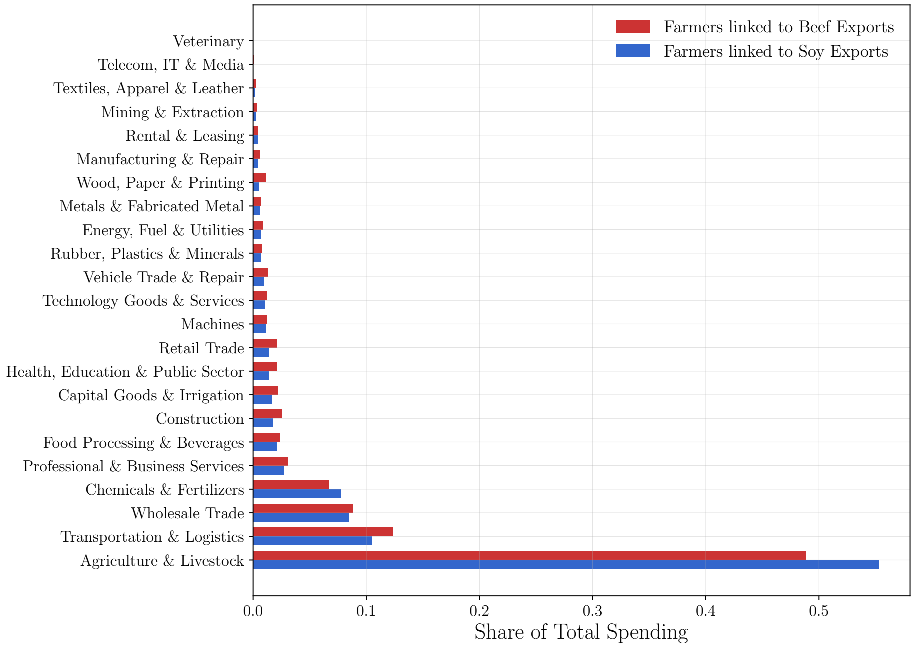


(c) Soybean Share

Notes: Each panel plots an exporter's log average yearly export value (USD) against the average land use share of its connected farmers between 2010 and 2020. Red dots represent beef exporters and blue dots represent soy exporters. The lines show linear fits with reported R^2. In all three panels, the R^2 is essentially zero, indicating no systematic correlation between exporter size and the land composition of connected farmers.

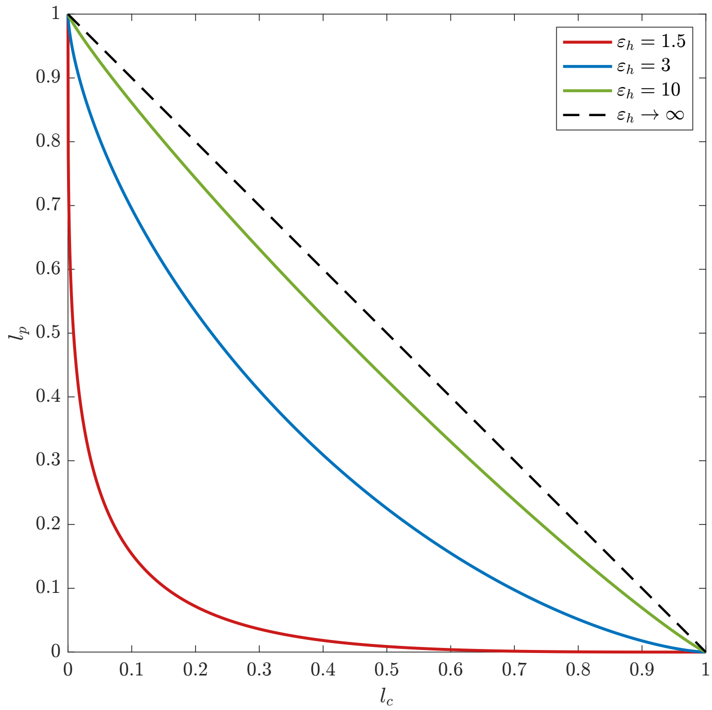
Figure F.4 decomposes total farm-level spending by the 2-digit cnae sector of the upstream supplier, separately for farmers linked to beef and soy exporters. For both commodities, the largest spending category is agriculture and livestock, which captures inter-farm transactions such as purchases of feed, seed, and live animals. Soy is a key input in cattle feed, linking the two commodity chains at the farm level. The next-largest categories are transportation and logistics, wholesale trade, and chemicals and fertilizers. Soy-linked farmers spend a notably larger share on chemicals and fertilizers, reflecting the input intensity of soybean cultivation. The three spending categories that we study in our empirical analysis, machines, capital goods and irrigation, and technology goods and services, are visible in the lower portion of the figure, each accounting for roughly 1 to 2 percent of total spending.

Figure F.4: Farm-Level Spending by Supplier Sector



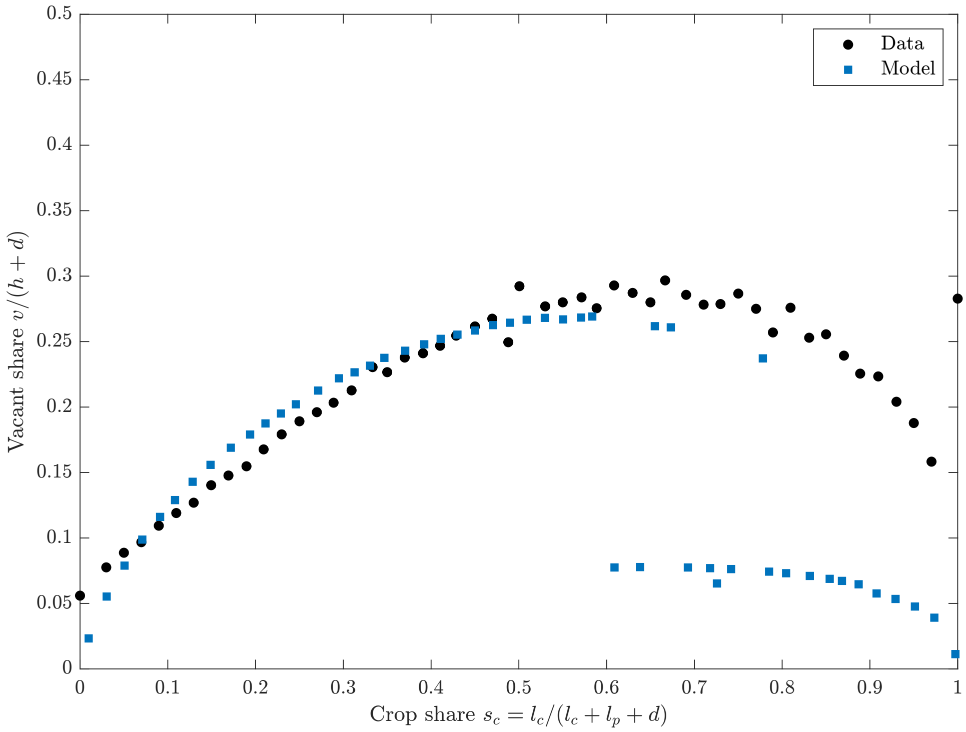
Notes: The figure displays the share of total farm-level spending directed to each supplier category (based on 2-digit cnae classification), separately for farmers linked to beef exporters (red) and soy exporters (blue). Categories are ranked by average spending share across both groups.

Figure F.1: The Plot Constraint for Different Values of ε_h



Notes: The figure plots the plot constraint (1) at equality, $\left(l_c^{(\varepsilon_h-1)/\varepsilon_h} + l_p^{(\varepsilon_h-1)/\varepsilon_h}\right)^{\varepsilon_h/(\varepsilon_h-1)} = 1$, with crop land l_c on the horizontal axis and pasture land l_p on the vertical axis, for $\varepsilon_h \in \{1.5, 3, 10\}$. The dashed line shows the limit $\varepsilon_h \to \infty$, where the constraint reduces to $l_c + l_p \leq 1$. Lower values of ε_h bow the constraint inward, so the sum $l_c + l_p$ is largest at full specialization.

Figure F.2: Vacant share of total property versus crop share of farmed land



Notes: The figure plots the median vacant share $v/(h+d)$ against the crop share of farmed land $s_c = l_c/(l_c + l_p + d)$ in 50 equal-width bins. Black dots are bin medians across 6.7 million CAR properties using MapBiomas land cover in 2003, all of Brazil. Blue squares are the calibrated model at $\varepsilon_h = 3.1$.

G Calibration Tables

Table G.1: Internal calibration (parameters and matched moments)

Parameter	Moment	Value	Model	Data	Source
ρ	Crop share of agricultural production value	0.75	0.34	0.34	FAO
$\varepsilon_h, \varepsilon_p$	Crop-share response, Legal Amazon (demeaned)	3.10, 1.95	-1.84	-1.84	Shift-share (this paper)
	Crop-share response, Non-Amazon (demeaned)		$+0.19$	$+0.37$	Shift-share (this paper)
$\bar{c}$	Pasture share of farm area, mean	0.018	0.66	0.60	CAR
$\underline{c}$	Pasture-to-crop land ratio	0.007	4.34	4.03	MapBiomas
ε	Pasture share of farm area, standard deviation	2.18	0.36	0.41	CAR
$\mathcal{D}_a^*$	Agricultural export / consumption ratio	0.009	0.019	0.021	OECD I-O
$\mathcal{D}_n^*$	Non-agricultural export / consumption ratio	61.4	0.33	0.26	OECD I-O
A_m	Employment size distributions, ag and non-ag	1.45	Figure 11		RAIS, IBGE-Censo
κ_{tn}	Employment size distribution, non-ag	0.041	Figure 11		RAIS
κ_{mn}	Employment size distribution, non-ag	0.232	Figure 11		RAIS
ζ	Employment size distributions, ag and non-ag	4.87	Figure 11		RAIS, IBGE-Censo
δ	Employment size distribution, non-ag	1.03	Figure 11		RAIS
κ_{ta}	Employment size distribution, ag	0.049	Figure 11		IBGE-Censo
κ_{ma}	Employment size distribution, ag	0.007	Figure 11		IBGE-Censo

Table G.2: Amazon propensity $\beta(s_c)$ estimated on the 2003 CAR farm cross section

	$\Pr\left(i \in \text{Legal Amazon}\right)$
s_c	-19.719^{***}
	(0.044)
s_c^2	40.747^{***}
	(0.115)
s_c^3	-23.789^{***}
	(0.077)
Constant	-0.764^{***}
	(0.001)
Observations	6,729,685
R^2	0.170

Notes. Logistic regression of an indicator for location in the Legal Amazon on a cubic polynomial in the crop land share $s_c = l_c/(l_c + l_p + d)$, estimated on the 2003 CAR farm cross section. Robust standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table G.3: External calibration

Parameter	Value	Source
η_a	4.1	Heresi (2024) Table 6 (agriculture, markup 1.32)
η_n	3.2	Heresi (2024) Table 6 (non-agriculture, markup 1.45)
σ	2	Feenstra et al. (2018)
θ	5	Qiu et al. (2026)
ν_c	0.44	Pellegrina (2022)
ν_p	0.17	Pellegrina (2022)
γ_c	0.50	Pellegrina (2022)
γ_p	0.53	Pellegrina (2022)
ν_n	0.543	OECD SNA Table 6A (labor share)
α_a	0.963	OECD I-O: home bias (agriculture)
α_n	0.928	OECD I-O: home bias (non-agriculture)
ϕ_a	0.035	OECD I-O: consumption share (agriculture)
ϕ_n	0.965	OECD I-O: consumption share (non-agriculture)
A_t	1	Normalization
h	1	Normalization (common plot size)
N	1	Normalization
$P^*_{f,a}$	1	Normalization
$P^*_{f,n}$	1	Normalization
r	0.45	Cavalcanti et al. (2023), credit-weighted firm spread
τ	0	Policy baseline

H Land Tenure in Brazilian Agriculture

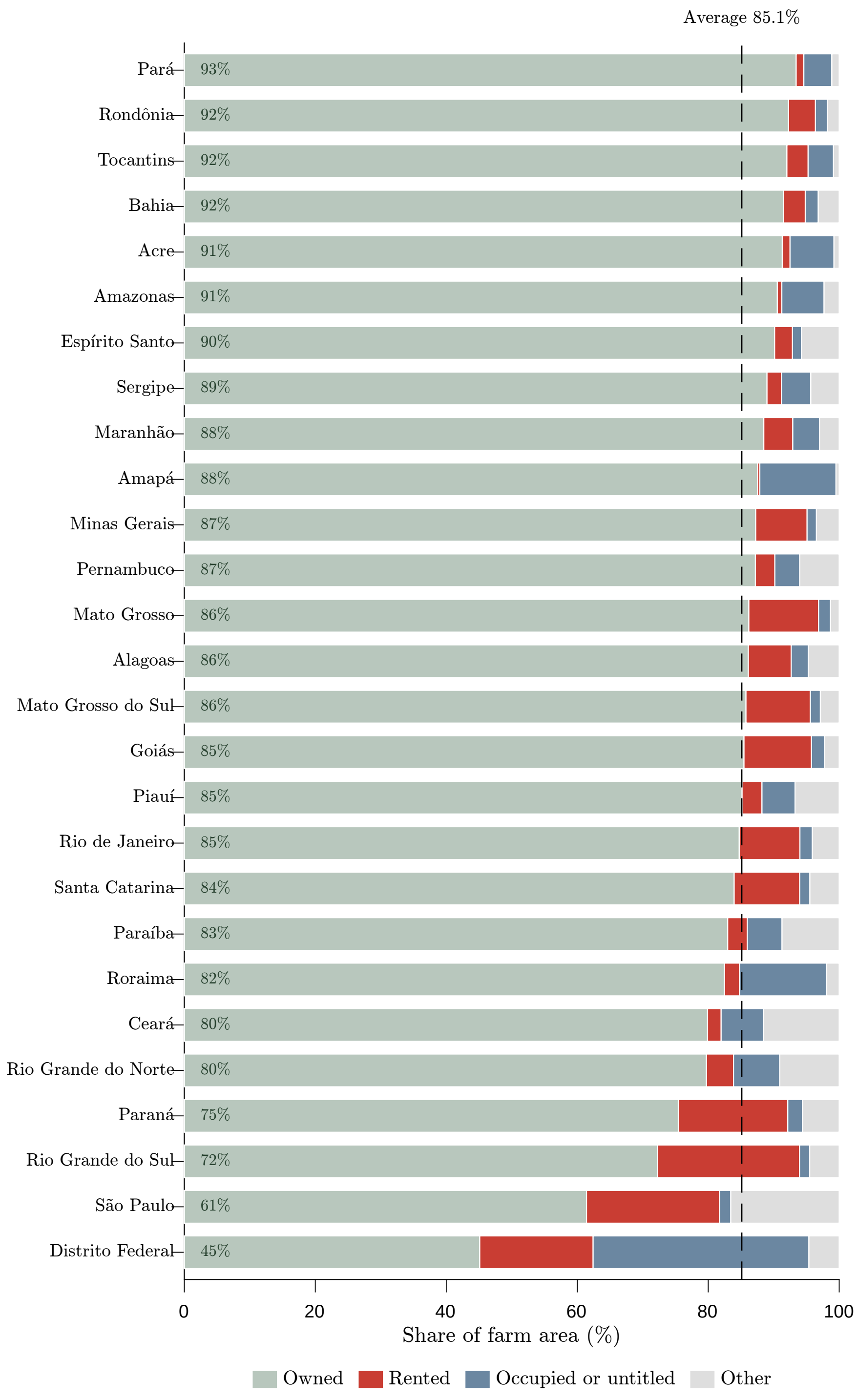


Figure H.3: Land tenure by state, Brazil 2017

Notes: Each bar shows the share of total agricultural establishment area by legal condition of the land, from the IBGE Censo Agropecuário 2017 (SIDRA Table 6754). Owned, rented, and occupied or untitled follow the census categories, and other combines sharecropped and loaned land. States are ordered by the owned share, and the dashed line marks the national average.

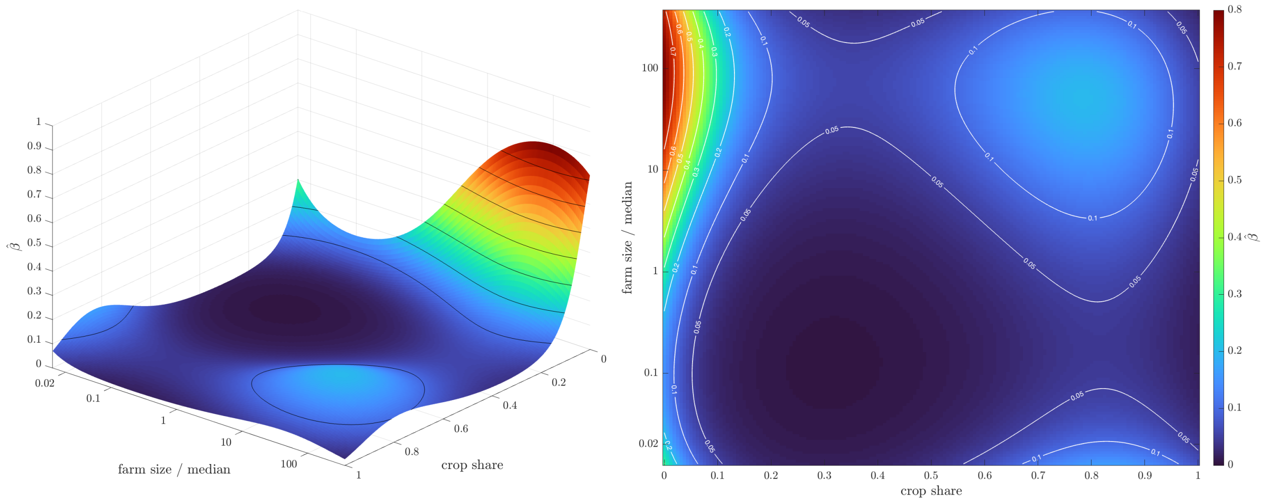
I Robustness: A Size-Augmented Amazon Bridge

The main text maps the model to the Legal Amazon through the propensity $\hat{\beta}(s_c)$, a logistic regression of Amazon location on a cubic polynomial in the crop share of farm area. In this appendix, we repeat the quantitative analysis under a richer bridge that adds farm size as a second observable. We estimate

$$\hat{\beta}(s_{c,i}, x_i) \equiv \widehat{\Pr}\big(j(i) \in \text{Legal Amazon} \,\big|\, s_{c,i},\, x_i\big), \qquad x_i \equiv \ln\big(\text{farm size}_i/\text{median farm size}\big), \tag{43}$$

by a logistic regression with cubic polynomials in s_c and x and their interaction on the same 6.7 million farms of the 2003 CAR cross section (R^2 of 0.242 against 0.170 for the univariate bridge). Farm size is farmed area, crops plus pasture; its model counterpart is total farmed land $h + d$, measured relative to the model's own mass-weighted median. We plot the resulting fit in Figure I.4. At the median size, a farm without crop land is in the Legal Amazon with probability 0.29, falling below five percent once crops cover a third of the farm, and conditional on the crop share larger farms are substantially more likely to lie in the Amazon. Large, pasture-dominated farms are therefore the most likely located within the legal Amazon.

Figure I.4: The Amazon propensity $\hat{\beta}(s_c, x)$ as a function of crop share and farm size



Notes. Both panels show the fitted probability $\hat{\beta}(s_c, x)$ that a farm lies in the Legal Amazon as a function of its crop share of farm area and its farm size relative to the median (farm size is farmed area, crops plus pasture; the size axis is in levels on log spacing). The left panel plots the fitted surface with contour lines at intervals of 0.1; the right panel shows the same surface as a heatmap with labeled contours. Estimated by a logistic regression with cubic polynomials in the crop share and in log relative farm size, plus their interaction, on 6.7 million farms in the 2003 CAR cross section.

We then recalibrate all internally calibrated parameters to the same targets as in the main text. Table I.4 compares the two calibrations. Importantly, the implied parameter space is close to our baseline calibration and both land-use elasticities are essentially unchanged: $\varepsilon_h = 3.10$ and $\varepsilon_p = 1.94$ against 3.10 and 1.95 in the main text.

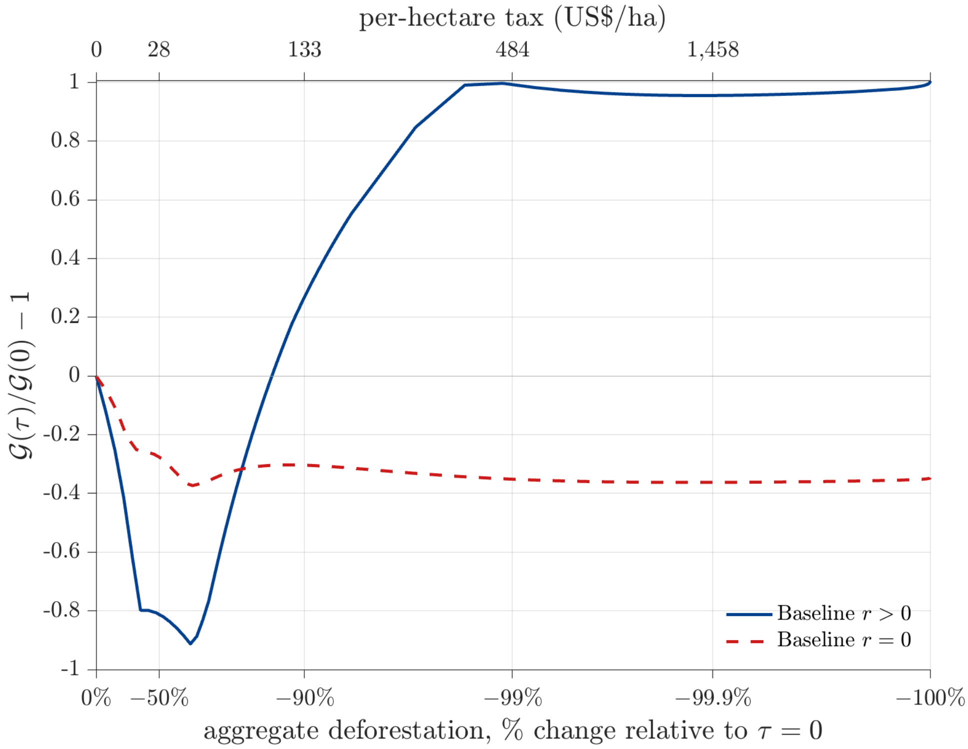
Table I.4: Internal calibration under the size-augmented bridge

	Value	
Parameter	Main text	Bivariate $\hat{\beta}$
ρ	0.75	0.76
ε_h	3.10	3.10
ε_p	1.95	1.94
$\bar{c}$	0.018	0.019
$\underline{c}$	0.007	0.007
ε	2.18	2.18
$\mathcal{D}_a^*$	0.009	0.009
$\mathcal{D}_n^*$	61.4	59.4
A_m	1.45	1.45
ζ	4.87	4.90
δ	1.03	1.03
κ_{tn}, κ_{mn}, κ_{ta}, κ_{ma}	0.041, 0.232, 0.049, 0.007	0.041, 0.232, 0.050, 0.007
Targeted moment	**Data**	**Model**
Crop-share response, Amazon vs. non-Amazon (pp)	-2.21	-1.95
Deforestation response, Legal Amazon (pp)	0.41	0.51

Notes. The upper panel compares the internally calibrated parameters of the main text (univariate bridge $\hat{\beta}(s_c)$) with the recalibration under the bivariate bridge $\hat{\beta}(s_c, x)$; targets and procedure are identical to the main text. The lower panel reports the two targeted general-equilibrium land-use responses per unit change in $\ln \mathcal{D}_a^*$ under the bivariate bridge; both lie within one standard error of the data.

Our policy conclusion survives. Converting the tax with agricultural value added per hectare, one unit of τ is worth about US\$3,700 per hectare in 2006 dollars: a tax of US\$28 per hectare halves aggregate deforestation, US\$133 removes ninety percent, and US\$484 ninety-nine percent. At the US\$42.5 tax studied by Souza-Rodrigues (2019) the model eliminates 63 percent of deforestation, against roughly 64 percent implied by his estimated land demand. Figure I.5 shows the gains from trade along the policy path: a moderate tax first lowers the gains by roughly ninety percent, and only once the policy is strong enough to trigger widespread adoption of the modern technology do the gains recover, ending 101 percent above the untaxed level when deforestation is fully eliminated.

Figure I.5: Environmental policy and the gains from trade



Notes. The vertical axis shows the change, relative to the $\tau = 0$ equilibrium, of the gains from trade $\mathcal{G}(\tau) = \Delta \ln C / \Delta \ln \mathcal{D}_a^*$, computed by doubling $\mathcal{D}_a^*$ at every level of the tax τ. The horizontal axis converts each tax into the change in aggregate deforestation on a logarithmic scale. The calibrated baseline economy is in blue and the frictionless economy ($r = 0$) in orange.

J Mathematical Appendix

J.1 Proof of Proposition 1

The Lagrangian of the farmer problem is

$$\mathcal{L} = p_c\, l_c^{\gamma_c} k^{\nu_c} n_c^{1-\gamma_c-\nu_c} + p_p\, \bar{l}_p^{\gamma_p}\, n_p^{1-\gamma_p} - r\, k - n_c - n_p - c\, d + \lambda\big(1 - l_c - l_p\big), \tag{44}$$

with first-order conditions

$$k: \quad r\, k = \nu_c\, p_c\, y_c, \qquad n_c: \quad n_c = (1 - \gamma_c - \nu_c)\, p_c\, y_c, \qquad n_p: \quad n_p = (1 - \gamma_p)\, p_p\, y_p, \tag{45}$$

$$l_c: \quad \gamma_c\, p_c\, \frac{y_c}{l_c} = \lambda, \qquad l_p: \quad \gamma_p\, p_p\, \frac{y_p}{\bar{l}_p} \left(\frac{\bar{l}_p}{l_p}\right)^{\frac{1}{\varepsilon_p}} = \lambda, \qquad d: \quad \gamma_p\, p_p\, \frac{y_p}{\bar{l}_p} \left(\frac{\bar{l}_p}{d}\right)^{\frac{1}{\varepsilon_p}} = c. \tag{46}$$

Since $\lambda = \gamma_c\, p_c\, y_c / l_c > 0$, the plot constraint binds, $l_c + l_p = 1$, and dividing the l_p condition by the d condition in (46) gives the shadow value of owned land, $\lambda = c\, (d/l_p)^{1/\varepsilon_p}$.

Substituting the input conditions (45) and the inverse demands into the technologies yields the revenue functions

$$p_c\, y_c = \left[\left(\frac{\nu_c}{r}\right)^{\nu_c} (1 - \gamma_c - \nu_c)^{1-\gamma_c-\nu_c}\right]^{\frac{\eta-1}{1+\gamma_c(\eta-1)}} (\rho\, p^w)^{\frac{\eta}{1+\gamma_c(\eta-1)}}\; l_c^{\frac{\gamma_c(\eta-1)}{1+\gamma_c(\eta-1)}}, \tag{47}$$

$$p_p\, y_p = (1-\gamma_p)^{\frac{(1-\gamma_p)(\eta-1)}{1+\gamma_p(\eta-1)}} \big((1-\rho)\, p^w\big)^{\frac{\eta}{1+\gamma_p(\eta-1)}}\, \bar{l}_p^{\frac{\gamma_p(\eta-1)}{1+\gamma_p(\eta-1)}}. \tag{48}$$

Totally differentiating the land conditions (46), the binding plot constraint, and the revenues (47)–(48) with respect to $\ln p^w$, and eliminating $d\ln \bar{l}_p$, $d\ln l_p^*$, and $d\ln d^*$, yields

$$\frac{d\ln l_c^*}{d\ln p^w} = \frac{\frac{\eta}{1+\gamma_p(\eta-1)} - \frac{\eta}{1+\gamma_c(\eta-1)} + \frac{\eta\,\varepsilon_p}{1+\gamma_c(\eta-1)}\Big(\frac{\gamma_p(\eta-1)}{1+\gamma_p(\eta-1)} - \frac{\varepsilon_p-1}{\varepsilon_p}\Big)\Big(\frac{d}{\bar{l}_p}\Big)^{\frac{\varepsilon_p-1}{\varepsilon_p}}}{-\frac{1}{1+\gamma_c(\eta-1)}\Big[1-\varepsilon_p\Big(\frac{\gamma_p(\eta-1)}{1+\gamma_p(\eta-1)} - \frac{\varepsilon_p-1}{\varepsilon_p}\Big)\Big(\frac{d}{\bar{l}_p}\Big)^{\frac{\varepsilon_p-1}{\varepsilon_p}}\Big] - \frac{l_c}{l_p}\cdot\frac{1}{1+\gamma_p(\eta-1)}}, \tag{49}$$

where $(d/\bar{l}_p)^{(\varepsilon_p-1)/\varepsilon_p} \in [0,1]$ is the cleared-land share of the pasture composite and

$$\frac{\gamma_p(\eta-1)}{1+\gamma_p(\eta-1)} - \frac{\varepsilon_p-1}{\varepsilon_p} = \frac{1+\gamma_p(\eta-1)-\varepsilon_p}{\varepsilon_p\big[1+\gamma_p(\eta-1)\big]} > 0 \qquad \text{if } \varepsilon_p < 1+\gamma_p(\eta-1). \tag{50}$$

It can be shown that the denominator of (49) is bounded away from zero,

$$\begin{aligned} &-\frac{1}{1+\gamma_c(\eta-1)}\Big[1-\varepsilon_p\Big(\frac{\gamma_p(\eta-1)}{1+\gamma_p(\eta-1)} - \frac{\varepsilon_p-1}{\varepsilon_p}\Big)\Big(\frac{d}{\bar{l}_p}\Big)^{\frac{\varepsilon_p-1}{\varepsilon_p}}\Big] - \frac{l_c}{l_p}\cdot\frac{1}{1+\gamma_p(\eta-1)} \\ &\le -\frac{\varepsilon_p}{\big[1+\gamma_c(\eta-1)\big]\big[1+\gamma_p(\eta-1)\big]} < 0 \qquad \text{if } \varepsilon_p < 1+\gamma_p(\eta-1) \end{aligned} \tag{51}$$

and that the equilibrium has the limiting behavior

$$\lim_{c\to 0}\Big(\frac{d^*}{\bar{l}_p^*}\Big)^{\frac{\varepsilon_p-1}{\varepsilon_p}} = 1, \qquad \lim_{c\to 0}\frac{l_c^*}{l_p^*} = 0, \qquad \lim_{c\to\infty}\Big(\frac{d^*}{\bar{l}_p^*}\Big)^{\frac{\varepsilon_p-1}{\varepsilon_p}} = 0 \qquad \text{(the last two if } \varepsilon_p < 1+\gamma_p(\eta-1)\text{)}. \tag{52}$$

Part (i). Since the denominator of (49) is strictly negative by (51), $d\ln l_c^*/d\ln p^w < 0$ iff the numerator of (49) is positive; rearranging the numerator, using (50),

$$\frac{d\ln l_c^*}{d\ln p^w} < 0 \quad \text{iff} \quad \Big(\frac{d^*}{\bar{l}_p^*}\Big)^{\frac{\varepsilon_p-1}{\varepsilon_p}} > \frac{(\eta-1)(\gamma_p-\gamma_c)}{1+\gamma_p(\eta-1)-\varepsilon_p}. \tag{53}$$

The constant on the right side of (53) satisfies

$$0 < \frac{(\eta-1)(\gamma_p-\gamma_c)}{1+\gamma_p(\eta-1)-\varepsilon_p} < 1 \qquad \text{if } \gamma_c < \gamma_p \text{ and } 1 < \varepsilon_p < 1+\gamma_c(\eta-1). \tag{54}$$

The share $(d^*/\bar{l}_p^*)^{(\varepsilon_p-1)/\varepsilon_p}$ is continuous and strictly decreasing in c and falls from 1 to 0 by (52); by continuity and monotonicity it crosses the constant in (53) exactly once, from above, at some $c^\star > 0$. Hence

$$\frac{d\ln l_c^*}{d\ln p^w} < 0 \ \text{ for } c < c^\star, \qquad \frac{d\ln l_c^*}{d\ln p^w} > 0 \ \text{ for } c > c^\star \qquad \text{if } \gamma_c < \gamma_p \text{ and } 1 < \varepsilon_p < 1+\gamma_c(\eta-1). \tag{55}$$

Fix $c < c^\star$. Differentiating the binding plot constraint $l_c + l_p = 1$ gives

$$\frac{d\ln l_p^*}{d\ln p^w} = -\frac{l_c}{l_p}\,\frac{d\ln l_c^*}{d\ln p^w} > 0. \tag{56}$$

Substituting (56) into the d condition in (46), totally differentiated with respect to $\ln p^w$, and solving for $d\ln d^*/d\ln p^w$,

$$\frac{d\ln d^*}{d\ln p^w} = \frac{\dfrac{\eta}{1+\gamma_p(\eta-1)} + \Big[\dfrac{\gamma_p(\eta-1)}{1+\gamma_p(\eta-1)} - \dfrac{\varepsilon_p-1}{\varepsilon_p}\Big]\Big(\dfrac{l_p}{\bar{l}_p}\Big)^{\frac{\varepsilon_p-1}{\varepsilon_p}} \dfrac{d\ln l_p^*}{d\ln p^w}}{\dfrac{1}{\varepsilon_p}\Big[1-\varepsilon_p\Big(\dfrac{\gamma_p(\eta-1)}{1+\gamma_p(\eta-1)} - \dfrac{\varepsilon_p-1}{\varepsilon_p}\Big)\Big(\dfrac{d}{\bar{l}_p}\Big)^{\frac{\varepsilon_p-1}{\varepsilon_p}}\Big]} > 0, \tag{57}$$

where the numerator is positive by (50) and (56), and it can be shown that the denominator satisfies $1-\varepsilon_p\big(\frac{\gamma_p(\eta-1)}{1+\gamma_p(\eta-1)} - \frac{\varepsilon_p-1}{\varepsilon_p}\big)(d/\bar{l}_p)^{(\varepsilon_p-1)/\varepsilon_p} \geq \varepsilon_p/\big[1+\gamma_p(\eta-1)\big] > 0$. Finally, differentiating $\ln s_c^* = \ln l_c^* - \ln(1+d^*)$,

$$\frac{d\ln s_c^*}{d\ln p^w} = \frac{d\ln l_c^*}{d\ln p^w} - \frac{d^*}{1+d^*}\frac{d\ln d^*}{d\ln p^w} < 0, \tag{58}$$

where the first term is negative by (55) and the second by (57). This proves part (i): if $\gamma_c < \gamma_p$ and $1 < \varepsilon_p < 1+\gamma_c(\eta-1)$, then

$$\frac{d\ln d^*}{d\ln p^w} > 0, \quad \frac{d\ln s_c^*}{d\ln p^w} < 0 \text{ for } c < c^\star, \qquad \text{and} \qquad \frac{d\ln l_c^*}{d\ln p^w} > 0 \text{ for } c > c^\star.$$

Part (ii). By the capital condition in (45) and the crop revenue (47), $rk^* = \nu_c p_c y_c$, so

$$\frac{d\ln(rk^*)}{d\ln p^w} = \frac{\eta}{1+\gamma_c(\eta-1)} + \frac{\gamma_c(\eta-1)}{1+\gamma_c(\eta-1)}\frac{d\ln l_c^*}{d\ln p^w}, \tag{59}$$

which is negative iff $d\ln l_c^*/d\ln p^w < -\eta/\gamma_c(\eta-1)$. Substituting (49) into this inequality and multiplying through by the denominator of (49), negative by (51), yields: $d\ln(rk^*)/d\ln p^w < 0$ iff

$$\frac{\eta}{1+\gamma_p(\eta-1)} - \frac{\eta}{\gamma_c(\eta-1)} + \frac{\eta\,\varepsilon_p}{\gamma_c(\eta-1)}\Big(\frac{\gamma_p(\eta-1)}{1+\gamma_p(\eta-1)} - \frac{\varepsilon_p-1}{\varepsilon_p}\Big)\Big(\frac{d}{\bar{l}_p}\Big)^{\frac{\varepsilon_p-1}{\varepsilon_p}}$$
$$- \frac{\eta}{\gamma_c(\eta-1)\big[1+\gamma_p(\eta-1)\big]}\frac{l_c}{l_p} > 0 \qquad \text{if } \varepsilon_p < 1+\gamma_p(\eta-1). \tag{60}$$

The left side of (60) is continuous and strictly decreasing in c: $(d^*/\bar{l}_p^*)^{(\varepsilon_p-1)/\varepsilon_p}$ is strictly decreasing and enters with a positive coefficient by (50), while l_c^*/l_p^* is strictly increasing and enters with a negative coefficient. Evaluating the left side of (60) at the limits (52), it can be shown that its limit as $c \to 0$ equals

$$\frac{\eta\,\big[\gamma_c(\eta-1)-\varepsilon_p\big]}{\gamma_c(\eta-1)\big[1+\gamma_p(\eta-1)\big]} > 0 \qquad \text{if } \varepsilon_p < \gamma_c(\eta-1), \tag{61}$$

while its limit as $c \to \infty$ is at most

$$\frac{\eta}{1+\gamma_p(\eta-1)} - \frac{\eta}{\gamma_c(\eta-1)} < 0 \qquad \text{if } \gamma_c(\eta-1) < 1+\gamma_p(\eta-1), \tag{62}$$

which holds if $\gamma_c < \gamma_p$. By continuity and monotonicity, the left side of (60) crosses zero exactly once, at some $\bar{c} > 0$, so that by (60)

$$\frac{d\ln(rk^*)}{d\ln p^w} < 0 \text{ for } c < \bar{c}, \qquad \frac{d\ln(rk^*)}{d\ln p^w} > 0 \text{ for } c > \bar{c} \qquad \text{if } \gamma_c < \gamma_p \text{ and } 1 < \varepsilon_p < \gamma_c(\eta-1). \tag{63}$$

For $c \geq c^\star$, $d\ln l_c^*/d\ln p^w > 0$ by (55), so $d\ln(rk^*)/d\ln p^w > 0$ by (59); then the left side of (60) is negative, by (60) with both inequalities reversed. If $\bar{c} \geq c^\star$, the left side of (60) would be both zero and negative at $c = \bar{c}$, a contradiction. Hence $\bar{c} < c^\star$. This proves part (ii): if $\gamma_c < \gamma_p$ and $1 < \varepsilon_p < \gamma_c(\eta - 1)$, then

$$\frac{d\ln(rk^*)}{d\ln p^w} < 0 \ \text{ for } c < \bar{c}, \qquad \frac{d\ln(rk^*)}{d\ln p^w} > 0 \ \text{ for } c > \bar{c}, \qquad 0 < \bar{c} < c^\star,$$

which completes the proof. □

J.2 General Equilibrium Model

J.2.1 Setup

Consider a farm with draws (z, c) and technology $A_i \in \{A_t, A_m\}$, taking Y_a, $P_{d,a}$, and P_n as given. We assume

$$\eta_a > 1, \qquad \varepsilon_p > 1, \qquad \gamma_c \leq \gamma_p, \qquad \gamma_p(\eta_a - 1) < \varepsilon_h - 1, \qquad \frac{1}{\varepsilon_h} + \frac{\gamma_c}{\varepsilon_p}(\eta_a - 1) > 1. \tag{64}$$

The last condition in (64) is equivalent to $\varepsilon_p \frac{\varepsilon_h - 1}{\varepsilon_h} < \gamma_c(\eta_a - 1)$. Combined with $\gamma_c(\eta_a - 1) \leq \gamma_p(\eta_a - 1) < \varepsilon_h - 1$, this implies

$$\varepsilon_p < \varepsilon_h, \qquad \varepsilon_p - 1 < \gamma_c(\eta_a - 1) \leq \gamma_p(\eta_a - 1) < \varepsilon_h - 1. \tag{65}$$

The Lagrangian of the farm problem is

$$\begin{aligned}
\mathcal{L} = {} & \rho\, p_d^c \Big(\frac{p_d^c}{P_{d,a}}\Big)^{-\eta_a} Y_a + (1-\rho)\, p_d^p \Big(\frac{p_d^p}{P_{d,a}}\Big)^{-\eta_a} Y_a - P_n(x_c + x_p) - (n_c + n_p) - (P_n c + \tau)\, d \\
& + \lambda^c \Big[z A_i\, l_c^{\gamma_c} x_c^{\nu_c} n_c^{1-\nu_c-\gamma_c} - \rho \Big(\frac{p_d^c}{P_{d,a}}\Big)^{-\eta_a} Y_a \Big] \\
& + \lambda^p \Big[z \Big(l_p^{\frac{\varepsilon_p - 1}{\varepsilon_p}} + d^{\frac{\varepsilon_p - 1}{\varepsilon_p}} \Big)^{\frac{\gamma_p \varepsilon_p}{\varepsilon_p - 1}} x_p^{\nu_p} n_p^{1-\nu_p-\gamma_p} - (1-\rho) \Big(\frac{p_d^p}{P_{d,a}}\Big)^{-\eta_a} Y_a \Big] \\
& + \mu \Big[h - \Big(l_c^{\frac{\varepsilon_h - 1}{\varepsilon_h}} + l_p^{\frac{\varepsilon_h - 1}{\varepsilon_h}} \Big)^{\frac{\varepsilon_h}{\varepsilon_h - 1}} \Big],
\end{aligned} \tag{66}$$

where λ^c and λ^p are Lagrange multipliers, $\mu \geq 0$ is the shadow value of plot land, and $\bar{l}_p = \big(l_p^{(\varepsilon_p - 1)/\varepsilon_p} + d^{(\varepsilon_p - 1)/\varepsilon_p}\big)^{\varepsilon_p/(\varepsilon_p - 1)}$. The FOCs are

$$\begin{aligned}
p_d^c :\quad & \lambda^c = \frac{\eta_a - 1}{\eta_a}\, p_d^c, & p_d^p :\quad & \lambda^p = \frac{\eta_a - 1}{\eta_a}\, p_d^p, \\
x_c :\quad & P_n x_c = \frac{\eta_a - 1}{\eta_a}\, \nu_c\, p_d^c\, y^c, & x_p :\quad & P_n x_p = \frac{\eta_a - 1}{\eta_a}\, \nu_p\, p_d^p\, y^p, \\
n_c :\quad & n_c = \frac{\eta_a - 1}{\eta_a}\, (1 - \nu_c - \gamma_c)\, p_d^c\, y^c, & n_p :\quad & n_p = \frac{\eta_a - 1}{\eta_a}\, (1 - \nu_p - \gamma_p)\, p_d^p\, y^p,
\end{aligned} \tag{67}$$

and the FOCs for l_c, l_p, and d are

$$l_c :\quad \frac{\eta_a - 1}{\eta_a}\, \gamma_c\, p_d^c\, y^c = \mu\, h^{\frac{1}{\varepsilon_h}}\, l_c^{\frac{\varepsilon_h - 1}{\varepsilon_h}}, \tag{68}$$

$$l_p :\quad \frac{\eta_a - 1}{\eta_a}\, \gamma_p\, p_d^p\, y^p \Big(\frac{l_p}{\bar{l}_p}\Big)^{\frac{\varepsilon_p - 1}{\varepsilon_p}} = \mu\, h^{\frac{1}{\varepsilon_h}}\, l_p^{\frac{\varepsilon_h - 1}{\varepsilon_h}}, \tag{69}$$

$$d: \quad \frac{\eta_a - 1}{\eta_a} \gamma_p \, p_d^p \, y^p \left(\frac{d}{\bar{l}_p}\right)^{\frac{\varepsilon_p - 1}{\varepsilon_p}} = (P_n c + \tau)\, d. \tag{70}$$

Since (68) gives $\mu = \frac{\eta_a - 1}{\eta_a} \gamma_c \, p_d^c \, y^c \, h^{-\frac{1}{\varepsilon_h}} l_c^{-\frac{\varepsilon_h - 1}{\varepsilon_h}} > 0$, the plot constraint binds, such that

$$l_c^{\frac{\varepsilon_h - 1}{\varepsilon_h}} + l_p^{\frac{\varepsilon_h - 1}{\varepsilon_h}} = h^{\frac{\varepsilon_h - 1}{\varepsilon_h}}, \qquad \bar{l}_p^{\frac{\varepsilon_p - 1}{\varepsilon_p}} = l_p^{\frac{\varepsilon_p - 1}{\varepsilon_p}} + d^{\frac{\varepsilon_p - 1}{\varepsilon_p}}. \tag{71}$$

Dividing (70) by (69) and rearranging,

$$\frac{d}{l_p} = \left[\frac{\mu \, (h/l_p)^{\frac{1}{\varepsilon_h}}}{P_n c + \tau}\right]^{\varepsilon_p}, \qquad \left(\frac{d}{\bar{l}_p}\right)^{\frac{\varepsilon_p - 1}{\varepsilon_p}} = \frac{1}{1 + \left(\frac{l_p}{d}\right)^{\frac{\varepsilon_p - 1}{\varepsilon_p}}}. \tag{72}$$

Substituting the FOCs for x_c and n_c and $y^c = \rho \, (p_d^c / P_{d,a})^{-\eta_a} Y_a$ into $y^c = z A_i \, l_c^{\gamma_c} x_c^{\nu_c} n_c^{1 - \nu_c - \gamma_c}$ and solving for $p_d^c \, y^c$ yields

$$p_d^c \, y^c = \left(\rho \, P_{d,a}^{\eta_a} Y_a\right)^{\frac{1}{1 + \gamma_c(\eta_a - 1)}} \left[z A_i \left(\frac{\eta_a - 1}{\eta_a}\right)^{1 - \gamma_c} \left(\frac{\nu_c}{P_n}\right)^{\nu_c} (1 - \nu_c - \gamma_c)^{1 - \nu_c - \gamma_c}\right]^{\frac{\eta_a - 1}{1 + \gamma_c(\eta_a - 1)}} l_c^{\frac{\gamma_c(\eta_a - 1)}{1 + \gamma_c(\eta_a - 1)}}, \tag{73}$$

and similarly for $p_d^p \, y^p$:

$$p_d^p \, y^p = \left((1 - \rho) \, P_{d,a}^{\eta_a} Y_a\right)^{\frac{1}{1 + \gamma_p(\eta_a - 1)}} \left[z \left(\frac{\eta_a - 1}{\eta_a}\right)^{1 - \gamma_p} \left(\frac{\nu_p}{P_n}\right)^{\nu_p} (1 - \nu_p - \gamma_p)^{1 - \nu_p - \gamma_p}\right]^{\frac{\eta_a - 1}{1 + \gamma_p(\eta_a - 1)}} \bar{l}_p^{\frac{\gamma_p(\eta_a - 1)}{1 + \gamma_p(\eta_a - 1)}}. \tag{74}$$

Substituting (67) and (68)–(71) into the objective yields operating profit

$$\pi_i^o = \frac{p_d^c \, y^c + p_d^p \, y^p}{\eta_a} + \mu \, h. \tag{75}$$

By the envelope theorem it can then be shown that

$$\frac{\partial \pi_i^o}{\partial \tau} = -d, \qquad \frac{\partial \pi_i^o}{\partial \ln A_i} = \frac{\eta_a - 1}{\eta_a} \, p_d^c \, y^c. \tag{76}$$

Integrating gives

$$\pi_m^o - \pi_t^o = \frac{\eta_a - 1}{\eta_a} \int_{\ln A_t}^{\ln A_m} p_d^c \, y^c(z, c, A) \, d\ln A. \tag{77}$$

Totally differentiating (68)–(71) and eliminating $d\ln(p_d^c \, y^c)$ and $d\ln(p_d^p \, y^p)$ by (73)–(74) yields

$$d\ln l_c = \frac{\varepsilon_h \left[d\ln\left(P_{d,a}^{\eta_a} Y_a\right) + (\eta_a - 1)\left(d\ln z + d\ln A_i\right) - M_c \, d\ln\mu\right]}{\varepsilon_h - M_c}, \tag{78}$$

$$d\ln l_p = \frac{\varepsilon_h}{\varepsilon_h - M_p} \Big[d\ln\left(P_{d,a}^{\eta_a} Y_a\right) + (\eta_a - 1)\, d\ln z - M_p \, d\ln\mu - \left(\frac{d}{\bar{l}_p}\right)^{\frac{\varepsilon_p - 1}{\varepsilon_p}} \left[\gamma_p(\eta_a - 1) - (\varepsilon_p - 1)\right] d\ln(P_n c + \tau)\Big], \tag{79}$$

$$d\ln d = \left(1 - \frac{\varepsilon_p}{\varepsilon_h}\right) d\ln l_p + \varepsilon_p \left[d\ln\mu - d\ln(P_n c + \tau)\right], \tag{80}$$

$$0 = \left(\frac{l_c}{h}\right)^{\frac{\varepsilon_h - 1}{\varepsilon_h}} d\ln l_c + \left(\frac{l_p}{h}\right)^{\frac{\varepsilon_h - 1}{\varepsilon_h}} d\ln l_p, \tag{81}$$

where

$$M_c \equiv 1+\gamma_c(\eta_a-1) \qquad \text{and} \qquad M_p \equiv 1+\gamma_p(\eta_a-1)\left(\tfrac{l_p}{l_p}\right)^{\frac{\varepsilon_p-1}{\varepsilon_p}}+(\varepsilon_p-1)\left(\tfrac{d}{l_p}\right)^{\frac{\varepsilon_p-1}{\varepsilon_p}}. \tag{82}$$

By (65), $1<M_c, M_p<\varepsilon_h$, so both denominators are strictly positive. Substituting (78)–(79) into (81) and multiplying by $\frac{1}{\varepsilon_h}(\varepsilon_h-M_c)(\varepsilon_h-M_p)>0$,

$$\begin{aligned}
0=&\left(\tfrac{l_c}{h}\right)^{\frac{\varepsilon_h-1}{\varepsilon_h}}(\varepsilon_h-M_p)\Big[d\ln\big(P_{d,a}^{\eta_a}Y_a\big)+(\eta_a-1)\big(d\ln z+d\ln A_i\big)-M_c\,d\ln\mu\Big]\\
&+\left(\tfrac{l_p}{h}\right)^{\frac{\varepsilon_h-1}{\varepsilon_h}}(\varepsilon_h-M_c)\Big[d\ln\big(P_{d,a}^{\eta_a}Y_a\big)+(\eta_a-1)\,d\ln z-M_p\,d\ln\mu\\
&\qquad-\left(\tfrac{d}{l_p}\right)^{\frac{\varepsilon_p-1}{\varepsilon_p}}\big[\gamma_p(\eta_a-1)-(\varepsilon_p-1)\big]\,d\ln(P_nc+\tau)\Big],
\end{aligned} \tag{83}$$

which pins down $d\ln\mu$. Since both coefficients are strictly positive, the two brackets in (83) have opposite signs, and by (78)–(79) $d\ln l_c$ and $d\ln l_p$ have the same sign as the first and the second bracket, respectively.

J.2.2 Hump-Shape Modernization Gains

Consider a common shift $d\ln\big(z^{\eta_a-1}P_{d,a}^{\eta_a}Y_a\big)$ at fixed A_i and $P_nc+\tau$. Solving (83),

$$\frac{d\ln\mu}{d\ln\big(z^{\eta_a-1}P_{d,a}^{\eta_a}Y_a\big)}=\frac{\left(\frac{l_c}{h}\right)^{\frac{\varepsilon_h-1}{\varepsilon_h}}(\varepsilon_h-M_p)+\left(\frac{l_p}{h}\right)^{\frac{\varepsilon_h-1}{\varepsilon_h}}(\varepsilon_h-M_c)}{\left(\frac{l_c}{h}\right)^{\frac{\varepsilon_h-1}{\varepsilon_h}}(\varepsilon_h-M_p)\,M_c+\left(\frac{l_p}{h}\right)^{\frac{\varepsilon_h-1}{\varepsilon_h}}(\varepsilon_h-M_c)\,M_p}>0. \tag{84}$$

Log-differentiating (68),

$$d\ln(p_d^c\,y^c)=d\ln\mu+\frac{\varepsilon_h-1}{\varepsilon_h}\,d\ln l_c, \tag{85}$$

and substituting (78) into (85),

$$\left.\frac{d\ln(p_d^c\,y^c)}{d\ln\big(z^{\eta_a-1}P_{d,a}^{\eta_a}Y_a\big)}\right|_{A_i}=\frac{\dfrac{\varepsilon_h-1}{\varepsilon_h}-\gamma_c(\eta_a-1)\,\dfrac{d\ln\mu}{d\ln\big(z^{\eta_a-1}P_{d,a}^{\eta_a}Y_a\big)}}{\dfrac{\varepsilon_h-M_c}{\varepsilon_h}}. \tag{86}$$

Differentiating (77) under the integral and substituting (86),

$$\frac{d\big[\pi_m^o-\pi_t^o\big]}{d\ln\big(z^{\eta_a-1}P_{d,a}^{\eta_a}Y_a\big)}=\frac{\eta_a-1}{\eta_a}\int_{\ln A_t}^{\ln A_m}p_d^c\,y^c\,\frac{\frac{\varepsilon_h-1}{\varepsilon_h}-\gamma_c(\eta_a-1)\,\frac{d\ln\mu}{d\ln\left(z^{\eta_a-1}P_{d,a}^{\eta_a}Y_a\right)}}{\frac{\varepsilon_h-M_c}{\varepsilon_h}}\,d\ln A. \tag{87}$$

Since $p_d^c\,y^c>0$ and $\varepsilon_h-M_c>0$, it can be shown that the gain is increasing (decreasing) whenever, for every $A_i\in[A_t,A_m]$,

$$\frac{d\ln\mu}{d\ln\big(z^{\eta_a-1}P_{d,a}^{\eta_a}Y_a\big)}\;<\;(>)\;\frac{\varepsilon_h-1}{\varepsilon_h\,\gamma_c(\eta_a-1)}. \tag{88}$$

Suppose $M_p\geq M_c$, then indeed by (84)

$$\frac{d\ln\mu}{d\ln\big(z^{\eta_a-1}P_{d,a}^{\eta_a}Y_a\big)}\;\leq\;\frac{1}{M_c}\;=\;\frac{1}{1+\gamma_c(\eta_a-1)}\;<\;\frac{\varepsilon_h-1}{\varepsilon_h\,\gamma_c(\eta_a-1)}, \tag{89}$$

where the last inequality follows from $\gamma_c(\eta_a - 1) < \varepsilon_h - 1$ in (65). Subtracting the two definitions in (82) and eliminating $(l_p/\bar{l}_p)^{(\varepsilon_p-1)/\varepsilon_p} = 1 - (d/\bar{l}_p)^{(\varepsilon_p-1)/\varepsilon_p}$ by (71),

$$M_p - M_c = (\gamma_p - \gamma_c)(\eta_a - 1) - \left(\tfrac{d}{\bar{l}_p}\right)^{\frac{\varepsilon_p - 1}{\varepsilon_p}} \big[\gamma_p(\eta_a - 1) - (\varepsilon_p - 1)\big], \tag{90}$$

so, given $\gamma_p(\eta_a - 1) - (\varepsilon_p - 1) > 0$ from (65),

$$M_p \geq M_c \quad \text{if and only if} \quad \left(\tfrac{d}{\bar{l}_p}\right)^{\frac{\varepsilon_p - 1}{\varepsilon_p}} \leq \frac{(\gamma_p - \gamma_c)(\eta_a - 1)}{\gamma_p(\eta_a - 1) - (\varepsilon_p - 1)}. \tag{91}$$

By (88), (89), and (91), the modernization gain is therefore strictly increasing whenever $(d/\bar{l}_p)^{(\varepsilon_p-1)/\varepsilon_p}$ lies below the right-hand side of (91).

Let $z \to \infty$, then it is straightforward to show that

$$\frac{d}{\bar{l}_p} \to \infty \qquad \text{and} \qquad l_c \to 0. \tag{92}$$

By (72), $(d/\bar{l}_p)^{(\varepsilon_p-1)/\varepsilon_p} \to 1$, such that by (82) $M_p \to \varepsilon_p$, and taking these limits in (84),

$$\frac{d \ln \mu}{d \ln\big(z^{\eta_a - 1} P_{d,a}^{\eta_a} Y_a\big)} \longrightarrow \frac{1}{\varepsilon_p} > \frac{\varepsilon_h - 1}{\varepsilon_h\, \gamma_c(\eta_a - 1)},$$

where the inequality follows from the last condition in (64). Hence, by (88), the modernization gain is strictly decreasing for all sufficiently large z. Moreover, the numerator on the right-hand side of (86) tends to $\frac{\varepsilon_h - 1}{\varepsilon_h} - \frac{\gamma_c(\eta_a - 1)}{\varepsilon_p} < 0$, so the right-hand side of (86) is eventually negative and bounded away from zero, such that $p_d^c y^c \to 0$, and, by (77),

$$\pi_m^o - \pi_t^o \longrightarrow 0 \qquad \text{as } z \to \infty. \tag{93}$$

It follows that the set $\{z : \pi_m^o - \pi_t^o \geq (1 + r) P_n \kappa_{ma}\}$ is bounded, so $\bar{z}_{ma}(c) < \infty$.

J.2.3 Proof of Proposition 2

Part (1). By the envelope theorem $\partial \pi_t^o / \partial \ln\big(P_{d,a}^{\eta_a} Y_a\big) = (p_d^c y^c + p_d^p y^p)/\eta_a > 0$. Totally differentiating $\pi_t^o(z_{ta}(c), c) = \kappa_{ta}$ therefore gives

$$\frac{\partial \ln z_{ta}(c)}{\partial \ln P_{d,a}} = -\frac{\eta_a}{\eta_a - 1} < 0 \qquad \text{for all } c. \tag{94}$$

We set $d \ln\big(P_{d,a}^{\eta_a} Y_a\big) = \eta_a\, d \ln P_{d,a} > 0$ and $d \ln z = d \ln A_i = d \ln(P_n c + \tau) = 0$ in (83). By (84), $d \ln \mu / d \ln P_{d,a} > 0$. The difference between the first and the second bracket of (83) equals $(M_p - M_c)\, d \ln \mu$, and since the two brackets have opposite signs, the first bracket, and hence $d \ln l_c / d \ln P_{d,a}$, is positive if and only if $M_p > M_c$. Thus (90)–(91),

$$\begin{aligned} \frac{d \ln l_c}{d \ln P_{d,a}} &< 0 \quad \text{if} \quad \left(\tfrac{d}{\bar{l}_p}\right)^{\frac{\varepsilon_p - 1}{\varepsilon_p}} > q, \\ \frac{d \ln l_c}{d \ln P_{d,a}} &> 0 \quad \text{if} \quad \left(\tfrac{d}{\bar{l}_p}\right)^{\frac{\varepsilon_p - 1}{\varepsilon_p}} < q, \end{aligned} \qquad \text{where} \quad q \equiv \frac{(\gamma_p - \gamma_c)(\eta_a - 1)}{\gamma_p(\eta_a - 1) - (\varepsilon_p - 1)}. \tag{95}$$

Note that $q > 0$ if $\gamma_c < \gamma_p$ and $q < 1$ by $\varepsilon_p - 1 < \gamma_c(\eta_a - 1)$ from (65). It can then be shown that $(d/\bar{l}_p)^{(\varepsilon_p-1)/\varepsilon_p}$ is strictly decreasing in c and that, at $\tau = 0$,

$$\left(\tfrac{d}{\bar{l}_p}\right)^{\frac{\varepsilon_p-1}{\varepsilon_p}} \longrightarrow 1 \quad \text{as } c \to 0 \qquad \text{and} \qquad \left(\tfrac{d}{\bar{l}_p}\right)^{\frac{\varepsilon_p-1}{\varepsilon_p}} \longrightarrow 0 \quad \text{as } c \to \infty.$$

If $\gamma_c < \gamma_p$, $(d/\bar{l}_p)^{(\varepsilon_p-1)/\varepsilon_p}$ therefore crosses q exactly once, from above, at some $c^\star \in (0, \infty)$ (by continuity and monotonicity). If $\gamma_c = \gamma_p$, it follows that $q = 0$ such that $(d/\bar{l}_p)^{(\varepsilon_p-1)/\varepsilon_p} > q$ for all c, and hence $c^\star = \infty$.

This means that for $c < c^\star$, $(d/\bar{l}_p)^{(\varepsilon_p-1)/\varepsilon_p} > q$, so by (95) and (81),

$$\frac{d\ln l_c}{d\ln P_{d,a}} < 0 \qquad \text{and} \qquad \frac{d\ln l_p}{d\ln P_{d,a}} > 0,$$

and vice versa for $c > c^\star$. For $c < c^\star$, by (80) with $d\ln(P_n c + \tau) = 0$ and $\varepsilon_p < \varepsilon_h$ from (65),

$$\frac{d\ln d}{d\ln P_{d,a}} = \left(1 - \frac{\varepsilon_p}{\varepsilon_h}\right)\frac{d\ln l_p}{d\ln P_{d,a}} + \varepsilon_p \frac{d\ln \mu}{d\ln P_{d,a}} \;>\; 0.$$

Finally, since $s_c = l_c/(l_c + l_p + d)$, the quotient rule gives

$$\frac{ds_c}{d\ln P_{d,a}} = \frac{l_c}{(l_c + l_p + d)^2}\left[(l_p + d)\frac{d\ln l_c}{d\ln P_{d,a}} - l_p \frac{d\ln l_p}{d\ln P_{d,a}} - d\,\frac{d\ln d}{d\ln P_{d,a}}\right] \;<\; 0.$$

This proves part (1).

Part (2). Using the envelope theorem it is straightforward to show that

$$\frac{\partial \ln z_{ma}(c)}{\partial \ln P_{d,a}} = \frac{\partial \ln \bar{z}_{ma}(c)}{\partial \ln P_{d,a}} = -\frac{\eta_a}{\eta_a - 1} < 0 \qquad \text{for all } c, \tag{96}$$

and since

$$e_a = \int_0^\infty z_{ta}(c)^{-\zeta}\, G(dc) \qquad \text{and} \qquad a_a = \int_0^\infty \left[z_{ma}(c)^{-\zeta} - \bar{z}_{ma}(c)^{-\zeta}\right] G(dc)$$

the partial derivative $\partial(a_a/e_a)/\partial P_{d,a} = 0$ follows immediately. This proves part (2).

Part (3). Implicit differentiation of $\pi_m^o - \pi_t^o = (1+r)P_n\kappa_{ma}$ gives

$$\frac{\partial z_{ma}}{\partial \kappa_{ma}} > 0, \qquad \frac{\partial \bar{z}_{ma}}{\partial \kappa_{ma}} < 0,$$

it follows that $\frac{\partial a_a}{\partial \kappa_{ma}} < 0$ and $\frac{\partial e_a}{\partial \kappa_{ma}} = 0$. Set $d\ln A_i \neq 0$ and all other changes to zero in (83) such that

$$\frac{d\ln \mu}{d\ln A_i} = \frac{\left(\frac{l_c}{h}\right)^{\frac{\varepsilon_h-1}{\varepsilon_h}}(\varepsilon_h - M_p)(\eta_a - 1)}{\left(\frac{l_c}{h}\right)^{\frac{\varepsilon_h-1}{\varepsilon_h}}(\varepsilon_h - M_p)M_c + \left(\frac{l_p}{h}\right)^{\frac{\varepsilon_h-1}{\varepsilon_h}}(\varepsilon_h - M_c)M_p} \;>\; 0.$$

Substituting into (78),

$$\frac{d\ln l_c}{d\ln A_i} = \frac{\varepsilon_h(\eta_a - 1)\left(\frac{l_p}{h}\right)^{\frac{\varepsilon_h-1}{\varepsilon_h}} M_p}{\left(\frac{l_c}{h}\right)^{\frac{\varepsilon_h-1}{\varepsilon_h}}(\varepsilon_h - M_p)M_c + \left(\frac{l_p}{h}\right)^{\frac{\varepsilon_h-1}{\varepsilon_h}}(\varepsilon_h - M_c)M_p} \;>\; 0,$$

so by (81), $d\ln l_p/d\ln A_i < 0$ follows immediately. Substituting $d\ln l_p/d\ln A_i = -\frac{\varepsilon_h M_p}{\varepsilon_h - M_p}\, d\ln\mu/d\ln A_i$ from (79) into (80),

$$\frac{d\ln d}{d\ln A_i} = \frac{\varepsilon_h(\varepsilon_p - M_p)}{\varepsilon_h - M_p}\,\frac{d\ln\mu}{d\ln A_i} < 0,$$

since $M_p - \varepsilon_p = \left(\frac{l_p}{l_p}\right)^{\frac{\varepsilon_p - 1}{\varepsilon_p}}\left[\gamma_p(\eta_a - 1) - (\varepsilon_p - 1)\right] > 0$ by (71) and (65). Hence, for farms that switch from A_m to A_t after a marginal increase in κ_{ma} ($d\ln A_i < 0$), l_c falls, l_p and d rise, and (as in part (1)) s_c falls. This proves part (3).

Part (4). Note that $d\ln(P_n c + \tau) = d\tau/(P_n c + \tau) > 0$, so from equation (83) (setting all other changes to zero) it follows that

$$\frac{d\ln\mu}{d\ln(P_n c + \tau)} = \frac{-\left(\frac{l_p}{h}\right)^{\frac{\varepsilon_h - 1}{\varepsilon_h}}(\varepsilon_h - M_c)\left(\frac{d}{l_p}\right)^{\frac{\varepsilon_p - 1}{\varepsilon_p}}\left[\gamma_p(\eta_a - 1) - (\varepsilon_p - 1)\right]}{\left(\frac{l_c}{h}\right)^{\frac{\varepsilon_h - 1}{\varepsilon_h}}(\varepsilon_h - M_p)\, M_c + \left(\frac{l_p}{h}\right)^{\frac{\varepsilon_h - 1}{\varepsilon_h}}(\varepsilon_h - M_c)\, M_p} < 0. \tag{97}$$

and by (78),

$$\frac{d\ln l_c}{d\ln(P_n c + \tau)} = -\frac{\varepsilon_h M_c}{\varepsilon_h - M_c}\,\frac{d\ln\mu}{d\ln(P_n c + \tau)} > 0,$$

and by (81), $d\ln l_p/d\ln(P_n c + \tau) < 0$. By (80), since $\varepsilon_p < \varepsilon_h$ from (65),

$$\frac{d\ln d}{d\ln(P_n c + \tau)} = \left(1 - \frac{\varepsilon_p}{\varepsilon_h}\right)\frac{d\ln l_p}{d\ln(P_n c + \tau)} + \varepsilon_p\left[\frac{d\ln\mu}{d\ln(P_n c + \tau)} - 1\right] < 0.$$

Since $s_c = l_c/(l_c + l_p + d)$, the quotient rule gives

$$\frac{ds_c}{d\ln(P_n c + \tau)} = \frac{l_c}{(l_c + l_p + d)^2}\left[(l_p + d)\,\frac{d\ln l_c}{d\ln(P_n c + \tau)} - l_p\,\frac{d\ln l_p}{d\ln(P_n c + \tau)} - d\,\frac{d\ln d}{d\ln(P_n c + \tau)}\right] > 0.$$

Substituting $d\ln l_c/d\ln(P_n c + \tau)$ into (85) gives

$$\frac{d\ln(p_d^c\, y^c)}{d\ln(P_n c + \tau)} = -\frac{\varepsilon_h\,\gamma_c(\eta_a - 1)}{\varepsilon_h - M_c}\cdot\frac{d\ln\mu}{d\ln(P_n c + \tau)} > 0, \tag{98}$$

so by (77) the modernization gain rises for a marginal increase in τ. Since, as in part (3), the gain crosses $(1 + r)P_n\kappa_{ma}$ from below at z_{ma} and from above at $\overline{z}_{ma}$,

$$\frac{\partial z_{ma}}{\partial\tau} < 0, \qquad \frac{\partial\overline{z}_{ma}}{\partial\tau} > 0,$$

so $\frac{\partial a_a}{\partial\tau} > 0$. Then, $\frac{\partial z_{ta}}{\partial\tau} > 0$ follows from $\frac{\partial\pi_t^o}{\partial\tau} = -d < 0$ which means that $\frac{\partial e_a}{\partial\tau} < 0$. Hence, $\frac{\partial a_a/e_a}{\partial\tau} > 0$. This proves part (4). □

J.2.4 Proof of Propositions 3 & 4

Recall that $P_c C = N + \Pi_a + \Pi_n + \tau D$ and $F(z) = 1 - z^{-\zeta}$, so $F(dz) = f(z)\,dz$ with $f(z) = \zeta\, z^{-\zeta-1}$. We can write $\Pi_a + \tau D$ as

$$\begin{aligned}\Pi_a + \tau D = &\int_0^\infty\left[\int_{z_{ta}}^{z_{ma}}\pi_t^o\, f\, dz + \int_{z_{ma}}^{\overline{z}_{ma}}\pi_m^o\, f\, dz + \int_{\overline{z}_{ma}}^\infty\pi_t^o\, f\, dz\right]G(dc)\\ &- \kappa_{ta}\int_0^\infty z_{ta}^{-\zeta}\, G(dc) - P_n\kappa_{ma}\int_0^\infty\left[z_{ma}^{-\zeta} - \overline{z}_{ma}^{-\zeta}\right]G(dc) + \tau D.\end{aligned} \tag{99}$$

Applying Leibniz's rule and $\partial \pi_i^o / \partial \tau = -d$ from (76) to each term gives

$$\frac{\partial(\Pi_a + \tau D)}{\partial \tau} = -D + r\, P_n \kappa_{ma} \frac{\partial a_a}{\partial \tau} + D + \tau \frac{\mathrm{d}D}{\mathrm{d}\tau} = r\, P_n \kappa_{ma} \frac{\partial a_a}{\partial \tau} + \tau \frac{\mathrm{d}D}{\mathrm{d}\tau}\,, \tag{100}$$

and hence

$$\frac{\partial C}{\partial \tau} = \frac{1}{P_c}\Big[\, r\, P_n \kappa_{ma} \frac{\partial a_a}{\partial \tau} \; + \; \tau \frac{\mathrm{d}D}{\mathrm{d}\tau} \Big]. \tag{101}$$

Proposition 4 follows immediately from the fundamental theorem of calculus. □

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